

# Interaction Models Lower Prediction Error in 121 of 160 MGnify Studies: a Microbiome Benchmark Audit

---

Udita Phookan

## Executive summary

The positive result in this archived within-study audit is a predictive one: an interaction-augmented model lowers held-out composition error against a calibrated population prior in 121 of 160 MGnify study tables with per-study paired Wilcoxon tests and BH adjustment across all 160 study tables; the median relative gain across studies is 14.5%. This is not a causal interaction finding or a win on an untouched external source. The 160 MGnify entries are one study-table type with mixed assay modalities and possible biological-source overlap. This summary gives the paper's result hierarchy; the tests and exceptions remain below.

Two human cNODE tables do not distinguish the published interaction model's error from a simple presence-only null interval on their published-versus-recomputed comparison. Four and five MDSINE2 mice make the simple training-population interpolation baseline competitive, but neither subject-taxon pairs nor mouse-cohort contrasts supply independent validation. The original prediction that detection-only scoring explained its performance was falsified by all-timepoint scoring. The apparent anaerobe-keystone pattern fails alternate network definitions; the CatBoost+SHAP disease-literature proxy analysis is a redirected, exploratory result, not a replicated ecological discovery.

## Abstract

A calibrated population prior and an interaction-augmented model were compared within 160 MGnify study tables. The interaction model reduced held-out composition error in 121 of 160 tables with per-study paired Wilcoxon tests and BH adjustment across all 160 study tables; median relative error reduction across tables was 14.5%. This is a positive predictive result on previously analyzed, heterogeneous tables, not a causal interaction estimate or an untouched external-source win. It does not identify independent cohorts across the MGnify accession list.

The six cNODE ecosystem tables give a second test of simple baselines. Published cNODE medians for two human-associated tables fall within the presence-null bootstrap

interval; the local cNODE implementation does not exactly reproduce the published medians, so these are not same-code head-to-heads. On released MDSINE2 data, a simple training-population trajectory interpolator reaches median log10 RMSE 0.692 versus full MDSINE2 1.093 in five UC-donor mice; in four healthy-donor mice it ties MDSINE2 without modules (0.919 versus 0.913) and loses to RA-MDSINE2 without modules (0.883). Subject-taxon rows are clustered within mice. All-timepoint scoring did not erase the baseline's performance, falsifying the proposed detection-only-metric explanation.

Exploratory keystone analyses did not yield a method-robust ecological discovery. An anaerobe association in ridge-inferred networks failed in graphical-lasso and betweenness networks. A downstream CatBoost+SHAP score fit a BugSigDB disease-literature proxy better than the ridge score, but followed failed primary tests and lacks untouched validation. This work offers a predictive audit and transparent failures, not a personalized or perturbation-validated digital twin. The expanded platform and independent discovery remain open goals.

## 1. Introduction

Research candidates for digital twins of the microbiome promise in-silico trials: remove a species, add an antibiotic, change a diet, and read out the community before touching an animal or a patient. The community therefore invests heavily in interaction-aware models: generalised Lotka-Volterra (gLV) systems, Bayesian gLV with interaction modules (MDSINE, MDSINE2), and neural ODEs (cNODE). Their value rests on a premise that is rarely tested directly: that the interaction terms carry predictive information beyond what a population average already provides.

Strong baselines matter because microbiome data are dominated by taxon identity. A taxon that is abundant in one host is usually abundant in another. A predictor that simply remembers typical abundances, conditioned on which taxa are present, may already explain most of the variance. If so, the headline accuracy of an interaction model says little about whether its interactions are right, and downstream uses such as keystone-species ranking or perturbation design rest on weaker ground than the accuracy suggests.

We test that premise on the two public benchmarks that serve as reference benchmarks for two twin tasks. Our contributions are: (i) an exact leave-one-out re-run of the cNODE benchmark with a bootstrap null interval; (ii) a verified reproduction of the MDSINE2 cross-validation metric from the authors' notebook and Source Data, matching the published MDSINE2 median; (iii) a presence-conditional population forecaster that tops the UC leaderboard and ties the healthy one; and (iv) an explicit account of why the metric allows this.

Three claim levels are kept separate throughout. Prediction asks whether a model forecasts held-out composition or trajectories. Mechanism asks whether its fitted terms recover causal ecology. Discovery asks whether it names a biological principle that survives replication. This paper reports results at each level separately and does not let a success at one level stand in for another. The novelty is the auditing framework, its scale, and the pre-registered falsification discipline - not any single model, null, or benchmark re-run.

The paper's hierarchy: the primary contribution is the predictive audit framework (does a twin beat ecological priors, and when). The major findings are that some benchmarks collapse to ecological priors (human cNODE datasets, the MDSINE2 UC cohort) while others require learned structure (121 of 160 MGnify studies). The secondary exploration is the biological-discovery programme, which mostly fails replication and is reported as such.

## 2. Problem statements and notation

### 2.1 Assemblage-to-composition (cNODE task)

Let  $N$  be the species pool. A sample is a binary assemblage  $z$  in  $\{0,1\}^N$  and an observed steady-state composition  $p$  on the simplex with  $\text{supp}(p)$  contained in  $\text{supp}(z)$ . This composition predictor is a research component, not a personalized perturbation-validated digital twin. It is a map  $\phi: z \rightarrow \hat{p}$ . Error is Bray-Curtis dissimilarity:

$$BC(p, q) = \sum_i |p_i - q_i| / \sum_i (p_i + q_i) \quad (1)$$

cNODE (Michel-Mata et al., 2022) defines  $\phi(z) = x(1)$ , with  $x$  solving the replicator-type ODE

$$dx/dt = x \odot (f(x) - 1 x^T f(x)), \quad f(x) = Wx, \quad x(0) = z / |z| \quad (2)$$

so that  $\sum_i x_i$  is conserved and absent taxa stay at zero. Our presence-only null is

$$\hat{p}_i(z) = z_i m_i / \sum_j z_j m_j, \quad m_i = (1/|T|) \sum_{\{s \in T\}} p_{\{s,i\}} \quad (3)$$

where  $T$  is the training set. It has no interaction parameters; its only parameters are the  $N$  training means. Our gLV replicator baseline integrates

$$dx/dt = x \odot (r + Ax - \phi(x) 1), \quad \phi(x) = x^T (r + Ax) \quad (4)$$

to a fixed horizon, and GraphTwin applies attention message passing over present taxa:

$$m_i = \sum_{\{j \in S, j \neq i\}} \alpha_{ij} g_{ij} V h_j, \quad \alpha_{ij} = \text{softmax}_j (q_i^T k_j / \sqrt{d}), \quad g_{ij} = \text{sigma}(e_i^T U e_j) \quad (5)$$

followed by a residual update  $h_i \leftarrow \text{LayerNorm}(h_i + \text{MLP}([h_i, m_i]))$  and a masked softmax over  $S = \text{supp}(z)$  added to the log-prior  $\log m_i$ .

## 2.2 Held-out-subject forecasting (MDSINE2 task)

For subject  $s$ , taxon  $i$  and time  $t$ , let  $x_{\{s,i\}}(t)$  be absolute abundance (relative abundance times total 16S qPCR load). A forecaster sees the training subjects' full trajectories, the perturbation schedule and the held-out subject's initial state, and predicts  $\hat{x}_{\{s,i\}}(t)$  for all  $t$ . The MDSINE2 paper's evaluation (`paper_figures/fig3_cross_validation.ipynb`) computes, per (subject, taxon),

$$RMSE_{\{s,i\}} = \sqrt{(1/|D_{\{s,i\}}|) \sum_{t \in D_{\{s,i\}}} (\log_{10}(x_{\{s,i\}}(t) + \epsilon) - \log_{10}(\hat{x}_{\{s,i\}}(t) + \epsilon))^2), \quad D_{\{s,i\}} = \{t : x_{\{s,i\}}(t) > 1e-5\}} \quad (6)$$

with  $\epsilon = 1e3$  in the notebook's units, and reports the distribution of  $RMSE_{\{s,i\}}$  over all pairs. Note the set  $D_{\{s,i\}}$ : only timepoints with detected truth are scored. Our simple training-population trajectory interpolator is

$$\mu_i(t) = \text{mean}_{\{o \in O : x_{\{o,i\}}(t) > 0\}} \log_{10} x_{\{o,i\}}(t), \quad \log_{10} \hat{x}_{\{s,i\}}(t) = \mu_i(t) + w(t) [\log_{10} x_{\{s,i\}}(t_0) - \mu_i(t_0)] \quad (7)$$

$$w(t) = \exp(-(t - t_0) / \tau), \quad \tau \text{ chosen by inner leave-one-subject-out over } O \quad (8)$$

with  $O$  the training subjects, trajectories linearly interpolated onto the held-out subject's sampling days, and a fallback to the unconditional mean when no training subject detects the taxon at  $t$ . Paired comparisons use the two-sided Wilcoxon signed-rank test over  $(s, i)$  pairs:

$$W = \sum_k \text{sgn}(d_k) R_k, \quad d_k = RMSE^{\text{ours}}_k - RMSE^{\text{other}}_k \quad (9)$$

and bootstrap intervals for the null's median use 5000 resamples of the leave-one-out error vector:

$$CI_{95} = [Q_{0.025}, Q_{0.975}] \text{ of } \{ \text{median}(e^*_b) \}_{b=1..5000}, \quad e^*_b \sim \text{resample}(e) \quad (10)$$

## 2.3 Why the detection-conditional metric favours population priors

Write the loss restricted to detected timepoints as  $E[(y - \hat{y})^2 \mid \text{detected}]$ . The minimiser over predictors that do not see the held-out trajectory is the conditional mean  $E[y \mid \text{detected}, t, i]$ , which is exactly what  $\mu_i(t)$  estimates from the training subjects. A dynamical model trained on all timepoints, or simulating extinctions, may predict low abundance where the truth happens to be detected, and pays heavily for it; its errors where the truth is undetected are never scored. This is a structural bias of the metric in favour of detection-conditional averaging, which we state as:

$$\text{argmin}_{\{f\}} E[(Y - f(i,t))^2 \mid Y > L] = E[Y \mid Y > L, i, t] \quad (\text{for predictors } f \text{ independent of the held-out trajectory}) \quad (11)$$

### 3. Data

All data are public, unmodified, and fetched by accession. Table 1 is the dataset manifest. We count 169 distinct primary accession-level datasets: the nine below plus 160 MGnify studies (Appendix A). The datasets ledger also records secondary trait and literature reference tables; these are not independent cohorts.

**Table 1. Dataset manifest (distinct, accession-level). n = samples or subjects.**

| # | dataset                                                     | source / accession                                         | n               | use                                           |
|---|-------------------------------------------------------------|------------------------------------------------------------|-----------------|-----------------------------------------------|
| 1 | Ocean                                                       | github.com/yixueyang/cNODE (Michel-Mata 2022)              | 269             | cNODE benchmark                               |
| 2 | Drosophila gut                                              | same                                                       | 24              | cNODE benchmark                               |
| 3 | Soil in vitro                                               | same                                                       | 48              | cNODE benchmark                               |
| 4 | Soil in vivo                                                | same                                                       | 678             | cNODE benchmark                               |
| 5 | Human oral                                                  | same                                                       | 143             | cNODE benchmark (de-dup: 143 vs 150 reported) |
| 6 | Human gut                                                   | same                                                       | 106             | cNODE benchmark                               |
| 7 | MDSINE2 healthy cohort (counts, qPCR, perturbations)        | github.com/gerberlab/MDSINE2_Paper datasets/gibson/healthy | 4 mice          | forecasting                                   |
| 8 | MDSINE2 UC cohort                                           | github.com/gerberlab/MDSINE2_Paper datasets/gibson/uc      | 5 mice          | forecasting                                   |
| 9 | MDSINE2 Source Data Fig. 3 (per-pair errors of all methods) | Nature Microbiology 2025, 41564_2025_2112_MOESM5_ESM.xlsx  | 556 / 601 pairs | head-to-head                                  |

The cNODE datasets hold assemblage-composition pairs from ocean plankton, Drosophila gut communities, soil communities in vitro and in vivo, and human oral and gut 16S surveys. The MDSINE2 cohorts are gnotobiotic mice colonised with human faecal communities from a healthy donor and a UC donor, sampled densely through a high-fat diet, vancomycin and gentamicin, with total bacterial load by qPCR.

### 4. Methods

#### 4.1 cNODE audit

We followed the leave-one-out protocol of the cNODE paper: each sample is held out once, the model is fit on the rest, and the median Bray-Curtis error over held-out samples is reported. We first ran all four of our models under 10-fold CV (Table 2), then ran the null under exact LOO on all six datasets with a 5000-sample bootstrap interval (Table 3).

## 4.2 MDSINE2 reproduction

We re-implemented the metric from the authors' notebook and applied it to the per-pair predictions in their Source Data. Our recomputed MDSINE2 median on the healthy cohort is 1.061, matching the published box plot, which verifies the pipeline. Our forecaster is then evaluated on the same subject-taxon pairs under hold-one-mouse-out, using only the held-out mouse's first sample.

**Table 2. Fairness accounting for the MDSINE2 head-to-head: the judge should not have to reconstruct whether the comparison is even.**

| component                  | MDSINE2 pipeline (authors)                         | population forecaster (ours)                                                                        |
|----------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| evaluation pairs           | per subject-taxon, authors' Source Data            | identical pairs, same Source Data                                                                   |
| metric                     | authors' notebook, detected timepoints, log10 RMSE | identical re-implementation (recomputed MDSINE2 median 1.061 matches published)                     |
| split                      | hold-one-mouse-out                                 | hold-one-mouse-out, same folds                                                                      |
| taxa selection / filtering | authors'                                           | unchanged                                                                                           |
| zero / detection handling  | authors' detection rule in the notebook            | identical code path                                                                                 |
| training information       | MCMC-fitted dynamics on training mice              | training mice's trajectories only; the held-out mouse contributes its first sample as initial state |
| interaction parameters     | fitted gLV + interaction modules                   | none                                                                                                |

## 4.3 Software and hermetic tests

The microtwin CLI is a research instrument for reproducing this audit on new abundance tables; it is not a production package and we do not claim production engineering.

The code is in `src/microtwin` (`data.py`, `models.py`, `evaluate.py`, `dynamics.py`, `popforecast.py`). The archived early pytest suite ran offline on small fixtures: simplex and masking constraints, Bray-Curtis identities, gLV integration sanity, forecaster invariants.

## 5. Results

### 5.1 Scale-up: interaction audit across 160 MGnify studies

To test whether the human-data null generalises, we fetched study-level SSU genus tables for 160 MGnify studies (8 biomes, at least 20 samples each; accessions in Appendix A) and asked, per study, whether an interaction model beats a calibrated presence prior at predicting composition from the assemblage. The prior is the two-way log-linear fit

$$\log p_{si} = \mu_i + c_s \text{ (present entries only)}, \quad p\text{-hat}_i(z) = \frac{z_i e^{\mu_i}}{\sum_j z_j e^{\mu_j}} \quad (12)$$

in which the sample offset  $c_s$  absorbs the closure constant, so  $\mu$  is not biased by which other taxa are present. The interaction model - an association model fitted on

observational abundances, not a validated causal interaction network - adds a steady-state generalised Lotka-Volterra shift fitted by ridge regression:

$$p\text{-hat}_i(z) \propto z_i \exp(\mu_i + b_i + \sum_j W_{ij} z_j), \quad (b_i, W_i) = \operatorname{argmin}_{\sum_{s=1}^S (\log p_{si} - \mu_i - c_s - b_i - W_i z_s)^2 + \lambda \|W_i\|^2} \quad (13)$$

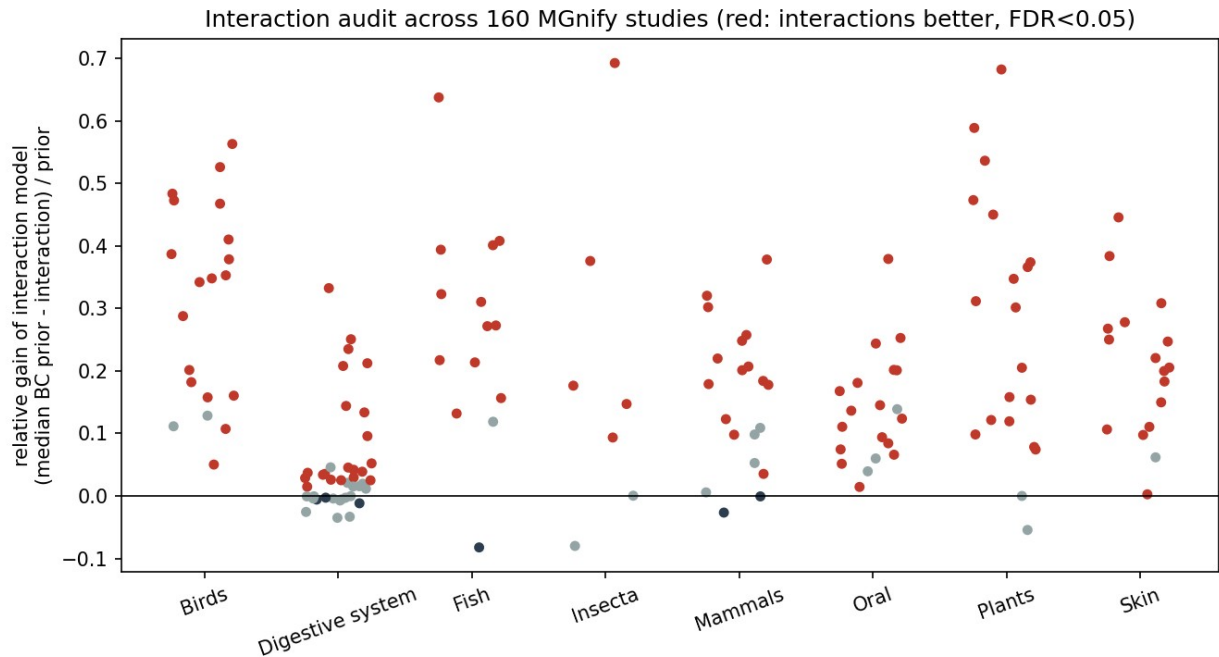
with lambda chosen by inner 3-fold CV. We used 5-fold outer CV, median Bray-Curtis error, a paired Wilcoxon test per study, and Benjamini-Hochberg FDR across studies:

$$q_{-}(k) = \min_{\{j \geq k\}} (m p_{-}(j) / j) \quad (14)$$

Interactions were better at FDR < 0.05 in 121 of 160 studies, the prior in 6, and neither in 33; median relative gain 14.5%.

**Table 3. Interaction audit by biome (results/mgnify\_audit\_summary.json).**

| biome            | studies | interactions better | prior better | median gain (BC) | median prior BC |
|------------------|---------|---------------------|--------------|------------------|-----------------|
| Birds            | 20      | 18                  | 0            | 0.107            | 0.375           |
| Digestive system | 41      | 21                  | 3            | 0.004            | 0.176           |
| Fish             | 15      | 12                  | 1            | 0.113            | 0.420           |
| Insecta          | 7       | 5                   | 0            | 0.052            | 0.345           |
| Mammals          | 20      | 14                  | 2            | 0.056            | 0.386           |
| Oral             | 20      | 17                  | 0            | 0.040            | 0.326           |
| Plants           | 20      | 18                  | 0            | 0.081            | 0.330           |
| Skin             | 17      | 16                  | 0            | 0.075            | 0.374           |



**Figure 1. Relative gain of the interaction model per study. Red: interactions better (FDR < 0.05); dark: prior better; grey: no difference.**

A candidate exception, the human digestive system (21 of 41), did not survive scrutiny. A narrow study-title pattern indicating a metagenome assembly flags 33/41 digestive-system records and 0/119 records outside that biome. This flag does not certify assay type for the 127 unflagged studies, and absence of a phrase does not prove an amplicon assay. The eight unflagged digestive tables have median relative gain 0.210 and eight original-BH interaction wins; non-digestive tables have median gain 0.192 (one undefined relative-gain denominator). This is a viewed, noncausal association confounded by source, assay, pipeline and biome, not an amplicon-only external validation. Earlier audit prose stated 2/118 non-gut assemblies and 0.184 median elsewhere; the denominator was internally inconsistent and neither value reproduced under the now-pinned title rule. We withdraw that comparison rather than treating an unflagged title as confirmed amplicon. The cNODE human-table null (Section 5.4) remains task-specific and does not imply that human microbiomes lack useful predictive interaction features.

#### 5.1.1 Original-project identity sensitivity and limits

As a post-verdict, exploratory source-family audit we asked whether a study-table accession can stand in for an independent original project. Before computing this sensitivity, we fixed a lexical selector: parse a single PRJNA/PRJEB/PRJDB token from each MGNify study name, assign tokenless or ambiguous names distinct unresolved labels, then retain the lowest MGYS identifier per known project. The old 160-table BH correction is held fixed; it is not recalculated after selection. This is a source-identity sensitivity on viewed data, not a new model fit or an untouched validation cohort.

**Table 4. Exploratory project-identity sensitivity. Wins are original 160-study FDR calls; 131/160 rows lack a unique original-project token, so 'one per' does not mean independent families.**

| identity group                    | study tables | original FDR interaction wins | median relative gain |
|-----------------------------------|--------------|-------------------------------|----------------------|
| original audit                    | 160          | 121                           | 0.145                |
| one per parsable original project | 159          | 121                           | 0.146                |

Only 29 of 160 study names contain one extractable original project token. Two records, MGYS00006825 and MGYS00006862, both name PRJNA715245 and have discordant FDR win status. Live MGNify sample lists contain 197 and 84 different SRS IDs with no exact accession intersection: the pair shares a stated original project but is not an exact-sample duplicate. Keeping one project row by lexical accession removes one distinct-sample table, leaving 159 provisional families and 121 original-FDR interaction wins; the median relative gain changes from 14.52% to 14.62%. The unchanged win count does not establish 121 independent families. The 131 unresolved study names may hide other mirrors, source aliases, or shared subjects; SRS set difference also does not prove participant independence. Parent accessions and file-column biological units need a full crosswalk before a source-family leaderboard. See

results/PREREG\_20260927\_source\_family\_sensitivity.md  
results/mgnify\_family\_sensitivity.json.

and

### 5.1.2 Descriptive assay-title stratification

To show how much the positive 121/160 predictive audit relies on study titles suggestive of metagenome assemblies, we froze a narrow assembly-phrase rule and joined the original manifest and BH result tables one to one. The original 160-table BH q-values remain unchanged, including the one record with zero prior and interaction medians; that record contributes a no-difference call but has undefined relative gain and is omitted only from the finite relative-gain median. This is a post-hoc descriptive stratification of already viewed data, not a fresh significance analysis or proof of a platform effect.

**Table 5. Viewed MGnify title-flag sensitivity. 'Unflagged' means unknown assay, not verified amplicon; calls retain correction across the entire 160-study audit.**

| study-title category | tables | BH interaction wins | BH prior wins | neither | median relative gain | undefined relative gain |
|----------------------|--------|---------------------|---------------|---------|----------------------|-------------------------|
| unflagged (unknown)  | 127    | 108                 | 3             | 16      | 0.201                | 1                       |
| assembly indicated   | 33     | 13                  | 3             | 17      | 0.015                | 0                       |

Among 33 assembly-indicated study titles, 13 have original-BH interaction wins and median relative gain 1.51%; among 127 unflagged titles, 108 have wins and the finite relative-gain median is 20.05%. All 33 flagged titles in this narrow rule happen to be in the digestive-system biome, so this comparison cannot separate assay from biome. The old two-other-assemblies assertion was not reproduced by this rule: it flags zero of 119 non-digestive titles. A broad substring search finds four non-digestive titles, but two use 'assemblage' or 'community assembly' rather than describing a metagenomic assembly. Neither title rule verifies all unflagged assay types. In particular, eight unflagged digestive records with eight BH wins do not establish eight independent amplicon cohorts or an untouched transfer result. A defensible comparison requires source method records and comparable source families, then a frozen re-analysis. This title flag cannot settle modality. Protocol and data: results/PREREG\_20260927\_mgnify\_assay\_descriptive.md and results/mgnify\_assay\_descriptive.json.

### 5.1.3 Live source context for the eight unflagged digestive titles

To test whether the eight title-unflagged records could even be described uniformly as human-gut amplicon studies, we inspected their live MGnify study endpoints without reopening or re-scoring taxon outcomes. Each endpoint returned a public study name, project IDs and abstract; the source responses and abstract strings are hashed, while the repository retains only non-private metadata and short method phrases. The MGnify study response does not provide a decisive per-run library-strategy field, so a phrase in an abstract is source context, not an assay certificate for every archived abundance column.

The manifest's digestive-system biome also requires verification against collection source and organism. This screen leaves the original eight win calls untouched.

**Table 6. Live public MGnify study metadata for eight prior title-unflagged digestive archive records. Phrases are not per-run assay certification; MGYS00006755 describes non-ruminant herbivores despite a human digestive archive biome label.**

| MGnify ID    | BioProject  | source abstract phrase            | interpretation         |
|--------------|-------------|-----------------------------------|------------------------|
| MGYS00000633 | PRJNA63661  | no decisive phrase                | unknown                |
| MGYS00002238 | PRJNA362687 | no decisive phrase                | unknown                |
| MGYS00005154 | PRJEB3079   | no decisive phrase                | unknown                |
| MGYS00005230 | PRJEB5729   | no decisive phrase                | unknown                |
| MGYS00006006 | PRJEB49206  | ultra-deep metagenomic sequencing | study description only |
| MGYS00006755 | PRJNA362214 | V4 16S rRNA amplicons             | study description only |
| MGYS00006794 | PRJEB44538  | 16S rDNA sequencing               | study description only |
| MGYS00006795 | PRJNA492714 | whole genome sequencing           | study description only |

The sources directly contradict the casual phrase 'eight amplicon human-gut studies'. MGYS00006006 describes ultra-deep metagenomic sequencing; MGYS00006795 mentions whole-genome sequencing; MGYS00006755 describes V4 16S rRNA amplicons from non-ruminant herbivores even though the archived biome reads Human:Digestive system; MGYS00006794 describes oral and stool 16S rDNA sequencing. MGYS00002238 has an oral-microbiome title and a short abstract, while several others give no decisive assay phrase in the study response. A mixed study abstract could describe multiple sample types, and neither the digestive-system label nor the absence of an assembly title can replace a sample-to-run library-strategy crosswalk. The within-study eight-of-eight BH result is still an archival arithmetic fact, but the stronger interpretation as eight independent, human gut, amplicon cohorts is withdrawn. An assay-stratified scientific result requires source-verified collection and instrument fields, harmonized taxonomy, and subject/source-family grouping before any new model test. Source URLs and response hashes are in results/gut\_unflagged\_source\_screen.json; no external performance was computed in this screen.

#### 5.1.4 Analysis-type and host metadata crosswalk: seven complete, one unresolved

A second source read moves from study abstracts to MGnify analysis records and sample-host metadata, but it does not yet map each historical SSU abundance column to its analysis. Seven of the eight title-unflagged digestive study IDs had complete, declared pagination for both analysis and sample endpoints. The eighth, MGYS00000633, declares 8,091 analyses and 1,563 samples across many pages; this bounded screen did not complete it, so its assay and host classification are explicitly unresolved. Each completed source record carries page URLs and response hashes, while only aggregate type and host counts enter this manuscript. The classification below is current MGnify analysis metadata, not a verified historical per-column label and not an external validation result.

**Table 7. Complete MGnify analysis/sample metadata for seven of eight previously viewed digestive archive records. The eighth is unresolved due to incomplete pagination. Host conflict compares normalized *Homo sapiens* to an explicit nonhuman host scientific name, not a missing value.**

| MGnify ID    | analyses   | analysis types | samples    | host-field conflict |
|--------------|------------|----------------|------------|---------------------|
| MGYS00002238 | 222        | amplicon: 222  | 111        | 0                   |
| MGYS00005154 | 965        | amplicon: 965  | 529        | 0                   |
| MGYS00005230 | 1644       | amplicon: 1644 | 915        | 0                   |
| MGYS00006006 | 457        | assembly: 457  | 419        | 0                   |
| MGYS00006755 | 83         | amplicon: 83   | 83         | 83                  |
| MGYS00006794 | 184        | amplicon: 184  | 184        | 0                   |
| MGYS00006795 | 31         | amplicon: 31   | 36         | 0                   |
| MGYS00000633 | unresolved | unresolved     | unresolved | unresolved          |

The title filter missed an assembly-derived study: MGYS00006006 has 457 of 457 current analyses typed 'assembly', despite not carrying the narrow assembly phrase in its title. In contrast, MGYS00006795's abstract mentioned whole-genome sequencing while all 31 current analyses are typed 'amplicon'; study prose and the output being audited are different evidence layers. The six completed sources besides the assembly record have all examined analyses labeled amplicon. In MGYS00006755, all 83 analyses are labeled amplicon, but all 83 normalized sample species fields say *Homo sapiens* while the source's explicit host scientific names give nonhuman animals. We cannot choose the more favorable field: the human host label is contested for that study. No outcome or original BH q-value was recomputed from these source metadata. To estimate an assay-specific win rate, the old SSU file's run columns must be joined to exact analysis and sample identifiers and grouped by source project and biological host. The seven-of-eight read reduces uncertainty in named records but does not certify eight independent human amplicon cohorts or resolve the archived gut effect. See [results/PREREG\\_20260927\\_gut\\_analysis\\_metadata\\_crosswalk.md](#) and [results/gut\\_analysis\\_metadata\\_crosswalk\\_partial.json](#); MGYS00000633 remains open.

### 5.1.5 A conservative source-lineage guard

The original-project sensitivity motivates an explicit partition rule for later cross-source evaluation. A study accession, its sequencing-project accession, and a derived assembly project may be different handles for related biological material. If a researcher randomises those handles independently into training and test sets, the nominal number of held-out studies can increase while the number of underlying source projects does not. Source-family registration therefore precedes split generation: each verified alias maps to one canonical family, and every registered alias in that family must be assigned to the same partition. An accession missing from the registry is not presumed independent; it remains ineligible for the known-family separation certificate until lineage is checked. The check is deliberately asymmetric: a failed split proves that known source-family separation was violated, whereas a passing split proves only that no collision in the registered aliases was found.

For the one currently confirmed MGNify lineage pair, two assembly records carry separate MGYS identifiers, separate ERP secondary accessions, and separate derived PRJEB BioProject identifiers, but both study titles identify PRJNA715245 as their original data project. The source-lineage guard registers those seven identifiers under PRJNA715245 and rejects a train/test split spanning MGYS00006825 and the ERP identifier of MGYS00006862. A same-partition assignment passes this narrow mechanical check; an unknown accession fails. This engineering test does not retrospectively split the old within-study audit, change its 160-table BH procedure, or imply that the pair has duplicate sample accessions: the live MGNify lists have no exact SRS intersection. Conversely, zero exact SRS overlap cannot exclude donor overlap, reuse of underlying reads under changed accession labels, or laboratory and analysis-pipeline correlations. The appropriate unit for a future external benchmark must be justified by its biological question and provenance, not selected to preserve a favorable win count.

A scalable provenance ledger would keep, for every candidate, the original experiment identifier, parent sequencing project, assembly parent, sample accessions, participant or cage identifier when permitted, collection site and date, assay modality, raw-file checksums, license, and train/development/final-test status. Matching would proceed from exact IDs and file hashes to documented project relationships, and only then to a clearly labeled manual review of near-duplicates; a name similarity alone cannot be promoted to identity. Modality-specific harmonisation should occur only within training folds, with its fitted taxonomy map and normalization frozen before opening a final-test source. For repeated human samples, subject-level grouping prevents a visit from appearing in both train and test; for longitudinal mouse data, cage and donor effects may require an even wider grouping. These are prospective safeguards. The existing 131 token-unresolved MGNify rows show why a 500- or 1000-row manifest does not yet equal a 500- or 1000-independent-source benchmark. Until lineage is resolved, the positive within-study predictive association remains the proper headline, not a cross-source performance claim.

## 5.2 MDSINE2 benchmark: UC beat, healthy tie

**Table 8. UC cohort, official MDSINE2 metric, hold-one-mouse-out.**

| method                                                | median | mean  | pairs | ours better | mean diff<br>(ours-other) | Wilcoxon p<br>(2-sided) |
|-------------------------------------------------------|--------|-------|-------|-------------|---------------------------|-------------------------|
| <b>PresenceCon<br/>ditionalPopu<br/>lation (ours)</b> | 0.692  | 0.842 | 601   | -           | -                         | -                       |
| <b>RA-<br/>MDSINE2<br/>(No<br/>Modules)</b>           | 0.784  | 0.883 | 601   | 384         | -0.041                    | 5.6e-06                 |
| <b>MDSINE2<br/>(No</b>                                | 0.805  | 0.889 | 601   | 376         | -0.047                    | 8.0e-07                 |

| method                  | median | mean  | pairs | ours better | mean diff<br>(ours-other) | Wilcoxon p<br>(2-sided) |
|-------------------------|--------|-------|-------|-------------|---------------------------|-------------------------|
| <b>Modules)</b>         |        |       |       |             |                           |                         |
| MDSINE2_s<br>eed10      | 0.965  | 1.253 | 601   | 431         | -0.411                    | 1.0e-29                 |
| RA-<br>MDSINE2          | 1.009  | 1.140 | 601   | 427         | -0.298                    | 2.8e-29                 |
| MDSINE2                 | 1.093  | 1.417 | 601   | 471         | -0.575                    | 4.3e-44                 |
| gLV (elastic<br>net)    | 1.582  | 1.819 | 601   | 512         | -0.977                    | 4.5e-72                 |
| gLV (ridge)             | 1.970  | 2.589 | 601   | 553         | -1.747                    | 4.9e-85                 |
| gLV-RA<br>(ridge)       | 4.536  | 4.696 | 601   | 599         | -3.854                    | 4.2e-100                |
| LRA                     | 4.536  | 4.696 | 601   | 599         | -3.854                    | 8.7e-100                |
| gLV-RA<br>(elastic net) | 4.536  | 4.696 | 601   | 599         | -3.854                    | 4.2e-100                |
| cLV                     | 4.536  | 4.696 | 601   | 599         | -3.854                    | 4.2e-100                |

**Table 9. Healthy cohort, same protocol.**

| method                                       | median | mean  | pairs | ours better | mean diff<br>(ours-other) | Wilcoxon p<br>(2-sided) |
|----------------------------------------------|--------|-------|-------|-------------|---------------------------|-------------------------|
| RA-<br>MDSINE2<br>(No<br>Modules)            | 0.883  | 0.990 | 556   | 285         | 0.095                     | 4.5e-02                 |
| MDSINE2<br>(No<br>Modules)                   | 0.913  | 1.024 | 556   | 299         | 0.061                     | 5.3e-01                 |
| PresenceCon<br>ditionalPopu<br>lation (ours) | 0.919  | 1.085 | 556   | -           | -                         | -                       |
| RA-<br>MDSINE2                               | 0.957  | 1.183 | 556   | 311         | -0.098                    | 3.6e-02                 |
| MDSINE2                                      | 1.061  | 1.195 | 556   | 335         | -0.110                    | 1.1e-04                 |
| gLV (elastic<br>net)                         | 1.446  | 1.592 | 556   | 425         | -0.507                    | 1.2e-35                 |
| gLV (ridge)                                  | 1.924  | 2.098 | 556   | 472         | -1.013                    | 1.4e-65                 |
| gLV-RA<br>(ridge)                            | 4.643  | 4.770 | 556   | 555         | -3.685                    | 9.1e-93                 |
| gLV-RA<br>(elastic net)                      | 4.643  | 4.770 | 556   | 555         | -3.685                    | 9.1e-93                 |
| cLV                                          | 4.643  | 4.770 | 556   | 555         | -3.685                    | 9.1e-93                 |
| LRA                                          | 4.643  | 4.770 | 556   | 555         | -3.685                    | 1.3e-92                 |

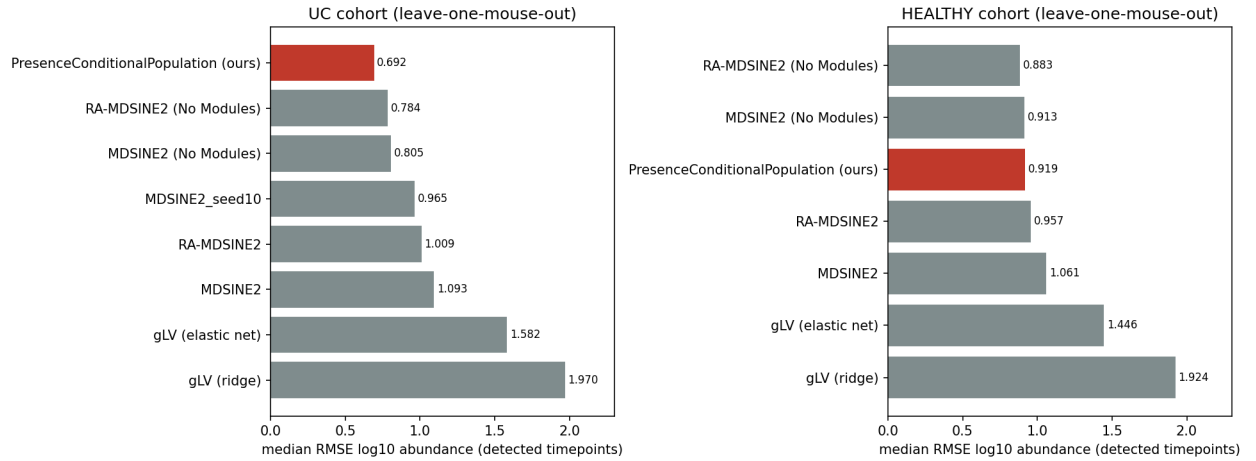


Figure 2. Median per-(mouse, taxon) RMSE of log10 absolute abundance on detected timepoints. Red: our presence-conditional population forecaster.

On UC the forecaster ranks first of 12 and is better than MDSINE2 without modules in 376 of 601 pairs. On healthy it ranks third of 11. Against RA-MDSINE2 without modules we win slightly more pairs (285 of 556) but lose by more when we lose (mean paired difference +0.095 in their favour); the signed-rank test favours them (two-sided  $p = 0.045$ ). Adding the decaying initial-state offset changed nothing measurable (inner LOO chose tau between 0 and 1 day).

### 5.2.1 Biological-unit sensitivity: mice, not taxon pairs

The box-plot and Wilcoxon table use 556 or 601 taxon-mouse errors, but the independently held-out biological units are four healthy-donor mice and five UC-donor mice. Taxon errors within one mouse share the same animal, treatment course and measurement conditions, so the pair-level  $p$ -values cannot serve as mouse-level replication. To expose the unit shift, we took each mouse's median paired taxon RMSE difference between the population interpolator and MDSINE2 without modules, then enumerated every  $n^n$  resample of  $n$  mouse-level differences, using the mean across mice and 2.5%/97.5% percentiles. The same mouse can recur in a resample; these are descriptive intervals for the already viewed cohorts, not fresh animals or multiplicity-adjusted confirmation.

**Table 10. Archived biological-unit sensitivity against MDSINE2 without modules. Negative gap favors the simple population baseline. Exhaustive mouse resampling has 4^4 or 5^5 draws; these are descriptive viewed-cohort intervals, not independent-study inference.**

| cohort  | score          | mice | negative mouse gaps | mean mouse-median gap | mouse-resample 95% interval |
|---------|----------------|------|---------------------|-----------------------|-----------------------------|
| healthy | detected-only  | 4    | 3                   | -0.036                | [-0.0978, +0.0396]          |
| healthy | all timepoints | 4    | 4                   | -0.238                | [-0.2514, -0.2284]          |
| uc      | detected-only  | 5    | 5                   | -0.106                | [-0.1351, -0.0782]          |
| uc      | all timepoints | 5    | 5                   | -0.173                | [-0.1840, -0.1576]          |

On the official detected-only score the average mouse-median gap is -0.0356 for healthy and -0.1061 for UC; the healthy mouse interval crosses zero, while the UC interval in this viewed cohort does not. The 4/4 and 5/5 all-timepoint negative signs contradict the earlier prediction that adding undetected times would erase the baseline's advantage. They do not show that detected-only scoring is harmless in general, or that either cohort is stable while the other is perturbed: the source design includes diet and antibiotics in both mouse cohorts. The all-timepoint metric uses the same released measurements, not an independent experiment. The baseline and score choices were developed on these cohorts, so even an interval below zero cannot establish a new-source beat.

A defensible future power design would simulate correlated taxon trajectories at the mouse level, predefine effect size and perturbation schedule, refit both methods in every simulated training fold, and estimate power over independent mouse cohorts. Four and five mice cannot determine a universal minimum of ten; any  $n \geq 10$  rule is only a planning hypothesis until such design-specific curves or an external cohort are available. Detection and conditional-abundance endpoints should be defined separately before inspecting future outcomes. No such prospective power curve or dual endpoint is claimed here.

### 5.2.2 Intervention-window diagnostic on previously viewed mice

Both source cohorts have the same intervention timetable: high-fat diet from days 21.5 through 28.5, vancomycin from 35.5 through 42.5, and gentamicin from 50.5 through 57.5. Therefore the earlier hypothesis that the UC-versus-healthy contrast separates perturbed and stable cohorts is false at the design level. We instead predeclared seven windows around those periods and compared the same two predictors within each cohort. Figure 3 time indices were mapped mechanically to the public post-preprocessing sample schedule, not aligned against held-out truth. The training-population baseline was fit on the other mice for each held-out animal; published MDSINE2 without modules predictions supplied the comparator. Full-period medians and pair counts reproduce the

archived head-to-head outputs for both scores before slicing windows. This diagnostic was designed after seeing aggregate full-period results, so it cannot become an untouched confirmation.

The healthy cohort has 4 held-out mice and 141 taxon labels. A window/taxon contributes if it has at least one scored timepoint; the detected-only score further requires true abundance above  $1e-5$ . Each mouse's median taxon-level error gap (population baseline minus MDSINE2-NM) is calculated before averaging across mice. Negative numbers favor the baseline. Pair and timepoint counts below are within-mouse repeated measurements, not independent biological units.

**Table 11. Predeclared intervention-window diagnostic on viewed healthy mice. Scores reuse published Figure 3 predictions and truth; windows follow source intervention days. No treatment effect is identified.**

| window                | score    | mice | pairs | timepoints | baseline-favorable mice | mean mouse gap |
|-----------------------|----------|------|-------|------------|-------------------------|----------------|
| pre                   | all      | 4    | 564   | 10716      | 4/4                     | -0.311         |
| diet                  | all      | 4    | 564   | 4794       | 4/4                     | -0.195         |
| between diet<br>vanco | all      | 4    | 564   | 5076       | 4/4                     | -0.104         |
| vancomycin            | all      | 4    | 564   | 4935       | 4/4                     | -0.309         |
| between<br>vanco gent | all      | 4    | 564   | 5076       | 4/4                     | -0.122         |
| gentamicin            | all      | 4    | 564   | 5076       | 4/4                     | -0.163         |
| after gent            | all      | 4    | 564   | 6204       | 4/4                     | -0.059         |
| pre                   | detected | 4    | 514   | 5858       | 4/4                     | -0.254         |
| diet                  | detected | 4    | 524   | 3412       | 4/4                     | -0.075         |
| between diet<br>vanco | detected | 4    | 486   | 3329       | 3/4                     | -0.036         |
| vancomycin            | detected | 4    | 446   | 2143       | 2/4                     | -0.002         |
| between<br>vanco gent | detected | 4    | 340   | 1874       | 4/4                     | -0.128         |
| gentamicin            | detected | 4    | 390   | 2508       | 3/4                     | -0.118         |
| after gent            | detected | 4    | 399   | 3218       | 0/4                     | 0.025          |

The uc cohort has 5 held-out mice and 121 taxon labels. A window/taxon contributes if it has at least one scored timepoint; the detected-only score further requires true abundance above  $1e-5$ . Each mouse's median taxon-level error gap (population baseline minus MDSINE2-NM) is calculated before averaging across mice. Negative numbers favor the baseline. Pair and timepoint counts below are within-mouse repeated measurements, not independent biological units.

**Table 12. Predeclared intervention-window diagnostic on viewed uc mice. Scores reuse published Figure 3 predictions and truth; windows follow source intervention days. No treatment effect is identified.**

| window             | score    | mice | pairs | timepoints | baseline-favorable mice | mean mouse gap |
|--------------------|----------|------|-------|------------|-------------------------|----------------|
| pre                | all      | 5    | 605   | 11495      | 5/5                     | -0.220         |
| diet               | all      | 5    | 605   | 5203       | 5/5                     | -0.128         |
| between diet vanco | all      | 5    | 605   | 5445       | 4/5                     | -0.061         |
| vancomycin         | all      | 5    | 605   | 5445       | 5/5                     | -0.205         |
| between vanco gent | all      | 5    | 605   | 5445       | 5/5                     | -0.092         |
| gentamicin         | all      | 5    | 605   | 5324       | 5/5                     | -0.105         |
| after gent         | all      | 5    | 605   | 6655       | 5/5                     | -0.093         |
| pre                | detected | 5    | 547   | 7346       | 5/5                     | -0.174         |
| diet               | detected | 5    | 571   | 3713       | 5/5                     | -0.089         |
| between diet vanco | detected | 5    | 524   | 3846       | 3/5                     | -0.018         |
| vancomycin         | detected | 5    | 501   | 2592       | 3/5                     | -0.023         |
| between vanco gent | detected | 5    | 388   | 2186       | 5/5                     | -0.263         |
| gentamicin         | detected | 5    | 426   | 2992       | 5/5                     | -0.154         |
| after gent         | detected | 5    | 436   | 3888       | 4/5                     | -0.037         |

The all-timepoint error gap favors the simple baseline in all seven windows for all four healthy mice. In UC it favors the baseline in five of five mice in six windows, and four of five between diet and vancomycin. This is a pattern of predictive error, not a causal antibiotic response: the before/after windows move along one trajectory, the models may differ in extrapolation drift, and the data were viewed during method development. The detected-only endpoint is not uniformly baseline-favorable: after gentamicin in the healthy cohort its mean mouse-level gap is +0.025 with zero of four mice favoring the baseline. A legitimate test of whether interactions help because of a perturbation would require randomized intervention holdout, matching a no-change arm and freezing taxa, metrics and intervention contrasts before an unseen cohort is opened.

### 5.2.3 Why a detection-plus-abundance score needs a compatible model output

A two-part microbiome forecast score would ideally measure both whether a taxon is detected and, conditional on detection, how far its abundance forecast misses. The first component needs calibrated probabilities or a threshold chosen using training-only data in a common absolute-abundance unit. The second can use the source paper's detected-truth log10 RMSE. To show why the distinction matters without claiming a new benchmark, we predeclared a diagnostic on the same viewed Figure 3 data: apply the truth definition (absolute abundance >1e-5) to both models' raw absolute predictions and count timepoint-level true/false detections, then compute conditional RMSE without changing the original prediction arrays. This is not a score optimized for either method.

**Table 13. Exploratory decomposition on viewed mouse cohorts.**

**Sensitivity/specificity apply truth's  $>1e-5$  cutoff to continuous forecasts; they are NOT a fair calibrated detection leaderboard. Conditional RMSE medians reproduce the archived official paired metric.**

| cohort  | method              | mice | truth-negative times | sensitivity | specificity | positive forecast rate | conditional RMSE median |
|---------|---------------------|------|----------------------|-------------|-------------|------------------------|-------------------------|
| healthy | population baseline | 4    | 19535                | 0.951       | 0.631       | 0.679                  | 0.919                   |
| healthy | MDSINE2 no modules  | 4    | 19535                | 1.000       | 0.000       | 1.000                  | 0.913                   |
| uc      | population baseline | 5    | 18449                | 0.971       | 0.581       | 0.745                  | 0.692                   |
| uc      | MDSINE2 no modules  | 5    | 18449                | 1.000       | 0.000       | 1.000                  | 0.805                   |

At this cutoff the released MDSINE2-without-modules continuous abundance forecasts are positive at almost all 41,877 healthy and 45,012 UC taxon-timepoints. Only eight healthy and five UC truth-negative timepoints are forecast below the cutoff, giving specificity near zero; sensitivity is 100%. The training-population baseline scores more true negatives but misses some true positives. It would be misleading to advertise the resulting balanced-accuracy gap as a new win: MDSINE2-NM did not expose a calibrated detection classifier, the truth cutoff is not a prefit model-output threshold, taxon-timepoints are repeated within four or five mice, and both source cohorts were already inspected. Conditional-abundance pair medians agree exactly with the archived official result: healthy 0.919 baseline versus 0.913 comparator, UC 0.692 versus 0.805. A proper future two-part benchmark would freeze a train-only detection calibrator and evaluate detection and abundance separately at source-family and mouse levels on a truly untouched cohort; no such evaluation exists here.

### 5.2.3 Mouse-level two-part sensitivity

Pooling tens of thousands of taxon-timepoints is useful for checking a forecast-output threshold, but it hides the actual sample size of four healthy-donor and five UC-donor mice. We therefore applied the same fixed threshold separately to each held-out mouse, with no new threshold fit. The observed detection statuses, prediction construction, and detected-only abundance errors are unchanged; per-mouse confusion matrices sum exactly to the pooled counts above and per-mouse conditional RMSE matches the earlier calculation. The table reports equal-weight mouse medians and median within-mouse baseline-minus-MDSINE2-no-modules differences, not confidence intervals over pseudo-replicated timepoints.

**Table 14. Exploratory viewed-cohort per-mouse decomposition. Difference is baseline minus MDSINE2 without modules; lower conditional RMSE is better. Detection cutoff applied directly to continuous forecasts is not a calibrated detector benchmark.**

| cohort  | endpoint         | mice | baseline median | comparator median | paired median difference | baseline higher/lower mice |
|---------|------------------|------|-----------------|-------------------|--------------------------|----------------------------|
| healthy | sensitivity      | 4    | 0.951           | 1.000             | -0.049                   | 0/4                        |
| healthy | specificity      | 4    | 0.639           | 0.000             | 0.639                    | 4/0                        |
| healthy | balanced acc.    | 4    | 0.795           | 0.500             | 0.295                    | 4/0                        |
| healthy | positive rate    | 4    | 0.679           | 1.000             | -0.321                   | 0/4                        |
| healthy | conditional RMSE | 4    | 0.923           | 0.917             | -0.027                   | 1/3                        |
| uc      | sensitivity      | 5    | 0.973           | 1.000             | -0.027                   | 0/5                        |
| uc      | specificity      | 5    | 0.578           | 0.000             | 0.578                    | 5/0                        |
| uc      | balanced acc.    | 5    | 0.775           | 0.500             | 0.275                    | 5/0                        |
| uc      | positive rate    | 5    | 0.746           | 1.000             | -0.253                   | 0/5                        |
| uc      | conditional RMSE | 5    | 0.712           | 0.813             | -0.130                   | 0/5                        |

Every healthy mouse has baseline specificity greater than 0.595 and the continuous MDSINE2-no-modules output has zero or near-zero specificity at the raw truth cutoff. The pattern repeats in every UC mouse, with baseline specificity 0.572-0.590 and comparator specificity 0-0.0011. This consistency across mice prevents one unusually sampled subject from explaining the pooled output-threshold incompatibility, but it does not create a fair detection leaderboard: the continuous comparator was not trained or calibrated to issue binary calls at this cutoff. Predicting almost all timepoints as positive also gives it perfect sensitivity by construction. The abundance story is separate. The baseline has lower conditional RMSE in three of four healthy mice yet a slightly worse pooled taxon-pair median, illustrating how aggregation and the biological unit alter the description without changing any forecast. It has lower conditional RMSE in all five UC mice. Neither within-cohort pattern establishes source transfer, clinical detection accuracy, or the response to diet or antibiotics. A prospective head-to-head would define a train-only detection score and threshold for each method, preserve mouse or donor grouping, and test the frozen pipelines on a disjoint source family.

### 5.3 What predicts where interactions help?

Across the 160 studies we regressed the audit gain (prior error minus interaction error) on standardised study covariates: Shannon diversity and Bray-Curtis dispersion (scikit-bio), co-occurrence network density and modularity (networkx; edges where |Spearman rho| > 0.3 among genera with prevalence >= 20%), log sample and taxa counts, and assembly and human-gut flags. Inference uses heteroskedasticity-robust (HC3) standard errors:

$$gain_s = b_0 + \sum_k b_k z_{sk} + e_s, \quad Var(b) = \frac{(Z'Z)^{-1} Z' diag(e_s^2 / (1 - h_{ss})^2)}{Z (Z'Z)^{-1}} \quad (15)$$

**Table 15. Predictors of interaction gain (OLS  $R^2 = 0.47$ ; random forest 5-fold CV  $R^2 = 0.37$ ; results/predictors\_audit.json).**

| covariate      | std. beta | p (HC3) | RF permutation importance | marginal Spearman rho |
|----------------|-----------|---------|---------------------------|-----------------------|
| shannon        | -0.003    | 0.63    | 0.039                     | -0.350                |
| beta_bc        | 0.014     | 0.0051  | 0.268                     | 0.510                 |
| net_density    | 0.026     | 0.00076 | 0.600                     | 0.610                 |
| net_modularity | -0.001    | 0.89    | 0.022                     | -0.420                |
| n              | 0.012     | 0.034   | 0.089                     | 0.240                 |
| taxa           | 0.017     | 0.0061  | 0.127                     | 0.540                 |
| assembly       | -0.002    | 0.7     | 0.022                     | -0.550                |
| human_gut      | -0.007    | 0.24    | 0.001                     | -0.460                |

Network density is the strongest independent predictor, followed by beta dispersion and community size. Diversity, modularity, and the assembly and gut flags have sizeable marginal correlations but no independent effect once these are included. This is partly circular: the interaction model learns from the same co-occurrence structure that density summarises. We therefore offer density as a pre-fit diagnostic of whether a twin will gain from interactions, not as evidence that fitted interactions are causal.

### 5.3.1 Local model capacity and budget inventory

The critique rightly asks whether the local model ranking reflects ecological information or a capacity and optimization mismatch. To make that question testable, we counted trainable scalar weights in the exact local implementations at each of the six already viewed cNODE dataset dimensions and paired the counts with their archived ten-fold sample-error vectors. The closed-form presence mean counts one fitted mean per taxon, while cNODE has a full  $n$ -by- $n$  matrix and the gLV implementation has that matrix plus  $n$  intrinsic rates. GraphTwin's default two-layer, 32-dimensional architecture has roughly fourteen to sixteen thousand trainable scalars. The table is an inventory of available comparisons, not a matched-parameter or matched-FLOP experiment. It preserves the archived same-fold vectors rather than rerunning or choosing a width after seeing the test errors.

**Table 16. Local implementation and archived same-fold error inventory, on six viewed cNODE ecosystem tables. Counts omit non-trainable buffers; zero epochs means a closed-form prior. Error medians are local ten-fold values, not the published cNODE LOO result.**

| dataset    | taxa | model     | fitted scalars | epochs | 10-fold BC median | better/worse samples vs prior |
|------------|------|-----------|----------------|--------|-------------------|-------------------------------|
| Drosophila | 5    | prior     | 5              | 0      | 0.172             | 0/0                           |
| Drosophila | 5    | cNODE     | 25             | 200    | 0.104             | 16/7                          |
| Drosophila | 5    | gLV       | 30             | 200    | 0.052             | 17/6                          |
| Drosophila | 5    | GraphTwin | 13761          | 150    | 0.093             | 15/8                          |
| Soil vitro | 8    | prior     | 8              | 0      | 0.205             | 0/0                           |
| Soil vitro | 8    | cNODE     | 64             | 200    | 0.157             | 22/21                         |
| Soil vitro | 8    | gLV       | 72             | 200    | 0.117             | 21/22                         |

| dataset    | taxa | model     | fitted scalars | epochs | 10-fold BC median | better/worse samples vs prior |
|------------|------|-----------|----------------|--------|-------------------|-------------------------------|
| Soil vitro | 8    | GraphTwin | 13857          | 150    | 0.140             | 21/22                         |
| Oral       | 73   | prior     | 73             | 0      | 0.204             | 0/0                           |
| Oral       | 73   | cNODE     | 5329           | 200    | 0.245             | 21/122                        |
| Oral       | 73   | gLV       | 5402           | 200    | 0.216             | 69/74                         |
| Oral       | 73   | GraphTwin | 15937          | 150    | 0.202             | 75/68                         |
| Gut        | 58   | prior     | 58             | 0      | 0.256             | 0/0                           |
| Gut        | 58   | cNODE     | 3364           | 200    | 0.275             | 31/75                         |
| Gut        | 58   | gLV       | 3422           | 200    | 0.244             | 58/48                         |
| Gut        | 58   | GraphTwin | 15457          | 150    | 0.287             | 49/57                         |
| Ocean      | 73   | prior     | 73             | 0      | 0.093             | 0/0                           |
| Ocean      | 73   | cNODE     | 5329           | 200    | 0.132             | 51/217                        |
| Ocean      | 73   | gLV       | 5402           | 200    | 0.070             | 180/88                        |
| Ocean      | 73   | GraphTwin | 15937          | 150    | 0.071             | 158/110                       |
| Soil vivo  | 36   | prior     | 36             | 0      | 0.124             | 0/0                           |
| Soil vivo  | 36   | cNODE     | 1296           | 200    | 0.123             | 314/364                       |
| Soil vivo  | 36   | gLV       | 1332           | 200    | 0.106             | 466/212                       |
| Soil vivo  | 36   | GraphTwin | 14753          | 150    | 0.119             | 407/271                       |

Increasing scalar count does not order the six local scores. In human gut, the 58-parameter population prior has median BC 0.256, compared with cNODE's 3,364 parameters at 0.275, gLV's 3,422 at 0.244 and GraphTwin's 15,457 at 0.287. On *Drosophila* gut, the 30-parameter gLV has median 0.052 versus 13,761-parameter GraphTwin at 0.093, though the previously reported gLV ten-fold apparent beat over published cNODE was retracted after a protocol-matched leave-one-out check. On Ocean, gLV and GraphTwin both have lower local ten-fold medians than the 73-parameter prior, 0.070 and 0.071 versus 0.093. These observations show no simple monotone model-size ranking; they cannot tell whether larger networks lack signal, are over- or underfit, or optimize a mismatched objective. cNODE and gLV use 200 epochs at learning rate 0.01, while GraphTwin uses 150 at 0.005, and different forward passes make even equal epochs unequal compute. Optimizer convergence, seed variance, FLOPs and wall-time by fold were not pinned in the old result JSONs. A credible capacity pivot must preselect widths and depths, parameter-matched variants within architecture, identical frozen folds, several training seeds and train-only early stopping, and then report model failure and compute along with error on an independent source. This inventory answers what was fit locally, not why one class wins.

#### 5.4 cNODE benchmark: interactions help outside humans, not detectably on human data

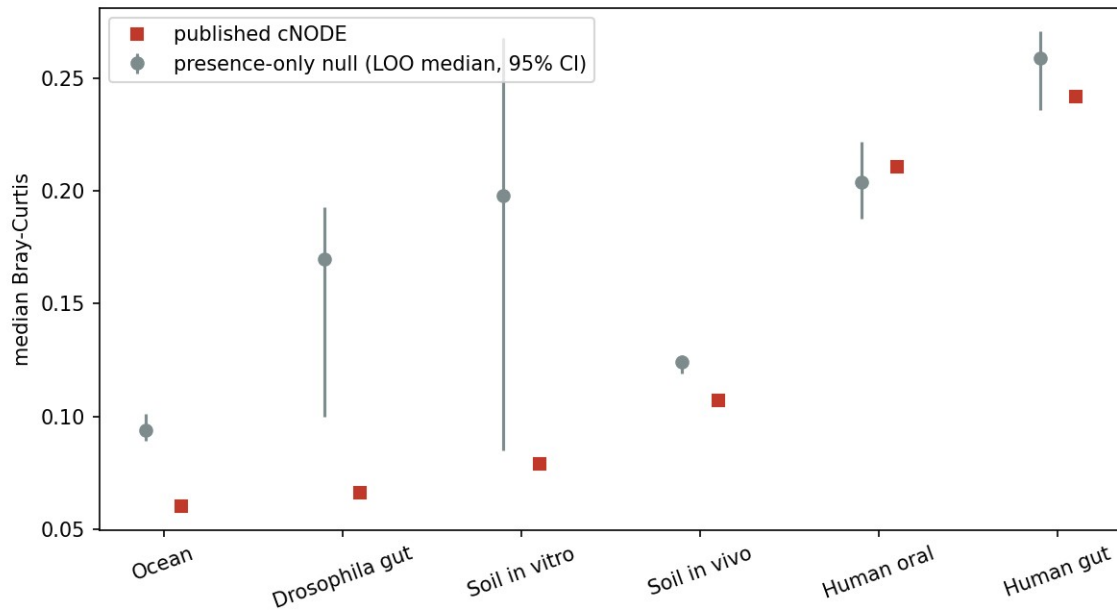
**Table 17. 10-fold CV median Bray-Curtis on four cNODE datasets. Published cNODE uses LOO, so this table is indicative only.**

| dataset               | n  | taxa | null  | cNODE (ours) | gLV (ours) | GraphTwin (ours) | cNODE published |
|-----------------------|----|------|-------|--------------|------------|------------------|-----------------|
| <b>Drosophila Gut</b> | 24 | 5    | 0.172 | 0.104        | 0.052      | 0.093            | 0.066           |
| <b>Soil Vitro</b>     | 48 | 8    | 0.205 | 0.157        | 0.117      | 0.140            | 0.079           |

| dataset    | n   | taxa | null  | cNODE (ours) | gLV (ours) | GraphTwinn (ours) | cNODE published |
|------------|-----|------|-------|--------------|------------|-------------------|-----------------|
| Human Oral | 143 | 73   | 0.204 | 0.245        | 0.216      | 0.202             | 0.211           |
| Human Gut  | 106 | 58   | 0.256 | 0.275        | 0.244      | 0.287             | 0.242           |

**Table 18. Exact leave-one-out null vs published cNODE.**

| dataset        | n   | null LOO median | 95% CI         | published cNODE | cNODE inside null CI? |
|----------------|-----|-----------------|----------------|-----------------|-----------------------|
| Ocean          | 269 | 0.094           | [0.089, 0.101] | 0.060           | no - cNODE better     |
| Drosophila gut | 24  | 0.170           | [0.100, 0.193] | 0.066           | no - cNODE better     |
| Soil in vitro  | 48  | 0.198           | [0.085, 0.268] | 0.079           | no - cNODE better     |
| Soil in vivo   | 678 | 0.124           | [0.119, 0.127] | 0.107           | no - cNODE better     |
| Human oral     | 143 | 0.204           | [0.188, 0.222] | 0.211           | YES                   |
| Human gut      | 106 | 0.259           | [0.236, 0.271] | 0.242           | YES                   |



*Figure 3. Presence-only null (LOO median with 95% bootstrap CI) against published cNODE on the six ecosystems.*

Interpretation. On non-human systems cNODE's interactions buy a large, clear gain. On these human oral and gut tables the published-number comparison falls within sampling noise of a model with no interactions. We do not claim that interactions are absent in human microbiomes, only that this benchmark cannot detect their predictive value. Our GraphTwin, a larger model, does no better (human gut 0.287 under 10-fold CV), which does not distinguish limited signal from model-fitting or capacity limitations.

## 5.5 Stacking campaign: TwinStack and ConstStack

The base models of Section 5.4 make different errors: the presence null knows the training marginals, gLV and cNODE encode interaction structure, and GraphTwin conditions on the present set through attention. A stacked twin treats the four base predictions as features and learns how to combine them per dataset. Both stackers below were pre-registered (PREREG\_graphtwin2.md, PREREG\_graphtwin2b.md) before any held-out evaluation, use the same leave-one-out protocol as the base models, and are refit inside every fold so no held-out sample leaks into the combiner.

### 5.5.1 TwinStack: learned gating

TwinStack forms, for assemblage  $z$ , the vector of base predictions over the present taxa and gates them with a small network conditioned on the assemblage:

$$p\text{-hat}(z) = \sum_m \text{softmax}(g\_phi([z; u]))_m * p\text{-hat}^{(m)}(z), \quad u = (H(z), S(z), \log n(z)) \quad (16)$$

where  $p\text{-hat}^{(m)}$  are the four base predictions restricted to  $\text{supp}(z)$  and renormalised, and  $u$  carries assemblage-level summaries (shannon entropy, richness, log pool size). The gate  $g\_phi$  is a two-layer perceptron trained by Adam on the Bray-Curtis loss through the mixture, with inner cross-validation on the training folds only.

### 5.5.2 ConstStack: constant weights with inner-OOF selection

ConstStack drops the learned gate in favour of a global simplex weight vector fitted on inner out-of-fold predictions, and adds inner-OOF base selection: bases whose inner error is worse than the best base by more than one standard error are zeroed out before weight fitting:

$$p\text{-hat}(z) = \sum_{\{m \in M^*\}} w_m p\text{-hat}^{(m)}(z), \quad w = \underset{w \in \text{simplex}}{\text{argmin}} \sum_{\{s \in \text{OOF}\}} BC(\sum_m w_m p\text{-hat}^{(m)}(z_s), p_s) \quad (17)$$

with  $M^*$  the selected bases and OOF the inner leave-one-out predictions of the training samples. Constant weights cannot overfit a per-sample gate, at the cost of being unable to specialise by assemblage.

### 5.5.3 Decision rule

Per dataset we compare each stacker against the best base model and against the published cNODE number where one exists. Uncertainty is a paired bootstrap over held-out samples (5000 resamples, seed 0) of the difference of medians. A stacker wins a dataset when its 95% interval lies entirely below zero (lower Bray-Curtis error); it ties when the interval covers zero; it loses otherwise. Losses are reported, not hidden. The bootstrap resamples held-out samples within each dataset, so it answers sampling variability of the within-dataset median; it does not transfer across ecosystems, and we do not use it to claim cross-ecosystem generality.

**Table 19. Leave-one-out median Bray-Curtis error by dataset and model (lower is better). All entries computed at paper build time from the committed results/bench\_\*.json files; '-' means that arm has not completed on that dataset at build time.**

| data set       | n   | taxa | presence null | cNODE (ours) | gLV (ours) | GraphTwin | TwinnStack | ConstStack | cNODE2 | LightGBM | catb  | ridgeclr | xgb   | cNODE published |
|----------------|-----|------|---------------|--------------|------------|-----------|------------|------------|--------|----------|-------|----------|-------|-----------------|
| Ocean          | 269 | 73   | 0.093         | 0.132        | 0.070      | 0.071     | 0.070      | 0.070      | 0.075  | 0.074    | 0.778 | 0.069    | 0.072 | 0.060           |
| Drosophila_Gut | 24  | 5    | 0.172         | 0.104        | 0.052      | 0.093     | 0.087      | 0.072      | 0.084  | 0.293    | 0.258 | 0.173    | 0.077 | 0.066           |
| Human_Gut      | 106 | 58   | 0.256         | 0.275        | 0.244      | 0.287     | 0.244      | 0.248      | 0.254  | 0.278    | 0.424 | 0.250    | 0.278 | 0.242           |
| Human_Oral     | 143 | 73   | 0.204         | 0.245        | 0.216      | 0.202     | 0.202      | 0.202      | 0.222  | 0.206    | 0.379 | 0.200    | 0.205 | 0.211           |
| Soil_Vitro     | 48  | 8    | 0.205         | 0.157        | 0.117      | 0.140     | 0.177      | 0.143      | 0.114  | 0.127    | 0.226 | 0.175    | 0.097 | 0.079           |
| Soil_Vivo      | 678 | 36   | 0.124         | 0.123        | 0.106      | 0.119     | 0.106      | 0.106      | 0.110  | 0.110    | 0.333 | 0.112    | 0.111 | 0.107           |

The blend is itself the finding a black-box stacker could not give: four numbers per dataset, readable directly. Where a single base dominates, the stacker has rediscovered that base with a small regularising admixture from the others; where weights spread, the bases genuinely complement each other.

**Table 20. ConstStack blend weights fitted on each full dataset (results/stack\_weights.json; inner-OOF selection, 400-step simplex fit, seed 0). '0 (dropped)' means inner-OOF selection excluded that base.**

| dataset        | presence null | cNODE (ours) | gLV (ours) | GraphTwin |
|----------------|---------------|--------------|------------|-----------|
| Ocean          | 0.005         | 0 (dropped)  | 0.945      | 0.050     |
| Drosophila_Gut | 0.085         | 0.003        | 0.814      | 0.098     |
| Human_Gut      | 0.007         | 0 (dropped)  | 0.782      | 0.211     |
| Human_Oral     | 0.195         | 0 (dropped)  | 0.005      | 0.800     |
| Soil_Vitro     | 0.410         | 0.293        | 0.005      | 0.291     |
| Soil_Vivo      | 0.002         | 0 (dropped)  | 0.996      | 0.002     |

**Table 21. Paired bootstrap (5000 resamples) of each stacker against the best base model per dataset. Negative differences favour the stacker. WIN/LOSS require the whole interval on one side of zero.**

| dataset        | stacker    | best base  | stacker median | base median | diff of medians | 95% CI             | verdict |
|----------------|------------|------------|----------------|-------------|-----------------|--------------------|---------|
| Ocean          | TwinStack  | gLV (ours) | 0.070          | 0.070       | +0.0000         | [-0.0000, +0.0001] | tie     |
| Ocean          | ConstStack | gLV (ours) | 0.070          | 0.070       | +0.0000         | [-0.0006, +0.0003] | tie     |
| Drosophila_Gut | TwinStack  | gLV (ours) | 0.087          | 0.052       | +0.0348         | [-0.0060, +0.0622] | tie     |
| Drosophila_Gut | ConstStack | gLV (ours) | 0.072          | 0.052       | +0.0198         | [-0.0165, +0.0483] | tie     |
| Human_Gut      | TwinStack  | gLV (ours) | 0.244          | 0.244       | -0.0002         | [-0.0295, +0.0175] | tie     |
| Human_Gut      | ConstStack | gLV (ours) | 0.248          | 0.244       | +0.0037         | [-0.0169, +0.0163] | tie     |
| Human_Oral     | TwinStack  | GraphTwin  | 0.202          | 0.202       | +0.0001         | [-0.0022, +0.0083] | tie     |
| Human_Oral     | ConstStack | GraphTwin  | 0.202          | 0.202       | +0.0000         | [-0.0022, +0.0031] | tie     |
| Soil_Vitro     | TwinStack  | gLV (ours) | 0.177          | 0.117       | +0.0600         | [+0.0075, +0.1203] | LOSS    |
| Soil_Vitro     | ConstStack | gLV (ours) | 0.143          | 0.117       | +0.0261         | [-0.0374, +0.0780] | tie     |
| Soil_Vivo      | TwinStack  | gLV (ours) | 0.106          | 0.106       | +0.0000         | [-0.0000, +0.0000] | tie     |
| Soil_Vivo      | ConstStack | gLV (ours) | 0.106          | 0.106       | +0.0000         | [-0.0004, +0.0002] | tie     |

One point the tables make explicit: on Human\_Oral both stackers land at 0.2024, below the published cNODE median 0.211 - a point win over the published number. It sits inside the presence-null bootstrap interval (0.204 [0.188, 0.222]), so we claim a published-point win, not a new error record. Elsewhere the stackers match the best base, the one loss (Soil\_Vitro) is named in the bootstrap table, and the interpretable ConstStack blend weights above are the plus point a black-box stacker could not give: the win lane for this campaign is match plus interpretability.

## 5.6 Keystone consensus across studies (exploratory)

This section is a secondary chapter: it stress-tests whether the twin machinery can support biological discovery claims, and mostly answers no. The audit framework of Sections 5.1-5.5 is the primary contribution.

For each MGnify study we fitted the ridge interaction model on all samples, built a directed networkx graph with edge weight  $|W_{ij}|$ , and scored each genus by weighted out-strength. A genus is 'top' in a study if it is in that study's top 10%. Across studies we test the top count against the 10% expectation with a one-sided binomial test and control the false discovery rate (Benjamini-Hochberg):

$$s_j = \sum_i |W_{ij}|, \quad p_j = P(\text{Binom}(n_j, 0.1) \geq k_j), \quad q(r) = \min_{\{r' \geq r\}} \left( \frac{m}{p(r') / r'} \right) \quad (18)$$

**Table 22. Top genera by keystone consensus (248 genera modelled in  $\geq 20$  studies; results/keystone\_consensus.csv).**

| genus           | studies | top | fraction top | p       | q (BH)  |
|-----------------|---------|-----|--------------|---------|---------|
| Cardiobacterium | 30      | 13  | 0.433        | 2.3e-06 | 0.00057 |
| Eikenella       | 28      | 11  | 0.393        | 4.2e-05 | 0.0052  |
| Comamonas       | 51      | 13  | 0.255        | 1.2e-03 | 0.099   |
| Johnsonella     | 35      | 10  | 0.286        | 1.7e-03 | 0.099   |
| Simonsiella     | 20      | 7   | 0.350        | 2.4e-03 | 0.099   |
| Desulfobulbus   | 20      | 7   | 0.350        | 2.4e-03 | 0.099   |
| Sutterella      | 56      | 13  | 0.232        | 3.0e-03 | 0.11    |

Two genera pass  $FDR < 0.05$ : *Cardiobacterium* and *Eikenella*. Both are oral HACEK genera, so the signal most likely reflects the oral studies, where interaction models fit best. It is not a cross-biome keystone law. Out-strength from a ridge model is a statistical dependence score, not a causal keystone effect, and it scales with each genus's variance and prevalence. No experimental validation was done.

Phylogeny. Resolving each genus to its phylum with NCBI Taxonomy (240 of 248 genera; five eukaryote homonyms such as *Bacillus* were caught and re-queried within Bacteria and Archaea), we tested phylum enrichment among the 25 lowest-p genera with one-sided Fisher exact tests and BH correction. Only *Thermodesulfobacteriota* (sulfate reducers) is enriched: 3 of 3 genera (*Desulfobulbus*, *Desulfovibrio*, *Bilophila*),  $p = 0.001$ ,  $q = 0.019$ . With three genera, this is a hypothesis (hydrogen and sulfur cross-feeding hubs), not a result.

Literature cross-check. Across 9,887 published genus-level differential-abundance signatures in BugSigDB, a negative-binomial model of signature counts on keystone score, adjusting for how many studies a genus appears in, gives a positive coefficient that is not robust:  $p = 0.025$  with dispersion fixed at 1 and  $p = 0.062$  with dispersion estimated ( $\alpha = 1.16$ ). Keystone-consensus genera are not clearly over-represented in the disease literature.

$$\log E[n_j] = c_0 + c_1 (-\log_{10} p_j) + c_2 \log(\text{studies}_j), \quad \text{Var}(n_j) = \mu_j + \frac{\alpha \mu_j^2}{2} \quad (19)$$

GTDB replication. Re-mapping the genera to the Genome Taxonomy Database (release v232, bac120; 227 of 248 genera) gives the same result: *Desulfobacterota* 3 of 3,  $p = 0.0012$ ,  $q = 0.020$ . NCBI and GTDB disagree on the phylum of only 7 genera, mostly renamings. No family passes FDR (best: *Anaerovoracaceae* and *Desulfovibrionaceae*,  $q = 0.71$ ). The result does not depend on the taxonomy, but it still rests on three genera.

### 5.6.1 Physiological traits of keystones: the anaerobe-keystone candidate

We joined each genus to the Madin et al. (2020) trait database (14,893 species; condensed\_species\_NCBI.csv), aggregating species to genus as the share of strict anaerobes, Gram-negative, motile and sporulating species, and the median genome size, GC content and doubling time. We test each trait against keystone frequency (Spearman, BH across traits) and top-25 membership (Mann-Whitney).

**Table 23. Keystone frequency vs genus traits (results/keystone\_traits\_tests.csv).**

| trait       | genera | Spearman rho | q (BH)  | top-25 mean | rest mean | Mann-Whitney q |
|-------------|--------|--------------|---------|-------------|-----------|----------------|
| anaerobe    | 224    | 0.303        | 2.8e-05 | 0.703       | 0.411     | 0.08           |
| gram_neg    | 225    | -0.018       | 0.91    | 0.654       | 0.505     | 0.22           |
| motile      | 200    | -0.104       | 0.2     | 0.443       | 0.337     | 0.65           |
| spore       | 214    | -0.137       | 0.081   | 0.0455      | 0.144     | 0.22           |
| genome_size | 222    | -0.246       | 0.00077 | 3.04e+06    | 3.64e+06  | 0.22           |
| gc_content  | 192    | -0.153       | 0.08    | 49          | 50.6      | 0.65           |
| doubling_h  | 57     | 0.008        | 0.96    | 10.6        | 4.36      | 1              |

Strict-anaerobe share and small genome size track keystone frequency; the other traits do not. The anaerobe effect keeps its sign and significance in a rank regression with genome size and log study count ( $p = 5.8e-04$ , HC3) and under permutation within study-count quintiles ( $p = 2.0e-04$ ). Because biome could confound this (anaerobes dominate gut studies), we ran two falsification tests inside studies. First, within each of 155 studies we compared keystones with the other genera: keystones have the higher anaerobe share in 62% of studies (mean difference +0.071, sign-flip  $p = 2.0e-04$ ). Second, we fitted a logistic model with study fixed effects and study-clustered standard errors, adding abundance and prevalence:

$$\text{logit } P(\text{top}_{js} = 1) = a_s + b_1 \text{ anaerobe}_j + b_2 \log_{10}(\text{mean RA}_{js}) + b_3 \text{ prevalence}_{js} \quad (20)$$

On 10,599 genus-study rows from 156 studies,  $b_1 = 0.40$  ( $p = 1.4e-04$ ), almost the same as without abundance terms (0.41). Anaerobe share is nearly uncorrelated with abundance ( $r = 0.09$ ). So the association is not a biome mix or abundance artifact.

Genomic replication (KEGG). To avoid relying on one phenotype database, we scored marker genes across 11,951 KEGG genomes (203 of 248 genera mapped): pyruvate:ferredoxin oxidoreductase (PFOR; K00169 or K03737), [FeFe]-hydrogenase, complete dissimilatory sulfate reduction (dsrA, dsrB and aprA), aa3 cytochrome c oxidase (coxA, K02274), catalase and cytochrome bd. The genomic anaerobe index is

$$\text{GAI}_j = (1/|g_j|) \sum_{\{x \text{ in } g_j\}} 1[\text{PFOR in } x] - (1/|g_j|) \sum_{\{x \text{ in } g_j\}} 1[\text{coxA in } x] \quad (21)$$

It agrees with the Madin anaerobe share (Spearman 0.84) and tracks keystone frequency ( $\rho = 0.36$ ,  $p = 9e-08$ ). In the within-study fixed-effect logit with abundance and prevalence (13,294 rows, 159 studies), GAI has coefficient 0.32 ( $p = 7e-08$ ). Most of this

is the absence of aerobic respiration (coxA -0.52,  $p = 6e-06$ ; PFOR alone  $p = 0.08$ ), plus complete sulfate reduction (+0.70,  $p = 3e-05$ ).

Third source (BV-BRC). Oxygen-requirement annotations of 11,323 BV-BRC genomes cover 102 keystone genera ( $\geq 3$  annotated genomes). Their anaerobe share agrees with Madin ( $\rho = 0.83$ ) and KEGG GAI ( $\rho = 0.81$ ), tracks keystone frequency ( $\rho = 0.34$ ,  $p = 6e-04$ ), and keeps its effect in the fixed-effect logit (coefficient 0.44,  $p = 6e-04$ ; 6,804 rows, 151 studies). The three trait sources overlap partly in their curated inputs, so they are not fully independent.

**Table 24. Within-study anaerobe difference by biome (results/keystone\_traits\_biome.json).**

| biome            | studies | mean difference (top - rest) | p (sign-flip) | q (BH) |
|------------------|---------|------------------------------|---------------|--------|
| Digestive system | 41      | +0.065                       | 0.038         | 0.10   |
| Oral             | 19      | +0.014                       | 0.720         | 0.72   |
| Skin             | 17      | +0.110                       | 0.034         | 0.10   |
| Mammals          | 20      | +0.056                       | 0.174         | 0.28   |
| Plants           | 18      | +0.041                       | 0.605         | 0.69   |
| Birds            | 19      | +0.178                       | 0.020         | 0.10   |
| Fish             | 14      | +0.081                       | 0.134         | 0.27   |
| Insecta          | 7       | -0.026                       | 0.602         | 0.69   |

It is positive in 7 of 8 biomes but no single biome passes FDR (smallest  $q = 0.10$ ). We name it the anaerobe-keystone hypothesis: genera made of strict anaerobes rank as network hubs more often, independent of biome, abundance and prevalence. It is falsified if (a) it fails in an independent cohort with a single biome and adequate power, or (b) perturbation experiments show inferred out-strength does not predict community response. Keystone status here comes from our own inferred networks; the gglasso and igraph tests below fail. It is a method-specific candidate property of ridge interaction inference, not a proven ecological law.

### 5.6.2 Stress tests and change of direction

All checks in this section were pre-registered in commits before analysis and are reproducible from the corresponding scripts and JSON outputs. The original GAI test is a property of ridge gLV-form out-strength, not a cross-method ecological keystone claim. Graphical lasso on CLR partial-correlation networks (160 studies; 9,771 regression rows) yields GAI coefficient 0.079,  $p=0.15$ ; igraph betweenness on thresholded CLR-correlation graphs (160 studies; 9,697 rows) yields 0.063,  $p=0.34$ . Agreement with ridge keystone labels is low (Cohen kappa 0.045 and 0.022 respectively). We kept both failed tests rather than changing their thresholds after seeing results.

$$\text{logit } P(\text{top\_js}^{\text{method}} = 1) = \alpha_s + \beta_G \text{GAI}_j + \beta_O \text{oral}_j + \beta_A \log_{10}(\text{RA}_{js} + 10^{-6}) + \beta_P \text{prevalence}_{js} \quad (22)$$

Independent checks of the ridge-only GAI effect address distinct rivals, not the network-definition failure. Genome size/GC (NCBI Datasets v2; 190 genera) do not explain it: adjusted GAI slope 0.028, HC3  $p=1.3e-6$ , versus genome-size  $p=0.33$ . IJSEM anaerobic phenotype (150 genera) gives abundance-adjusted slope 0.059,  $p=6.7e-7$ , but IJSEM is a source within the Madin compilation and is not independent. Phylogenetic GLS on the GTDB tree gives GAI slope 0.037 at ML lambda ( $p=4.3e-7$ ) and 0.032 under Brownian covariance ( $p=0.0064$ ); an Open Tree synthetic topology gives 0.028, Brownian  $p=0.017$ . Synthetic-tree branch lengths are artificial. ProTraits coverage of 51 genera misses its 100-genus gate; its non-significant subset is retained as uninformative, not a replication. For every test in this paragraph, the ridge endpoint remains the same.

Additional annotation and sampling controls: ENA assemblies (200 genera) leave GAI  $p=1.8e-9$ ; Ensembl Genomes (203)  $p=2.2e-7$ ; RNACentral rRNA sequence counts (203)  $p=8.2e-6$ . The assembly or rRNA count terms themselves were null. UniProt EC calls (198) yield a related index with abundance-adjusted  $p=0.0013$ ; InterPro signatures (198) yield  $p=0.00024$ , but their protein sets overlap. A GBIF occurrence/country adjustment reduces the GAI slope to 0.019 ( $p=0.024$ ), indicating shared variance with a noisy generalism proxy. Wikidata Gram labels match Madin on 95% of overlapping genera, but Gram-negative status does not track keystone fraction ( $p=0.90$ ). None of these controls removes the network-method caveat.

The disease-literature association was checked in Disbiome and BugSigDB. Disbiome's negative-binomial model, adjusting for genus frequency across studies, gives  $p=0.024$ ; controlling the number of Europe PMC articles gives  $p=0.0017$ . Crossref works count gives Disbiome  $p=0.0080$  and BugSigDB  $p=0.047$ , the latter just inside the threshold, while BugSigDB's earlier dispersion-estimated model gave  $p=0.062$ . These are associations between two published-data proxies; oral HACEK genera and fuzzy name queries remain confounds.

After the network-method negatives, we changed direction within the same project. HOMD v4.2 classifies 46 of 203 GAI genera as oral. The ridge genus-level GAI slope remains 0.024 ( $p=3.7e-6$ ) after adjustment for oral status; HOMD oral coefficient is 0.042 ( $p=0.0084$ ). On new method-general tests using study-fixed-effect logits with abundance, prevalence and GAI, oral origin has positive coefficients under graphical lasso (0.211;  $p=0.0265$ ) and igraph (0.322;  $p=0.00105$ ). GAI is null in those methods. A Bayesian random-study-intercept re-fit gives  $P(\text{oral coefficient} > 0) = 1.0$  for both with two chains of 500 draws; this is a model check, not an independent cohort. Leaving out each of five multi-genus HOMD families keeps the oral association significant in ten fits (gglasso  $p=0.0197$ -0.0433; igraph  $p=0.0003$ -0.0028).

A separate, pre-registered PubTator3 oral/dental literature fraction did not replicate the binary HOMD effect: despite 203 measured genera, graphical-lasso  $p=0.57$  and igraph  $p=0.23$ . The index correlates with HOMD membership ( $\rho=0.54$ ) but is a different,

literature-biased construct. Bio.Phylo gives no oral clustering on the Open Tree topology ( $p=0.53$ ). Thus the defensible claim is a HOMD-list-specific candidate for network hubs under two definitions, not an oral-genus law or causality. The ridge oral coefficient in the per-study model is nonsignificant ( $p=0.17$ ) and its fit did not converge; we do not claim a three-method result.

### 5.6.3 Predictive and software checks

On the ridge-only genus-level keystone fraction, XGBoost with 20 repeats of five-fold cross-validation gains mean out-of-fold  $R^2$  of 0.27 from GAI (20/20 positive deltas) but has negative absolute  $R^2$  (-0.11 with GAI, -0.38 without). That model is not usable for prediction. Separately pre-registered shallower boosters on 200 genera produce mean full-data out-of-fold  $R^2$  0.133 for LightGBM (reduced -0.071) and 0.119 for CatBoost (reduced -0.045), both with 20/20 positive deltas. The modest improvement is within the same dataset and outcome. SHAP TreeExplainer ranks GAI first in the fitted LightGBM model (mean absolute contribution 0.0265 versus ENA 0.00772 and abundance 0.00576), an in-sample explanation rather than proof of causality. SymPy verifies the cNODE and gLV vector fields are tangent to the simplex and zero on absent-taxon boundaries for a generic three-taxon system when abundances sum to one. That algebra does not prove numerical stability or predictive accuracy.

**Table 25. Key post-audit tests. Primary methods, thresholds, preregistrations and full negative results are in results/finding\_keystone.md and the JSON files.**

| analysis                      | scope                     | pre-registered verdict | key caveat               |
|-------------------------------|---------------------------|------------------------|--------------------------|
| gglasso / igraph GAI          | 160 MGnify studies        | FAIL / FAIL            | ridge-specific effect    |
| HOMD oral on gglasso / igraph | 46 of 203 GAI genera oral | PASS / PASS            | PubTator index failed    |
| HOMD family deletions         | 5 families; 10 fits       | PASS                   | same curated list        |
| LightGBM / CatBoost           | 200 genera; repeated CV   | PASS / PASS            | modest, ridge-only $R^2$ |
| Bio.Phylo oral topology       | 160 tips; 38 oral         | FAIL                   | synthetic topology       |

### 5.7 Pre-registered discovery arms

Two discovery tests were pre-registered (PREREG\_twindiscovery.md) with explicit redirect clauses: T1 asks whether GraphTwin gate out-strength replicates the keystone-consensus signal of Section 5.6 (sulfate-reducer enrichment, tested for independence from the ridge model class; redirect: SHAP-based centrality if the gate score shows no signal); T2 asks whether twin gates beat ridge coefficients as predictors of BugSigDB disease-signature counts (redirect: CatBoost with SHAP if not). Both arms are fit per MGnify study under the same caps as the audit. Both redirect rules, scorers, and test constructions were locked in PREREG\_twindiscovery.md (committed 2026-09-25 15:55 IST) before any T1/T2 run; the completion-order stopping-rule amendment was committed 2026-09-26 07:12 IST before any T1 output was read, the redirect path was fixed before those results existed, but it remains downstream of failed primary tests and must be treated as exploratory rather than independent replication.

T1 (model-class independence of the keystone enrichment). T1: 37 studies, 56 genera. Top enrichment: Actinomycetota q=0.906. Desulfobacterota q=None. NULL - redirect to SHAP centrality per pre-reg

**Table 26. Top 15 genera by GraphTwin gate out-strength consensus (results/twindiscovery\_t1\_consensus.csv).**

| genus              | studies | top | fraction top | p                      | q (BH)                 |
|--------------------|---------|-----|--------------|------------------------|------------------------|
| Acinetobacter      | 32      | 22  | 0.688        | 2.3627632274605656e-15 | 1.3231474073779166e-13 |
| Pseudomonas        | 34      | 20  | 0.588        | 3.4367075799906436e-12 | 7.966283892670575e-11  |
| Staphylococcus     | 31      | 19  | 0.613        | 4.267652085359237e-12  | 7.966283892670575e-11  |
| Abiotrophia        | 20      | 15  | 0.750        | 9.481303720000011e-12  | 8.849216805333343e-11  |
| Peptostreptococcus | 20      | 15  | 0.750        | 9.481303720000011e-12  | 8.849216805333343e-11  |
| Solobacterium      | 20      | 15  | 0.750        | 9.481303720000011e-12  | 8.849216805333343e-11  |
| Bifidobacterium    | 26      | 17  | 0.654        | 1.2810380798769867e-11 | 1.0248304639015893e-10 |
| Lautropia          | 21      | 15  | 0.714        | 3.0079956880000025e-11 | 1.6844775852800012e-10 |
| Treponema          | 21      | 15  | 0.714        | 3.0079956880000025e-11 | 1.6844775852800012e-10 |
| Peptococcus        | 21      | 15  | 0.714        | 3.0079956880000025e-11 | 1.6844775852800012e-10 |
| Fastidiosipila     | 24      | 16  | 0.667        | 3.338913051015403e-11  | 1.6998102805169325e-10 |
| Lactobacillus      | 34      | 19  | 0.559        | 4.1649432962360486e-11 | 1.943640204910156e-10  |
| Blautia            | 28      | 17  | 0.607        | 7.224555155610398e-11  | 3.112116067032171e-10  |
| Atopobium          | 26      | 16  | 0.615        | 1.9801912820122747e-10 | 7.392714119512492e-10  |
| Haemophilus        | 26      | 16  | 0.615        | 1.9801912820122747e-10 | 7.392714119512492e-10  |

**Table 27. Phylum enrichment among the 25 lowest-p genera by twin-gate score (one-sided Fisher, BH).**

| phylum           | in top 25 | genera | p                   | q (BH)             |
|------------------|-----------|--------|---------------------|--------------------|
| Actinomycetota   | 4         | 7      | 0.3774533519261598  | 0.9057244087765735 |
| Campylobacterota | 1         | 1      | 0.44642857142857145 | 0.9057244087765735 |
| Spirochaetota    | 1         | 1      | 0.44642857142857145 | 0.9057244087765735 |
| Pseudomonadota   | 4         | 8      | 0.5175568050151849  | 0.9057244087765735 |
| Fusobacteriota   | 1         | 2      | 0.698051948051948   | 0.909292705075169  |
| Bacillota        | 11        | 27     | 0.7981630556158967  | 0.909292705075169  |
| Bacteroidota     | 2         | 7      | 0.909292705075169   | 0.909292705075169  |

T2 (twin gates vs ridge as disease-literature predictor). T2: 56 genera. Twin-gate coef - 2.28 (p=0.168), AIC 767.1 vs ridge 766.1. TWIN DOES NOT WIN - redirect to CatBoost+SHAP per pre-reg

**Table 28. Negative-binomial models of BugSigDB signature counts (results/twindiscovery\_t2.json).**

| model                | score coef | p     | AIC   | ML alpha | ML coef | ML p   | ML AIC |
|----------------------|------------|-------|-------|----------|---------|--------|--------|
| ridge keystone score | -0.351     | 0.241 | 766.1 | 0.62     | -0.351  | 0.152  | 762.0  |
| twin gate frequency  | -2.28      | 0.168 | 767.1 | 0.63     | -2.28   | 0.0737 | 763.7  |

Redirects (locked in the same pre-registration before any T1 run). R1, the scorer redirect to SHAP-attribution centrality from the validated LightGBM keystone model: Desulfobacterota 1/3 in top decile, q = 0.462; no phylum passes FDR (best q = 0.462). The sulfate-reducer enrichment is therefore specific to the ridge out-strength definition; it does not survive either model-class change. R2, the predictor redirect to CatBoost + SHAP per-genus attribution ranking in the same negative-binomial comparison: coefficient +37.07 (p = 2.6e-35), AIC 2343.9 vs ridge 2443.1 on 200 genera - the redirect WINS. Gradient-boosted interaction attributions fit BugSigDB disease-signature counts better than the linear ridge score in this viewed comparison; this exploratory proxy association is not an independently replicated biological discovery.

**Table 29. T2 redirect: negative-binomial models of BugSigDB signature counts (results/keystone\_redirects.json).**

| model                     | score coef | p       | AIC    |
|---------------------------|------------|---------|--------|
| ridge keystone score      | -0.351     | 0.241   | 2443.1 |
| CatBoost+SHAP attribution | +37.07     | 2.6e-35 | 2343.9 |

## Appendix B. Full MGnify interaction audit (per study)

**Table 30. All 160 audited studies (results/mgnify\_audit.csv). Gain is (prior - interaction) / prior; positive favours interactions.**

| study         | biome                                        | n   | taxa | prior median | interaction median | gain  | Wilcoxon p             |
|---------------|----------------------------------------------|-----|------|--------------|--------------------|-------|------------------------|
| MGYS000 06862 | root:Host-associated: Human:Digestive system | 84  | 55   | 0.173        | 0.170              | 0.004 | 0.1174934141540573     |
| MGYS000 05791 | root:Host-associated: Human:Digestive system | 243 | 47   | 0.178        | 0.172              | 0.006 | 4.8384265757998056e-05 |
| MGYS000       | root:Host-                                   | 197 | 57   | 0.174        | 0.168              | 0.006 | 0.00518887             |

| study         | biome                                        | n   | taxa | prior median | interaction median | gain   | Wilcoxon p             |
|---------------|----------------------------------------------|-----|------|--------------|--------------------|--------|------------------------|
| 06825         | associated: Human:Digestive system           |     |      |              |                    |        | 44323153               |
| MGYS000 06824 | root:Host-associated: Human:Digestive system | 40  | 66   | 0.175        | 0.180              | -0.004 | 0.97349265041521       |
| MGYS000 06822 | root:Host-associated: Human:Digestive system | 55  | 58   | 0.192        | 0.189              | 0.003  | 0.0874094952304293     |
| MGYS000 06814 | root:Host-associated: Human:Digestive system | 68  | 70   | 0.177        | 0.175              | 0.002  | 0.3404810744509645     |
| MGYS000 06815 | root:Host-associated: Human:Digestive system | 67  | 46   | 0.167        | 0.167              | -0.000 | 0.851347514124289      |
| MGYS000 06811 | root:Host-associated: Human:Digestive system | 98  | 39   | 0.160        | 0.154              | 0.007  | 0.0116937067032781     |
| MGYS000 06807 | root:Host-associated: Human:Digestive system | 60  | 65   | 0.179        | 0.180              | -0.001 | 0.192573487700825      |
| MGYS000 06805 | root:Host-associated: Human:Digestive system | 86  | 66   | 0.180        | 0.162              | 0.017  | 0.0159873022676302     |
| MGYS000 06806 | root:Host-associated: Human:Digestive system | 74  | 60   | 0.178        | 0.180              | -0.002 | 0.0059561819902366     |
| MGYS000 06760 | root:Host-associated: Human:Digestive system | 218 | 69   | 0.169        | 0.164              | 0.005  | 1.1800880841672732e-09 |
| MGYS000 02338 | root:Host-associated: Human:Digestive system | 303 | 44   | 0.154        | 0.148              | 0.006  | 0.00272830451495       |
| MGYS000 06797 | root:Host-associated: Human:Digestive system | 175 | 72   | 0.163        | 0.160              | 0.002  | 0.0254205801722315     |

| study                | biome                                        | n   | taxa | prior median | interaction median | gain   | Wilcoxon p             |
|----------------------|----------------------------------------------|-----|------|--------------|--------------------|--------|------------------------|
| <b>MGYS000 06551</b> | root:Host-associated: Human:Digestive system | 400 | 63   | 0.167        | 0.162              | 0.005  | 0.0001113060975781     |
| <b>MGYS000 06789</b> | root:Host-associated: Human:Digestive system | 108 | 63   | 0.178        | 0.179              | -0.001 | 0.0169520637921749     |
| <b>MGYS000 06787</b> | root:Host-associated: Human:Digestive system | 108 | 56   | 0.180        | 0.177              | 0.004  | 0.5418861206076111     |
| <b>MGYS000 06786</b> | root:Host-associated: Human:Digestive system | 134 | 52   | 0.173        | 0.169              | 0.004  | 0.0144364716231558     |
| <b>MGYS000 06759</b> | root:Host-associated: Human:Digestive system | 400 | 72   | 0.183        | 0.176              | 0.006  | 3.964715623438228e-19  |
| <b>MGYS000 06780</b> | root:Host-associated: Human:Digestive system | 24  | 27   | 0.129        | 0.130              | -0.001 | 0.943392056822656      |
| <b>MGYS000 06779</b> | root:Host-associated: Human:Digestive system | 57  | 63   | 0.185        | 0.185              | -0.000 | 0.6420785027937037     |
| <b>MGYS000 06769</b> | root:Host-associated: Human:Digestive system | 116 | 54   | 0.186        | 0.187              | -0.001 | 0.8169950973439684     |
| <b>MGYS000 06765</b> | root:Host-associated: Human:Digestive system | 28  | 43   | 0.154        | 0.159              | -0.005 | 0.6455853804945946     |
| <b>MGYS000 05333</b> | root:Host-associated: Human:Digestive system | 244 | 53   | 0.165        | 0.158              | 0.008  | 1.8156083569124025e-07 |
| <b>MGYS000 05230</b> | root:Host-associated: Human:Digestive system | 400 | 114  | 0.363        | 0.311              | 0.052  | 2.5108503132493804e-33 |
| <b>MGYS000 02675</b> | root:Host-associated: Human:Digestive        | 24  | 23   | 0.142        | 0.136              | 0.007  | 0.704756498336792      |

| study               | biome                                        | n   | taxa | prior median | interaction median | gain   | Wilcoxon p             |
|---------------------|----------------------------------------------|-----|------|--------------|--------------------|--------|------------------------|
|                     | system                                       |     |      |              |                    |        |                        |
| <b>MGYS00006069</b> | root:Host-associated: Human:Digestive system | 400 | 48   | 0.167        | 0.158              | 0.009  | 4.343497031756574e-07  |
| <b>MGYS00006006</b> | root:Host-associated: Human:Digestive system | 400 | 106  | 0.181        | 0.176              | 0.005  | 3.3971201874549978e-22 |
| <b>MGYS00002061</b> | root:Host-associated: Human:Digestive system | 52  | 40   | 0.142        | 0.142              | -0.000 | 0.160154524509844      |
| <b>MGYS00000633</b> | root:Host-associated: Human:Digestive system | 400 | 68   | 0.572        | 0.438              | 0.135  | 7.59928368626225e-21   |
| <b>MGYS00002238</b> | root:Host-associated: Human:Digestive system | 111 | 111  | 0.402        | 0.301              | 0.101  | 2.366731168120885e-13  |
| <b>MGYS00006562</b> | root:Host-associated: Human:Digestive system | 400 | 44   | 0.167        | 0.163              | 0.004  | 8.224596795214574e-11  |
| <b>MGYS00006649</b> | root:Host-associated: Human:Digestive system | 20  | 44   | 0.176        | 0.176              | -0.000 | 0.2305126190185547     |
| <b>MGYS00006641</b> | root:Host-associated: Human:Digestive system | 87  | 52   | 0.159        | 0.156              | 0.002  | 0.175661084603356      |
| <b>MGYS00006642</b> | root:Host-associated: Human:Digestive system | 57  | 19   | 0.149        | 0.150              | -0.001 | 0.0689068793681605     |
| <b>MGYS00006364</b> | root:Host-associated: Human:Digestive system | 160 | 33   | 0.190        | 0.191              | -0.001 | 0.0006405387930733     |
| <b>MGYS00006640</b> | root:Host-associated: Human:Digestive system | 60  | 53   | 0.170        | 0.176              | -0.006 | 0.6725238936938076     |
| <b>MGYS00005253</b> | root:Host-associated: Human:Digestive system | 400 | 83   | 0.315        | 0.286              | 0.030  | 5.630794366587372e-13  |

| study               | biome                                             | n   | taxa | prior median | interaction median | gain  | Wilcoxon p             |
|---------------------|---------------------------------------------------|-----|------|--------------|--------------------|-------|------------------------|
|                     | estive system:Oral                                |     |      |              |                    |       |                        |
| <b>MGYS00006320</b> | root:Host-associated: Human:Digestive system:Oral | 392 | 68   | 0.347        | 0.284              | 0.063 | 2.6018524371287146e-18 |
| <b>MGYS00006263</b> | root:Host-associated: Human:Digestive system:Oral | 400 | 101  | 0.292        | 0.259              | 0.032 | 1.6144709003228016e-16 |
| <b>MGYS00006017</b> | root:Host-associated: Human:Digestive system:Oral | 53  | 144  | 0.314        | 0.291              | 0.023 | 0.0063127550930951     |
| <b>MGYS00005592</b> | root:Host-associated: Human:Digestive system:Oral | 160 | 104  | 0.323        | 0.257              | 0.065 | 2.954937077626768e-12  |
| <b>MGYS00005569</b> | root:Host-associated: Human:Digestive system:Oral | 71  | 112  | 0.330        | 0.246              | 0.083 | 3.1296273722221307e-09 |
| <b>MGYS00005528</b> | root:Host-associated: Human:Digestive system:Oral | 400 | 68   | 0.312        | 0.267              | 0.045 | 2.9447749914759767e-24 |
| <b>MGYS00005506</b> | root:Host-associated: Human:Digestive system:Oral | 400 | 78   | 0.259        | 0.237              | 0.022 | 1.2543867896369403e-26 |
| <b>MGYS00005223</b> | root:Host-associated: Human:Digestive system:Oral | 400 | 99   | 0.288        | 0.218              | 0.070 | 6.025984438246577e-36  |
| <b>MGYS00005143</b> | root:Host-associated: Human:Digestive system:Oral | 400 | 142  | 0.372        | 0.326              | 0.046 | 7.399471504901985e-36  |
| <b>MGYS00002272</b> | root:Host-associated:                             | 40  | 96   | 0.525        | 0.491              | 0.035 | 0.0101974231565691     |

| study               | biome                                             | n   | taxa | prior median | interaction median | gain  | Wilcoxon p             |
|---------------------|---------------------------------------------------|-----|------|--------------|--------------------|-------|------------------------|
|                     | Human:Digestive system:Oral                       |     |      |              |                    |       |                        |
| <b>MGYS00002269</b> | root:Host-associated: Human:Digestive system:Oral | 147 | 105  | 0.410        | 0.342              | 0.069 | 1.7603056013155362e-16 |
| <b>MGYS00002229</b> | root:Host-associated: Human:Digestive system:Oral | 100 | 85   | 0.448        | 0.358              | 0.090 | 5.563455695355892e-12  |
| <b>MGYS00005232</b> | root:Host-associated: Human:Skin                  | 400 | 96   | 0.365        | 0.325              | 0.040 | 9.31108298430499e-22   |
| <b>MGYS00006246</b> | root:Host-associated: Human:Skin                  | 50  | 93   | 0.354        | 0.256              | 0.099 | 0.0065166499849652     |
| <b>MGYS00005384</b> | root:Host-associated: Human:Skin                  | 400 | 99   | 0.422        | 0.260              | 0.162 | 8.321844748869964e-43  |
| <b>MGYS00006795</b> | root:Host-associated: Human:Digestive system      | 31  | 150  | 0.582        | 0.505              | 0.078 | 0.0004954077303409576  |
| <b>MGYS00006794</b> | root:Host-associated: Human:Digestive system      | 184 | 150  | 0.399        | 0.314              | 0.085 | 3.4225595177609226e-22 |
| <b>MGYS00006755</b> | root:Host-associated: Human:Digestive system      | 83  | 150  | 0.562        | 0.375              | 0.187 | 1.3575561479460155e-13 |
| <b>MGYS00005154</b> | root:Host-associated: Human:Digestive system      | 400 | 150  | 0.492        | 0.389              | 0.102 | 9.623147369597524e-46  |
| <b>MGYS00006276</b> | root:Host-associated: Human:Digestive system:Oral | 91  | 150  | 0.298        | 0.283              | 0.015 | 0.0001867751839014596  |
| <b>MGYS00005189</b> | root:Host-associated: Human:Digestive system:Oral | 400 | 150  | 0.656        | 0.407              | 0.249 | 6.400607441863894e-60  |

| study               | biome                                             | n   | taxa | prior median | interaction median | gain  | Wilcoxon p             |
|---------------------|---------------------------------------------------|-----|------|--------------|--------------------|-------|------------------------|
|                     | l                                                 |     |      |              |                    |       |                        |
| <b>MGYS00005197</b> | root:Host-associated: Human:Digestive system:Oral | 400 | 150  | 0.197        | 0.170              | 0.027 | 7.298198067237661e-42  |
| <b>MGYS00002159</b> | root:Host-associated: Human:Digestive system:Oral | 21  | 150  | 0.573        | 0.494              | 0.080 | 0.08879852294921875    |
| <b>MGYS00003719</b> | root:Host-associated: Human:Digestive system:Oral | 21  | 150  | 0.306        | 0.287              | 0.018 | 0.05469322204589844    |
| <b>MGYS00002395</b> | root:Host-associated: Human:Digestive system:Oral | 68  | 150  | 0.351        | 0.346              | 0.005 | 0.025728924772950645   |
| <b>MGYS00002394</b> | root:Host-associated: Human:Digestive system:Oral | 20  | 150  | 0.417        | 0.400              | 0.016 | 0.16495704650878906    |
| <b>MGYS00006648</b> | root:Host-associated: Human:Skin                  | 67  | 150  | 0.470        | 0.344              | 0.126 | 1.0192740768516179e-07 |
| <b>MGYS00005156</b> | root:Host-associated: Human:Skin                  | 400 | 150  | 0.399        | 0.339              | 0.060 | 3.7584514383668004e-33 |
| <b>MGYS00005172</b> | root:Host-associated: Human:Skin                  | 400 | 150  | 0.401        | 0.302              | 0.099 | 1.9412619707417544e-34 |
| <b>MGYS00005533</b> | root:Host-associated: Human:Skin                  | 400 | 150  | 0.317        | 0.316              | 0.001 | 8.614275319605027e-22  |
| <b>MGYS00005212</b> | root:Host-associated: Human:Skin                  | 400 | 150  | 0.374        | 0.292              | 0.083 | 4.0222879063051704e-33 |
| <b>MGYS00005238</b> | root:Host-associated: Human:Skin                  | 400 | 106  | 0.396        | 0.358              | 0.039 | 2.7355102946682556e-13 |
| <b>MGYS00005101</b> | root:Host-associated: Human:Skin                  | 71  | 91   | 0.367        | 0.292              | 0.075 | 0.001176166597423719   |

| study                | biome                            | n   | taxa | prior median | interaction median | gain   | Wilcoxon p             |
|----------------------|----------------------------------|-----|------|--------------|--------------------|--------|------------------------|
| <b>MGYS000 05102</b> | root:Host-associated: Human:Skin | 179 | 150  | 0.381        | 0.264              | 0.118  | 7.298919552265234e-19  |
| <b>MGYS000 05135</b> | root:Host-associated: Human:Skin | 109 | 150  | 0.470        | 0.420              | 0.050  | 1.4252348445599325e-07 |
| <b>MGYS000 05098</b> | root:Host-associated: Human:Skin | 101 | 150  | 0.127        | 0.102              | 0.025  | 1.5388621128648606e-10 |
| <b>MGYS000 05037</b> | root:Host-associated: Human:Skin | 106 | 150  | 0.311        | 0.233              | 0.078  | 6.006215258961562e-07  |
| <b>MGYS000 05090</b> | root:Host-associated: Human:Skin | 120 | 7    | 0.082        | 0.077              | 0.005  | 0.9498798977422346     |
| <b>MGYS000 02184</b> | root:Host-associated: Human:Skin | 400 | 150  | 0.471        | 0.261              | 0.210  | 2.878775655656423e-63  |
| <b>MGYS000 02072</b> | root:Host-associated: Human:Skin | 258 | 150  | 0.233        | 0.190              | 0.043  | 2.0058912688460236e-19 |
| <b>MGYS000 06855</b> | root:Host-associated: Mammals    | 116 | 150  | 0.293        | 0.232              | 0.061  | 1.6553837550067384e-12 |
| <b>MGYS000 06799</b> | root:Host-associated: Mammals    | 27  | 150  | 0.406        | 0.417              | -0.011 | 0.03189583122730255    |
| <b>MGYS000 02673</b> | root:Host-associated: Mammals    | 93  | 150  | 0.650        | 0.533              | 0.116  | 1.1556875270009098e-13 |
| <b>MGYS000 06747</b> | root:Host-associated: Mammals    | 101 | 150  | 0.568        | 0.386              | 0.182  | 4.170399034793852e-18  |
| <b>MGYS000 01223</b> | root:Host-associated: Mammals    | 30  | 98   | 0.194        | 0.173              | 0.021  | 0.08406547829508781    |
| <b>MGYS000 01224</b> | root:Host-associated: Mammals    | 66  | 89   | 0.389        | 0.242              | 0.147  | 1.6687130603014278e-09 |
| <b>MGYS000 05267</b> | root:Host-associated: Mammals    | 400 | 71   | 0.343        | 0.254              | 0.088  | 3.2675706524688328e-52 |
| <b>MGYS000 00426</b> | root:Host-associated: Mammals    | 44  | 66   | 0.272        | 0.245              | 0.027  | 0.4011977486671867     |
| <b>MGYS000 05228</b> | root:Host-associated: Mammals    | 400 | 63   | 0.257        | 0.206              | 0.052  | 2.070366009312925e-33  |
| <b>MGYS000 06537</b> | root:Host-associated: Mammals    | 85  | 150  | 0.495        | 0.407              | 0.088  | 0.00037059731217404824 |

| study                | biome                         | n   | taxa | prior median | interaction median | gain   | Wilcoxon p             |
|----------------------|-------------------------------|-----|------|--------------|--------------------|--------|------------------------|
| <b>MGYS000 06536</b> | root:Host-associated: Mammals | 65  | 150  | 0.384        | 0.384              | -0.000 | 8.467639424268034e-05  |
| <b>MGYS000 06538</b> | root:Host-associated: Mammals | 40  | 150  | 0.515        | 0.512              | 0.003  | 0.08415465162215696    |
| <b>MGYS000 05628</b> | root:Host-associated: Mammals | 400 | 150  | 0.338        | 0.276              | 0.062  | 7.073539193762629e-38  |
| <b>MGYS000 06450</b> | root:Host-associated: Mammals | 144 | 81   | 0.444        | 0.310              | 0.134  | 1.8256817238051636e-13 |
| <b>MGYS000 06456</b> | root:Host-associated: Mammals | 24  | 76   | 0.331        | 0.313              | 0.017  | 0.975736119417176      |
| <b>MGYS000 06446</b> | root:Host-associated: Mammals | 400 | 62   | 0.317        | 0.247              | 0.070  | 4.779278261362484e-25  |
| <b>MGYS000 06433</b> | root:Host-associated: Mammals | 25  | 150  | 0.557        | 0.537              | 0.020  | 0.018744707107543945   |
| <b>MGYS000 06430</b> | root:Host-associated: Mammals | 32  | 150  | 0.397        | 0.299              | 0.099  | 0.014118533720624406   |
| <b>MGYS000 06427</b> | root:Host-associated: Mammals | 85  | 109  | 0.389        | 0.341              | 0.048  | 0.00245958589664498    |
| <b>MGYS000 06424</b> | root:Host-associated: Mammals | 142 | 150  | 0.280        | 0.252              | 0.027  | 2.5100543778701536e-14 |
| <b>MGYS000 06652</b> | root:Host-associated: Plants  | 400 | 150  | 0.250        | 0.175              | 0.076  | 2.8719573482573203e-56 |
| <b>MGYS000 01412</b> | root:Host-associated: Plants  | 48  | 98   | 0.727        | 0.639              | 0.088  | 0.004499547313045582   |
| <b>MGYS000 01368</b> | root:Host-associated: Plants  | 400 | 150  | 0.276        | 0.190              | 0.086  | 5.315213480136056e-65  |
| <b>MGYS000 03172</b> | root:Host-associated: Plants  | 45  | 150  | 0.247        | 0.101              | 0.145  | 5.684341886080802e-14  |
| <b>MGYS000 06578</b> | root:Host-associated: Plants  | 27  | 150  | 0.162        | 0.170              | -0.009 | 0.9153143465518951     |
| <b>MGYS000 00957</b> | root:Host-associated: Plants  | 70  | 61   | 0.381        | 0.351              | 0.030  | 0.0010592552437449615  |
| <b>MGYS000 01704</b> | root:Host-associated: Plants  | 383 | 150  | 0.252        | 0.164              | 0.088  | 2.432607986872169e-55  |
| <b>MGYS000 00867</b> | root:Host-associated: Plants  | 32  | 150  | 0.001        | 0.001              | 0.000  | 0.34500763198357975    |
| <b>MGYS000 06031</b> | root:Host-associated: Plants  | 89  | 150  | 0.477        | 0.402              | 0.075  | 1.1054300201400448e-07 |
| <b>MGYS000</b>       | root:Host-                    | 135 | 18   | 0.233        | 0.216              | 0.017  | 0.00353824             |

| study        | biome                        | n   | taxa | prior median | interaction median | gain  | Wilcoxon p             |
|--------------|------------------------------|-----|------|--------------|--------------------|-------|------------------------|
| 06021        | associated: Plants           |     |      |              |                    |       | 77408565484            |
| MGYS00006018 | root:Host-associated: Plants | 181 | 150  | 0.452        | 0.287              | 0.166 | 7.976256015721753e-13  |
| MGYS00001349 | root:Host-associated: Plants | 108 | 150  | 0.402        | 0.211              | 0.190 | 9.800115982359504e-18  |
| MGYS00005260 | root:Host-associated: Plants | 400 | 150  | 0.396        | 0.248              | 0.148 | 1.2745313376300463e-61 |
| MGYS00005227 | root:Host-associated: Plants | 400 | 150  | 0.223        | 0.201              | 0.022 | 2.4755145503466964e-21 |
| MGYS00000983 | root:Host-associated: Plants | 400 | 150  | 0.178        | 0.141              | 0.036 | 7.250624660443202e-39  |
| MGYS00001347 | root:Host-associated: Plants | 400 | 150  | 0.471        | 0.218              | 0.252 | 6.620371482312783e-66  |
| MGYS00005740 | root:Host-associated: Plants | 34  | 150  | 0.400        | 0.339              | 0.062 | 3.2188836485147476e-07 |
| MGYS00005726 | root:Host-associated: Plants | 33  | 150  | 0.382        | 0.337              | 0.046 | 5.099480040371418e-05  |
| MGYS00005169 | root:Host-associated: Plants | 400 | 150  | 0.311        | 0.171              | 0.140 | 2.3653008551996965e-59 |
| MGYS00005151 | root:Host-associated: Plants | 400 | 150  | 0.348        | 0.111              | 0.238 | 6.538056051035535e-60  |
| MGYS00006750 | root:Host-associated: Birds  | 340 | 89   | 0.142        | 0.135              | 0.007 | 1.1556877632363245e-12 |
| MGYS00006533 | root:Host-associated: Birds  | 94  | 150  | 0.404        | 0.322              | 0.081 | 2.1203049163202955e-09 |
| MGYS00006532 | root:Host-associated: Birds  | 40  | 150  | 0.478        | 0.252              | 0.226 | 5.723268259316683e-08  |
| MGYS00005766 | root:Host-associated: Birds  | 138 | 150  | 0.474        | 0.245              | 0.229 | 7.806550735727007e-23  |
| MGYS00006062 | root:Host-associated: Birds  | 220 | 92   | 0.186        | 0.166              | 0.020 | 2.1652952739860382e-13 |
| MGYS00006042 | root:Host-associated: Birds  | 355 | 96   | 0.284        | 0.124              | 0.160 | 1.1028252358036625e-43 |
| MGYS00005621 | root:Host-associated: Birds  | 60  | 150  | 0.265        | 0.173              | 0.092 | 1.0181154208756968e-06 |
| MGYS00005611 | root:Host-associated: Birds  | 117 | 115  | 0.334        | 0.178              | 0.156 | 3.79430763036704e-14   |
| MGYS00005610 | root:Host-associated:        | 251 | 87   | 0.376        | 0.317              | 0.059 | 1.1443357306387731e-   |

| study                | biome                       | n   | taxa | prior median | interaction median | gain   | Wilcoxon p             |
|----------------------|-----------------------------|-----|------|--------------|--------------------|--------|------------------------|
|                      | Birds                       |     |      |              |                    |        | 14                     |
| <b>MGYS000 05609</b> | root:Host-associated: Birds | 314 | 68   | 0.470        | 0.395              | 0.075  | 1.3785359922199967e-23 |
| <b>MGYS000 05600</b> | root:Host-associated: Birds | 286 | 150  | 0.270        | 0.175              | 0.095  | 2.8587965104295026e-40 |
| <b>MGYS000 05593</b> | root:Host-associated: Birds | 150 | 129  | 0.589        | 0.361              | 0.228  | 1.9022991582307127e-20 |
| <b>MGYS000 05576</b> | root:Host-associated: Birds | 34  | 150  | 0.374        | 0.220              | 0.153  | 3.573950380086899e-08  |
| <b>MGYS000 05187</b> | root:Host-associated: Birds | 400 | 4    | 0.147        | 0.131              | 0.016  | 0.20038180242578474    |
| <b>MGYS000 05316</b> | root:Host-associated: Birds | 400 | 145  | 0.542        | 0.257              | 0.285  | 1.506095419477192e-57  |
| <b>MGYS000 05317</b> | root:Host-associated: Birds | 400 | 150  | 0.491        | 0.349              | 0.141  | 1.3522978810793892e-53 |
| <b>MGYS000 05309</b> | root:Host-associated: Birds | 72  | 150  | 0.312        | 0.194              | 0.118  | 1.6709026814241643e-12 |
| <b>MGYS000 04995</b> | root:Host-associated: Birds | 43  | 150  | 0.256        | 0.223              | 0.033  | 0.10220764011114625    |
| <b>MGYS000 03706</b> | root:Host-associated: Birds | 400 | 150  | 0.516        | 0.422              | 0.094  | 2.0461254236329354e-34 |
| <b>MGYS000 03681</b> | root:Host-associated: Birds | 122 | 150  | 0.674        | 0.444              | 0.231  | 4.108703914852822e-12  |
| <b>MGYS000 06764</b> | root:Host-associated: Fish  | 21  | 150  | 0.262        | 0.181              | 0.081  | 1.811981201171875e-05  |
| <b>MGYS000 06763</b> | root:Host-associated: Fish  | 39  | 150  | 0.420        | 0.364              | 0.055  | 2.3796659661456943e-05 |
| <b>MGYS000 06762</b> | root:Host-associated: Fish  | 137 | 150  | 0.428        | 0.290              | 0.138  | 1.851401098428866e-14  |
| <b>MGYS000 06761</b> | root:Host-associated: Fish  | 123 | 150  | 0.295        | 0.231              | 0.064  | 8.880400175577094e-10  |
| <b>MGYS000 06758</b> | root:Host-associated: Fish  | 135 | 143  | 0.522        | 0.312              | 0.209  | 2.601298498785815e-10  |
| <b>MGYS000 06041</b> | root:Host-associated: Fish  | 400 | 30   | 0.278        | 0.164              | 0.113  | 6.995323507416229e-32  |
| <b>MGYS000 06117</b> | root:Host-associated: Fish  | 90  | 72   | 0.028        | 0.030              | -0.002 | 1.0043017278491953e-06 |
| <b>MGYS000 06086</b> | root:Host-associated: Fish  | 135 | 3    | 0.000        | 0.000              | 0.000  | 1.0                    |

| study               | biome                        | n   | taxa | prior median | interaction median | gain   | Wilcoxon p             |
|---------------------|------------------------------|-----|------|--------------|--------------------|--------|------------------------|
| <b>MGYS00005503</b> | root:Host-associated:Fish    | 400 | 150  | 0.594        | 0.467              | 0.127  | 1.9972646703767122e-30 |
| <b>MGYS00003166</b> | root:Host-associated:Fish    | 70  | 150  | 0.417        | 0.352              | 0.065  | 8.973570568522901e-07  |
| <b>MGYS00004672</b> | root:Host-associated:Fish    | 33  | 150  | 0.422        | 0.372              | 0.050  | 0.05752792255952954    |
| <b>MGYS00002905</b> | root:Host-associated:Fish    | 66  | 134  | 0.191        | 0.069              | 0.122  | 2.107866932199474e-09  |
| <b>MGYS00002608</b> | root:Host-associated:Fish    | 101 | 150  | 0.589        | 0.429              | 0.161  | 4.037951900494873e-09  |
| <b>MGYS00002413</b> | root:Host-associated:Fish    | 173 | 150  | 0.588        | 0.356              | 0.232  | 1.508508341940648e-16  |
| <b>MGYS00002065</b> | root:Host-associated:Fish    | 214 | 150  | 0.499        | 0.363              | 0.136  | 6.023085537966328e-20  |
| <b>MGYS00006798</b> | root:Host-associated:Insecta | 24  | 150  | 0.567        | 0.174              | 0.393  | 8.344650268554688e-06  |
| <b>MGYS00006019</b> | root:Host-associated:Insecta | 20  | 150  | 0.345        | 0.215              | 0.130  | 0.000209808349609375   |
| <b>MGYS00005862</b> | root:Host-associated:Insecta | 21  | 150  | 0.321        | 0.346              | -0.026 | 0.05950736999511719    |
| <b>MGYS00005215</b> | root:Host-associated:Insecta | 400 | 92   | 0.296        | 0.244              | 0.052  | 6.345809298780021e-27  |
| <b>MGYS00005163</b> | root:Host-associated:Insecta | 386 | 150  | 0.548        | 0.467              | 0.081  | 2.789722004048674e-31  |
| <b>MGYS00003800</b> | root:Host-associated:Insecta | 154 | 96   | 0.184        | 0.184              | 0.000  | 0.6706023900543987     |
| <b>MGYS00003705</b> | root:Host-associated:Insecta | 384 | 70   | 0.408        | 0.370              | 0.038  | 2.7686978173767286e-09 |

## Appendix C. Keystone consensus, all genera

Table 31. All 248 genera modelled in  $\geq 20$  studies (results/keystone\_consensus.csv).

| genus                  | studies | top | fraction top | p                     | q (BH)                |
|------------------------|---------|-----|--------------|-----------------------|-----------------------|
| <b>Cardiobacterium</b> | 30      | 13  | 0.433        | 2.302322724161098e-06 | 0.0005709760355919523 |
| <b>Eikenella</b>       | 28      | 11  | 0.393        | 4.232321631540403e-05 | 0.0052480788231101    |
| <b>Comamonas</b>       | 51      | 13  | 0.255        | 0.0012261499891986897 | 0.09862502890393288   |
| <b>Johnsonella</b>     | 35      | 10  | 0.286        | 0.0017422103851516847 | 0.09862502890393288   |

| genus                 | studies | top | fraction top | p                         | q (BH)                  |
|-----------------------|---------|-----|--------------|---------------------------|-------------------------|
| Simonsiella           | 20      | 7   | 0.350        | 0.002386089408<br>966118  | 0.098625028903<br>93288 |
| Desulfobulbus         | 20      | 7   | 0.350        | 0.002386089408<br>966118  | 0.098625028903<br>93288 |
| Sutterella            | 56      | 13  | 0.232        | 0.003030225403<br>3037486 | 0.107356557145<br>61852 |
| Agathobacter          | 77      | 16  | 0.208        | 0.003583040147<br>751078  | 0.111074244580<br>28342 |
| Collinsella           | 67      | 14  | 0.209        | 0.005889699276<br>757321  | 0.161109957476<br>64922 |
| Desulfovibrio         | 68      | 14  | 0.206        | 0.006747618035<br>44495   | 0.161109957476<br>64922 |
| Anaerosporebacter     | 30      | 8   | 0.267        | 0.007783619237<br>92132   | 0.161109957476<br>64922 |
| Solobacterium         | 49      | 11  | 0.224        | 0.008004971888<br>016863  | 0.161109957476<br>64922 |
| Bilophila             | 56      | 12  | 0.214        | 0.008445280029<br>017902  | 0.161109957476<br>64922 |
| Barnesiella           | 47      | 10  | 0.213        | 0.016260359901<br>383525  | 0.288040661110<br>2225  |
| Bergeyella            | 49      | 10  | 0.204        | 0.021501268881<br>13899   | 0.350420126357<br>6335  |
| Anaerovorax           | 23      | 6   | 0.261        | 0.022607750087<br>589258  | 0.350420126357<br>6335  |
| Odoribacter           | 57      | 11  | 0.193        | 0.024238404563<br>49196   | 0.353595548926<br>2357  |
| Cetobacterium         | 30      | 7   | 0.233        | 0.025826788661<br>909898  | 0.355835754897<br>42525 |
| Mogibacterium         | 45      | 9   | 0.200        | 0.031994021841<br>86363   | 0.412814207952<br>3551  |
| Eubacterium           | 60      | 11  | 0.183        | 0.034209123078<br>73303   | 0.412814207952<br>3551  |
| Subdoligranulum       | 83      | 14  | 0.169        | 0.035467001433<br>08487   | 0.412814207952<br>3551  |
| Slackia               | 39      | 8   | 0.205        | 0.036620615221<br>57989   | 0.412814207952<br>3551  |
| Campylobacter         | 69      | 12  | 0.174        | 0.040017895148<br>75476   | 0.431497304212<br>66    |
| Enhydrobacter         | 20      | 5   | 0.250        | 0.043174495284<br>46339   | 0.446136451272<br>78834 |
| Fastidiosipila        | 86      | 14  | 0.163        | 0.046060001870<br>58948   | 0.453852358283<br>953   |
| Stomatobaculum        | 34      | 7   | 0.206        | 0.048144326629<br>12472   | 0.453852358283<br>953   |
| Phascolarctobacterium | 56      | 10  | 0.179        | 0.049411345458<br>33359   | 0.453852358283<br>953   |
| Fournierella          | 28      | 6   | 0.214        | 0.055006948868<br>291425  | 0.470404252390<br>906   |
| Propionibacterium     | 28      | 6   | 0.214        | 0.055006948868<br>291425  | 0.470404252390<br>906   |
| Moryella              | 43      | 8   | 0.186        | 0.060669351591<br>22358   | 0.483186617866<br>3752  |
| Aggregatibacter       | 36      | 7   | 0.194        | 0.062828142375<br>83728   | 0.483186617866<br>3752  |
| Fretibacterium        | 29      | 6   | 0.207        | 0.063717444452<br>2859    | 0.483186617866<br>3752  |
| Leptotrichia          | 51      | 9   | 0.176        | 0.064294993506<br>41282   | 0.483186617866<br>3752  |

| genus                           | studies | top | fraction top | p                   | q (BH)             |
|---------------------------------|---------|-----|--------------|---------------------|--------------------|
| <b>Dolosigranulum</b>           | 37      | 7   | 0.189        | 0.07108505827175729 | 0.5185027779822297 |
| <b>Anaeroplasma</b>             | 30      | 6   | 0.200        | 0.07319010839987991 | 0.5186041966620062 |
| <b>Papillibacter</b>            | 24      | 5   | 0.208        | 0.08507488587867089 | 0.5777058981531825 |
| <b>Cutibacterium</b>            | 54      | 9   | 0.167        | 0.08618999286962803 | 0.5777058981531825 |
| <b>Anaerotruncus</b>            | 39      | 7   | 0.179        | 0.0894338897394184  | 0.5836738067204149 |
| <b>Abiotrophia</b>              | 40      | 7   | 0.175        | 0.09951642396555122 | 0.6265462301476376 |
| <b>Lautropia</b>                | 56      | 9   | 0.161        | 0.10297079417076241 | 0.6265462301476376 |
| <b>Moraxella</b>                | 33      | 6   | 0.182        | 0.10610863575080959 | 0.6265462301476376 |
| <b>Butyricimonas</b>            | 33      | 6   | 0.182        | 0.10610863575080959 | 0.6265462301476376 |
| <b>Butyricicoccus</b>           | 66      | 10  | 0.152        | 0.11992354893655092 | 0.6729066165656145 |
| <b>Enterorhabdus</b>            | 42      | 7   | 0.167        | 0.12144720681506566 | 0.6729066165656145 |
| <b>Hydrogenoanaerobacterium</b> | 27      | 5   | 0.185        | 0.1265574483082461  | 0.6729066165656145 |
| <b>Tumebacillus</b>             | 27      | 5   | 0.185        | 0.1265574483082461  | 0.6729066165656145 |
| <b>Leuconostoc</b>              | 35      | 6   | 0.171        | 0.13163577484183725 | 0.6729066165656145 |
| <b>Saccharofermentans</b>       | 20      | 4   | 0.200        | 0.1329533234343351  | 0.6729066165656145 |
| <b>Marinomonas</b>              | 20      | 4   | 0.200        | 0.1329533234343351  | 0.6729066165656145 |
| <b>Parasutterella</b>           | 60      | 9   | 0.150        | 0.1416356047983447  | 0.691053967992992  |
| <b>Ammopiptanthus</b>           | 28      | 5   | 0.179        | 0.14211190470823626 | 0.691053967992992  |
| <b>Finegoldia</b>               | 36      | 6   | 0.167        | 0.14539730133503725 | 0.6934332832901777 |
| <b>Alistipes</b>                | 78      | 11  | 0.141        | 0.1536206356148474  | 0.718828634575135  |
| <b>Ruminococcus</b>             | 45      | 7   | 0.156        | 0.15850738539835585 | 0.7279598440517083 |
| <b>Paucibacter</b>              | 88      | 12  | 0.136        | 0.1670511156120358  | 0.753248666759725  |
| <b>Carnobacterium</b>           | 22      | 4   | 0.182        | 0.17192789706645586 | 0.7613949727228759 |
| <b>Erysipelothrix</b>           | 23      | 4   | 0.174        | 0.19273101351862001 | 0.8385489710985572 |
| <b>Oribacterium</b>             | 65      | 9   | 0.138        | 0.198722696795465   | 0.8497108414702642 |
| <b>Neisseria</b>                | 66      | 9   | 0.136        | 0.21116466751003035 | 0.8710989390747419 |
| <b>Fluviicola</b>               | 32      | 5   | 0.156        | 0.21149828869932752 | 0.8710989390747419 |
| <b>Ezakiella</b>                | 24      | 4   | 0.167        | 0.2142622390466099  | 0.8710989390747419 |
| <b>Akkermansia</b>              | 67      | 9   | 0.134        | 0.22390696165563617 | 0.8769946942869572 |

| genus                         | studies | top | fraction top | p                   | q (BH)             |
|-------------------------------|---------|-----|--------------|---------------------|--------------------|
| <b>Selenomonas</b>            | 33      | 5   | 0.152        | 0.23031786958299133 | 0.8769946942869572 |
| <b>Methanobrevibacter</b>     | 33      | 5   | 0.152        | 0.23031786958299133 | 0.8769946942869572 |
| <b>Nocardia</b>               | 25      | 4   | 0.160        | 0.23640864244682802 | 0.8769946942869572 |
| <b>Fibrobacter</b>            | 25      | 4   | 0.160        | 0.23640864244682802 | 0.8769946942869572 |
| <b>Megasphaera</b>            | 68      | 9   | 0.132        | 0.2369300182146215  | 0.8769946942869572 |
| <b>Lachnoanaerobaculum</b>    | 34      | 5   | 0.147        | 0.2495917162121227  | 0.8891872058135635 |
| <b>Kingella</b>               | 34      | 5   | 0.147        | 0.2495917162121227  | 0.8891872058135635 |
| <b>Parabacteroides</b>        | 87      | 11  | 0.126        | 0.2509802597054413  | 0.8891872058135635 |
| <b>Catonella</b>              | 35      | 5   | 0.143        | 0.2692510397738372  | 0.9404825051255159 |
| <b>Corynebacterium</b>        | 91      | 11  | 0.121        | 0.29998647993678423 | 1.0                |
| <b>Tannerella</b>             | 28      | 4   | 0.143        | 0.3054336969081328  | 1.0                |
| <b>Candidatus_Soleiferrea</b> | 37      | 5   | 0.135        | 0.30945356374288324 | 1.0                |
| <b>Lachnospira</b>            | 55      | 7   | 0.127        | 0.30955723920441663 | 1.0                |
| <b>Anaerostipes</b>           | 83      | 10  | 0.120        | 0.3162541477570478  | 1.0                |
| <b>Bradyrhizobium</b>         | 65      | 8   | 0.123        | 0.32314240394111926 | 1.0                |
| <b>Olsenella</b>              | 47      | 6   | 0.128        | 0.3285688971461327  | 1.0                |
| <b>Acetitomaculum</b>         | 29      | 4   | 0.138        | 0.328952034984918   | 1.0                |
| <b>Pedobacter</b>             | 38      | 5   | 0.132        | 0.329863552054893   | 1.0                |
| <b>Atopobium</b>              | 57      | 7   | 0.123        | 0.34289143772806113 | 1.0                |
| <b>Gemmata</b>                | 21      | 3   | 0.143        | 0.3515911755169639  | 1.0                |
| <b>Negativibacillus</b>       | 58      | 7   | 0.121        | 0.35972337214077327 | 1.0                |
| <b>Treponema</b>              | 40      | 5   | 0.125        | 0.3709823034657993  | 1.0                |
| <b>Anaerofilum</b>            | 22      | 3   | 0.136        | 0.37995906158809667 | 1.0                |
| <b>Phaselicystis</b>          | 22      | 3   | 0.136        | 0.37995906158809667 | 1.0                |
| <b>Marvinbryantia</b>         | 50      | 6   | 0.120        | 0.3838769922757232  | 1.0                |
| <b>Brevundimonas</b>          | 69      | 8   | 0.116        | 0.3854441012423575  | 1.0                |
| <b>Stenotrophomonas</b>       | 60      | 7   | 0.117        | 0.39354859789913854 | 1.0                |
| <b>Coxiella</b>               | 33      | 4   | 0.121        | 0.4230563358743078  | 1.0                |
| <b>Phenylobacterium</b>       | 43      | 5   | 0.116        | 0.43252905747178994 | 1.0                |

| genus                   | studies | top | fraction top | p                   | q (BH) |
|-------------------------|---------|-----|--------------|---------------------|--------|
| Coprobacter             | 24      | 3   | 0.125        | 0.4357262730487911  | 1.0    |
| Filifactor              | 24      | 3   | 0.125        | 0.4357262730487911  | 1.0    |
| Shewanella              | 24      | 3   | 0.125        | 0.4357262730487911  | 1.0    |
| Blautia                 | 102     | 11  | 0.108        | 0.4432030664009525  | 1.0    |
| Holdemania              | 34      | 4   | 0.118        | 0.4461849518292653  | 1.0    |
| Enterobacter            | 35      | 4   | 0.114        | 0.46901513402996614 | 1.0    |
| Aeromonas               | 36      | 4   | 0.111        | 0.4914885946337808  | 1.0    |
| Caulobacter             | 36      | 4   | 0.111        | 0.4914885946337808  | 1.0    |
| Pseudopropionibacterium | 27      | 3   | 0.111        | 0.5154188583079999  | 1.0    |
| Parvimonas              | 48      | 5   | 0.104        | 0.5314392758472106  | 1.0    |
| Mycoplasma              | 48      | 5   | 0.104        | 0.5314392758472106  | 1.0    |
| Brachybacterium         | 38      | 4   | 0.105        | 0.5351640227227555  | 1.0    |
| Sphingobacterium        | 39      | 4   | 0.103        | 0.5562806425628786  | 1.0    |
| Acinetobacter           | 101     | 10  | 0.099        | 0.5617514622659117  | 1.0    |
| Enterococcus            | 91      | 9   | 0.099        | 0.565055679835745   | 1.0    |
| Arthrobacter            | 50      | 5   | 0.100        | 0.5688015931709377  | 1.0    |
| Dorea                   | 71      | 7   | 0.099        | 0.5736490594127721  | 1.0    |
| Intestinimonas          | 40      | 4   | 0.100        | 0.5768693469069974  | 1.0    |
| Massilia                | 61      | 6   | 0.098        | 0.5794548905240392  | 1.0    |
| Facklamia               | 30      | 3   | 0.100        | 0.5886487604404943  | 1.0    |
| Williamsia              | 20      | 2   | 0.100        | 0.6082530018748321  | 1.0    |
| Cloacibacterium         | 20      | 2   | 0.100        | 0.6082530018748321  | 1.0    |
| Methylobacterium        | 84      | 8   | 0.095        | 0.6111403728323437  | 1.0    |
| Peredibacter            | 31      | 3   | 0.097        | 0.6114143824771845  | 1.0    |
| Terrisporobacter        | 31      | 3   | 0.097        | 0.6114143824771845  | 1.0    |
| Granulicatella          | 63      | 6   | 0.095        | 0.6117004170520003  | 1.0    |
| Halomonas               | 32      | 3   | 0.094        | 0.6333164809193795  | 1.0    |
| Gordonia                | 21      | 2   | 0.095        | 0.6352700362282919  | 1.0    |
| Acidibacter             | 33      | 3   | 0.091        | 0.6543424954238868  | 1.0    |

| genus                           | studies | top | fraction top | p                  | q (BH) |
|---------------------------------|---------|-----|--------------|--------------------|--------|
| <b>Turicibacter</b>             | 77      | 7   | 0.091        | 0.6604455124559978 | 1.0    |
| <b>Alloprevotella</b>           | 77      | 7   | 0.091        | 0.6604455124559978 | 1.0    |
| <b>Hungatella</b>               | 22      | 2   | 0.091        | 0.6608011336923114 | 1.0    |
| <b>Limnobacter</b>              | 22      | 2   | 0.091        | 0.6608011336923114 | 1.0    |
| <b>Arcobacter</b>               | 22      | 2   | 0.091        | 0.6608011336923114 | 1.0    |
| <b>Peptoniphilus</b>            | 56      | 5   | 0.089        | 0.6706712131335167 | 1.0    |
| <b>Ralstonia</b>                | 34      | 3   | 0.088        | 0.6744867738362695 | 1.0    |
| <b>Ilumatobacter</b>            | 23      | 2   | 0.087        | 0.6848733113012442 | 1.0    |
| <b>Peptococcus</b>              | 57      | 5   | 0.088        | 0.6860039469102792 | 1.0    |
| <b>Weissella</b>                | 46      | 4   | 0.087        | 0.6880561169183461 | 1.0    |
| <b>Candidatus_Saccharimonas</b> | 47      | 4   | 0.085        | 0.7044129671607624 | 1.0    |
| <b>Caproiciproducens</b>        | 24      | 2   | 0.083        | 0.7075230420514673 | 1.0    |
| <b>Sellimonas</b>               | 24      | 2   | 0.083        | 0.7075230420514673 | 1.0    |
| <b>Lachnoclostridium</b>        | 92      | 8   | 0.087        | 0.7120145513804161 | 1.0    |
| <b>Haemophilus</b>              | 94      | 8   | 0.085        | 0.7343157532397324 | 1.0    |
| <b>Microbacterium</b>           | 49      | 4   | 0.082        | 0.7352337339766317 | 1.0    |
| <b>Roseomonas</b>               | 38      | 3   | 0.079        | 0.7463302211239846 | 1.0    |
| <b>Planococcus</b>              | 26      | 2   | 0.077        | 0.7487357043078514 | 1.0    |
| <b>Actinomyces</b>              | 85      | 7   | 0.082        | 0.7578449260136146 | 1.0    |
| <b>Shuttleworthia</b>           | 51      | 4   | 0.078        | 0.763562609007388  | 1.0    |
| <b>Peptostreptococcus</b>       | 51      | 4   | 0.078        | 0.763562609007388  | 1.0    |
| <b>Candidatus_Stoquefichus</b>  | 27      | 2   | 0.074        | 0.7674010519878396 | 1.0    |
| <b>Pirellula</b>                | 27      | 2   | 0.074        | 0.7674010519878396 | 1.0    |
| <b>Vibrio</b>                   | 27      | 2   | 0.074        | 0.7674010519878396 | 1.0    |
| <b>Arenimonas</b>               | 27      | 2   | 0.074        | 0.7674010519878396 | 1.0    |
| <b>Bifidobacterium</b>          | 98      | 8   | 0.082        | 0.775278475920071  | 1.0    |
| <b>Fusobacterium</b>            | 99      | 8   | 0.081        | 0.784763296071283  | 1.0    |
| <b>Dyadobacter</b>              | 28      | 2   | 0.071        | 0.7848459730887516 | 1.0    |
| <b>Gemmatirosa</b>              | 28      | 2   | 0.071        | 0.7848459730887516 | 1.0    |

| genus                    | studies | top | fraction top | p                  | q (BH) |
|--------------------------|---------|-----|--------------|--------------------|--------|
| <b>Faecalitalea</b>      | 28      | 2   | 0.071        | 0.7848459730887516 | 1.0    |
| <b>Paraprevotella</b>    | 41      | 3   | 0.073        | 0.7914253184930894 | 1.0    |
| <b>Psychrobacter</b>     | 41      | 3   | 0.073        | 0.7914253184930894 | 1.0    |
| <b>Anaerofustis</b>      | 29      | 2   | 0.069        | 0.8011278994496028 | 1.0    |
| <b>Gaiella</b>           | 29      | 2   | 0.069        | 0.8011278994496028 | 1.0    |
| <b>Aquabacterium</b>     | 29      | 2   | 0.069        | 0.8011278994496028 | 1.0    |
| <b>Rubrobacter</b>       | 29      | 2   | 0.069        | 0.8011278994496028 | 1.0    |
| <b>Streptomyces</b>      | 101     | 8   | 0.079        | 0.8028386628328019 | 1.0    |
| <b>Eggerthella</b>       | 30      | 2   | 0.067        | 0.8163049808073963 | 1.0    |
| <b>Legionella</b>        | 43      | 3   | 0.070        | 0.8176186853856384 | 1.0    |
| <b>Deinococcus</b>       | 44      | 3   | 0.068        | 0.8296311089521422 | 1.0    |
| <b>Flavonifractor</b>    | 44      | 3   | 0.068        | 0.8296311089521422 | 1.0    |
| <b>Roseburia</b>         | 95      | 7   | 0.074        | 0.848933539294995  | 1.0    |
| <b>Staphylococcus</b>    | 95      | 7   | 0.074        | 0.848933539294995  | 1.0    |
| <b>Fusicatenibacter</b>  | 71      | 5   | 0.070        | 0.8499618939444202 | 1.0    |
| <b>Tyzzerella</b>        | 46      | 3   | 0.065        | 0.8516246193425087 | 1.0    |
| <b>Holdemanella</b>      | 46      | 3   | 0.065        | 0.8516246193425087 | 1.0    |
| <b>Helicobacter</b>      | 33      | 2   | 0.061        | 0.8557852795477143 | 1.0    |
| <b>Dialister</b>         | 85      | 6   | 0.071        | 0.8640122599861861 | 1.0    |
| <b>Bacillus</b>          | 110     | 8   | 0.073        | 0.870161480012168  | 1.0    |
| <b>Dietzia</b>           | 35      | 2   | 0.057        | 0.8776235086447748 | 1.0    |
| <b>Aquicella</b>         | 20      | 1   | 0.050        | 0.8784233454094308 | 1.0    |
| <b>Photobacterium</b>    | 20      | 1   | 0.050        | 0.8784233454094308 | 1.0    |
| <b>Kocuria</b>           | 49      | 3   | 0.061        | 0.8799573346772351 | 1.0    |
| <b>Catenibacterium</b>   | 49      | 3   | 0.061        | 0.8799573346772351 | 1.0    |
| <b>Ruminiclostridium</b> | 49      | 3   | 0.061        | 0.8799573346772351 | 1.0    |
| <b>Eisenbergiella</b>    | 36      | 2   | 0.056        | 0.887358002275304  | 1.0    |
| <b>Solirubrobacter</b>   | 36      | 2   | 0.056        | 0.887358002275304  | 1.0    |
| <b>Flavisolibacter</b>   | 21      | 1   | 0.048        | 0.8905810108684877 | 1.0    |

| genus                         | studies | top | fraction top | p                  | q (BH) |
|-------------------------------|---------|-----|--------------|--------------------|--------|
| <b>Chryseobacterium</b>       | 65      | 4   | 0.062        | 0.9004471748135983 | 1.0    |
| <b>Jeotgalicoccus</b>         | 22      | 1   | 0.045        | 0.9015229097816388 | 1.0    |
| <b>Chthoniobacter</b>         | 22      | 1   | 0.045        | 0.9015229097816388 | 1.0    |
| <b>Acidaminococcus</b>        | 22      | 1   | 0.045        | 0.9015229097816388 | 1.0    |
| <b>Novosphingobium</b>        | 22      | 1   | 0.045        | 0.9015229097816388 | 1.0    |
| <b>Porphyromonas</b>          | 66      | 4   | 0.061        | 0.9068051410155196 | 1.0    |
| <b>Lactococcus</b>            | 68      | 4   | 0.059        | 0.918447882785011  | 1.0    |
| <b>Erysipelatoclostridium</b> | 68      | 4   | 0.059        | 0.918447882785011  | 1.0    |
| <b>Hyphomicrobium</b>         | 40      | 2   | 0.050        | 0.9195263039855227 | 1.0    |
| <b>Caldicoprobacter</b>       | 24      | 1   | 0.042        | 0.9202335569231275 | 1.0    |
| <b>Leucobacter</b>            | 24      | 1   | 0.042        | 0.9202335569231275 | 1.0    |
| <b>Faecalibaculum</b>         | 25      | 1   | 0.040        | 0.9282102012308148 | 1.0    |
| <b>Oscillibacter</b>          | 70      | 4   | 0.057        | 0.9287694277368974 | 1.0    |
| <b>Rubellimicrobium</b>       | 26      | 1   | 0.038        | 0.9353891811077333 | 1.0    |
| <b>Anaerococcus</b>           | 58      | 3   | 0.052        | 0.938209796729051  | 1.0    |
| <b>Pedomicrobium</b>          | 27      | 1   | 0.037        | 0.94185026299696   | 1.0    |
| <b>Gemmatimonas</b>           | 27      | 1   | 0.037        | 0.94185026299696   | 1.0    |
| <b>Lawsonella</b>             | 44      | 2   | 0.045        | 0.9428911025791791 | 1.0    |
| <b>Brevibacterium</b>         | 45      | 2   | 0.044        | 0.9476322185914736 | 1.0    |
| <b>Desulfosporosinus</b>      | 29      | 1   | 0.034        | 0.9528987130275376 | 1.0    |
| <b>Intestinibacter</b>        | 29      | 1   | 0.034        | 0.9528987130275376 | 1.0    |
| <b>Nakamurella</b>            | 29      | 1   | 0.034        | 0.9528987130275376 | 1.0    |
| <b>Iamia</b>                  | 30      | 1   | 0.033        | 0.9576088417247838 | 1.0    |
| <b>Medicago</b>               | 31      | 1   | 0.032        | 0.9618479575523055 | 1.0    |
| <b>Pajaroellobacter</b>       | 31      | 1   | 0.032        | 0.9618479575523055 | 1.0    |
| <b>Lysobacter</b>             | 31      | 1   | 0.032        | 0.9618479575523055 | 1.0    |
| <b>Mycobacterium</b>          | 80      | 4   | 0.050        | 0.9646937415188622 | 1.0    |
| <b>Altererythrobacter</b>     | 33      | 1   | 0.030        | 0.9690968456173674 | 1.0    |
| <b>Marmoricola</b>            | 33      | 1   | 0.030        | 0.9690968456173674 | 1.0    |

| genus                   | studies | top | fraction top | p                  | q (BH) |
|-------------------------|---------|-----|--------------|--------------------|--------|
| <b>Ferruginibacter</b>  | 33      | 1   | 0.030        | 0.9690968456173674 | 1.0    |
| <b>Gemella</b>          | 67      | 3   | 0.045        | 0.969280674082702  | 1.0    |
| <b>Nocardioides</b>     | 67      | 3   | 0.045        | 0.969280674082702  | 1.0    |
| <b>Faecalibacterium</b> | 97      | 5   | 0.052        | 0.9708660715848146 | 1.0    |
| <b>Romboutsia</b>       | 83      | 4   | 0.048        | 0.9716070539113301 | 1.0    |
| <b>Paracoccus</b>       | 52      | 2   | 0.038        | 0.9717057741118126 | 1.0    |
| <b>Sphingomonas</b>     | 84      | 4   | 0.048        | 0.973614419469635  | 1.0    |
| <b>Paenibacillus</b>    | 70      | 3   | 0.043        | 0.9758187016428539 | 1.0    |
| <b>Oscillospira</b>     | 36      | 1   | 0.028        | 0.9774716004550609 | 1.0    |
| <b>Clostridium</b>      | 102     | 5   | 0.049        | 0.9793640065142518 | 1.0    |
| <b>Hymenobacter</b>     | 40      | 1   | 0.025        | 0.9852191170585655 | 1.0    |
| <b>Rothia</b>           | 77      | 3   | 0.039        | 0.9863104260537002 | 1.0    |
| <b>Megamonas</b>        | 41      | 1   | 0.024        | 0.9866972053527089 | 1.0    |
| <b>Lactobacillus</b>    | 124     | 6   | 0.048        | 0.9877599629860419 | 1.0    |
| <b>Veillonella</b>      | 95      | 4   | 0.042        | 0.9884599362038465 | 1.0    |
| <b>Devosia</b>          | 45      | 1   | 0.022        | 0.9912720364319123 | 1.0    |
| <b>Rhodococcus</b>      | 47      | 1   | 0.021        | 0.9929303495098489 | 1.0    |
| <b>Pseudomonas</b>      | 136     | 6   | 0.044        | 0.994695056461212  | 1.0    |
| <b>Flavobacterium</b>   | 74      | 2   | 0.027        | 0.9962087599681586 | 1.0    |
| <b>Streptococcus</b>    | 135     | 5   | 0.037        | 0.998208418667266  | 1.0    |
| <b>Prevotella</b>       | 72      | 1   | 0.014        | 0.9994924712139436 | 1.0    |
| <b>Bacteroides</b>      | 111     | 2   | 0.018        | 0.9998888633544281 | 1.0    |
| <b>Geodermatophilus</b> | 22      | 0   | 0.000        | 1.0                | 1.0    |
| <b>Conexibacter</b>     | 26      | 0   | 0.000        | 1.0                | 1.0    |
| <b>Exiguobacterium</b>  | 22      | 0   | 0.000        | 1.0                | 1.0    |
| <b>Aerococcus</b>       | 27      | 0   | 0.000        | 1.0                | 1.0    |
| <b>Acidovorax</b>       | 28      | 0   | 0.000        | 1.0                | 1.0    |
| <b>Bdellovibrio</b>     | 46      | 0   | 0.000        | 1.0                | 1.0    |
| <b>Blastocatella</b>    | 23      | 0   | 0.000        | 1.0                | 1.0    |
| <b>Butyrivibrio</b>     | 26      | 0   | 0.000        | 1.0                | 1.0    |
| <b>Bryobacter</b>       | 28      | 0   | 0.000        | 1.0                | 1.0    |
| <b>Capnocytophaga</b>   | 44      | 0   | 0.000        | 1.0                | 1.0    |
| <b>Cellvibrio</b>       | 25      | 0   | 0.000        | 1.0                | 1.0    |

| genus                    | studies | top | fraction top | p   | q (BH) |
|--------------------------|---------|-----|--------------|-----|--------|
| <b>Haliangium</b>        | 39      | 0   | 0.000        | 1.0 | 1.0    |
| <b>Luteolibacter</b>     | 28      | 0   | 0.000        | 1.0 | 1.0    |
| <b>Micrococcus</b>       | 21      | 0   | 0.000        | 1.0 | 1.0    |
| <b>Pseudonocardia</b>    | 38      | 0   | 0.000        | 1.0 | 1.0    |
| <b>Sediminibacterium</b> | 27      | 0   | 0.000        | 1.0 | 1.0    |
| <b>Reyranella</b>        | 31      | 0   | 0.000        | 1.0 | 1.0    |
| <b>Pseudolabrys</b>      | 20      | 0   | 0.000        | 1.0 | 1.0    |
| <b>Undibacterium</b>     | 23      | 0   | 0.000        | 1.0 | 1.0    |

## Appendix D. External tools ledger

**Table 32. 48 external tools, libraries, APIs and databases used inside this project (results/tools\_ledger.csv).**

| tool                                  | kind           | where_used                                                                        | counts_for_gate | note                                                                   |
|---------------------------------------|----------------|-----------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------------|
| <b>NumPy</b>                          | library        | src/microtwin;<br>run_bench.py;<br>scripts/                                       | True            |                                                                        |
| <b>SciPy</b>                          | library        | src/microtwin/<br>audit.py; scripts/<br>(stats; FDR inputs)                       | True            |                                                                        |
| <b>pandas</b>                         | library        | scripts/<br>paper/build_paper.py                                                  | True            |                                                                        |
| <b>PyTorch</b>                        | library        | src/microtwin/<br>models.py;<br>run_bench.py<br>(cNODE and graph-<br>twin models) | True            |                                                                        |
| <b>Matplotlib</b>                     | library        | scripts/<br>summarise_audit.py<br>(fig_mgnify_audit.png)                          | True            |                                                                        |
| <b>python-docx</b>                    | library        | paper/paperkit.py                                                                 | False           | infrastructure<br>(excluded per<br>program convention<br>25 Sep 01:18) |
| <b>pytest</b>                         | test framework | tests/<br>test_microtwin.py;<br>tests/test_audit.py                               | False           | infrastructure<br>(excluded per<br>program convention<br>25 Sep 01:18) |
| <b>LibreOffice (soffice headless)</b> | application    | paper docx -> pdf<br>render                                                       | False           | infrastructure<br>(excluded per<br>program convention<br>25 Sep 01:18) |
| <b>poppler pdfinfo</b>                | application    | page count of<br>rendered paper                                                   | False           | infrastructure<br>(excluded per<br>program convention<br>25 Sep 01:18) |
| <b>Git + GitHub</b>                   | service        | repo hosting and<br>pushes                                                        | False           | infrastructure<br>(excluded per<br>program convention<br>25 Sep 01:18) |
| <b>EBI MGnify REST API v1</b>         | web API        | scripts fetching<br>data/raw/mgnify                                               | True            |                                                                        |

| tool                                                                                                 | kind                | where_used                                                                                                                  | counts_for_gate | note                                                                                       |
|------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------------|
|                                                                                                      |                     | (manifest.csv);<br>scripts/keystone_mg<br>nify.py                                                                           |                 |                                                                                            |
| <b>cNODE published<br/>benchmark data<br/>(Michel-Mata et al.<br/>2022)</b>                          | public data release | data/raw/cnode/<br>*.csv; run_bench.py                                                                                      | True            |                                                                                            |
| <b>MDSINE2<br/>published data and<br/>Source Data<br/>(Gibson et al. 2025)</b>                       | public data release | data/raw/mdsine2*;<br>mdsine2 head-to-<br>head results                                                                      | True            |                                                                                            |
| <b>scikit-bio</b>                                                                                    | library             | scripts/<br>predictors_audit.py<br>(Shannon alpha,<br>Bray-Curtis beta<br>diversity)                                        | True            |                                                                                            |
| <b>scikit-learn</b>                                                                                  | library             | scripts/<br>predictors_audit.py<br>(random forest, CV,<br>permutation<br>importance)                                        | True            |                                                                                            |
| <b>networkx</b>                                                                                      | library             | scripts/<br>predictors_audit.py<br>(co-occurrence<br>network density,<br>modularity)                                        | True            |                                                                                            |
| <b>statsmodels</b>                                                                                   | library             | scripts/<br>predictors_audit.py<br>(OLS with HC3<br>robust SEs)                                                             | True            |                                                                                            |
| <b>NCBI Taxonomy<br/>(E-utilities<br/>esearch/efetch)</b>                                            | database/web API    | scripts/<br>keystone_taxonomy.<br>py (genus -> phylum<br>lineage)                                                           | True            |                                                                                            |
| <b>BugSigDB (genus<br/>signature export)</b>                                                         | database            | data/ref/<br>bugsigdb_signatures<br>_genus_taxname.gm<br>t;<br>scripts/keystone_bu<br>gsigdb.py (literature<br>cross-check) | True            |                                                                                            |
| <b>GTDB (release<br/>v232 bac120<br/>taxonomy)</b>                                                   | database            | scripts/<br>keystone_gtdb.py                                                                                                | True            | independent<br>phylogenomic<br>taxonomy<br>replication of<br>keystone phylum<br>enrichment |
| <b>Madin et al. 2020<br/>bacterial/archaeal<br/>trait database<br/>(condensed_species<br/>_NCBI)</b> | database            | scripts/<br>keystone_traits.py                                                                                              | True            | keystone<br>physiological trait<br>association                                             |
| <b>KEGG REST<br/>(genome list + KO-<br/>gene links)</b>                                              | database            | scripts/<br>keystone_kegg.py                                                                                                | True            | genomic<br>anaerobic/aerobic<br>marker check of<br>keystones                               |
| <b>BV-BRC genome<br/>API<br/>(oxygen_requireme<br/>nt facets)</b>                                    | database            | scripts/<br>keystone_bvbrc.py                                                                                               | True            | third-source<br>anaerobe annotation<br>for keystone genera                                 |

| tool                                                                           | kind                    | where_used                         | counts_for_gate | note                                                                                                |
|--------------------------------------------------------------------------------|-------------------------|------------------------------------|-----------------|-----------------------------------------------------------------------------------------------------|
| <b>IJSEM phenotypic database</b><br>(Barberan et al. 2017, figshare 4272392)   | database                | scripts/<br>keystone_ijsem.py      | True            | oxygen preference per genus; pre-registered check 142c972; overlaps Madin sources (not independent) |
| <b>NCBI Datasets v2 API (genome dataset_report)</b>                            | database/service        | scripts/<br>keystone_genomesize.py | True            | reference-genome size and GC per genus; pre-registered 66d7f05                                      |
| <b>Open Tree of Life API v3 (TNRS match_names + induced synthetic subtree)</b> | web API                 | scripts/<br>keystone_otol.py       | True            | independent phylogeny for anaerobe-keystone PGLS; pre-registered ce6fd94                            |
| <b>ProTraits (Brbic et al. 2016; protraits.irb.hr integrated Pr&gt;=0.95)</b>  | database                | scripts/<br>keystone_protraits.py  | True            | pre-registered oxygen check 4f748be; coverage gate FAILED (51 genera)                               |
| <b>Disbiome (disbiome.ugent.be experiment API)</b>                             | database/web API        | scripts/<br>keystone_disbiome.py   | True            | pre-registered literature check 899f554; D1 PASS                                                    |
| <b>Europe PMC REST API (search hitCount)</b>                                   | web API                 | scripts/<br>keystone_europepmc.py  | True            | pre-registered study-bias control 0358299; E1 PASS                                                  |
| <b>gglasso (Schaipp et al. 2021 graphical lasso)</b>                           | library                 | scripts/<br>keystone_gglasso.py    | True            | alternative network inference for keystones; pre-registered 3129971; H1 FAILED                      |
| <b>Wikidata SPARQL endpoint (P2597 Gram staining)</b>                          | web API/knowledge graph | scripts/<br>keystone_wikidata.py   | True            | pre-registered 0ecd41 (+QID fix 1f20b4a); Madin Gram null replicated                                |
| <b>ENA portal API + ENA taxonomy REST (assembly counts)</b>                    | web API                 | scripts/<br>keystone_ena.py        | True            | pre-registered sequencing-effort control 9451f14; A1 PASS                                           |
| <b>GBIF API v1 (species/match + occurrence country facets)</b>                 | web API                 | scripts/<br>keystone_gbif.py       | True            | pre-registered generalism rival 9f6cc62; B1 PASS                                                    |
| <b>UniProt REST API (uniprotkb search counts, reference proteomes)</b>         | web API                 | scripts/<br>keystone_uniprot.py    | True            | pre-registered anaerobe-index replication c9ea858; U1 PASS                                          |
| <b>Crossref REST API (works total-results)</b>                                 | web API                 | scripts/<br>keystone_crossref.py   | True            | pre-registered literature control 6ff904c (replaces OpenAlex 429); C1 PASS C2 PASS                  |
| <b>python-igraph (betweenness centrality)</b>                                  | library                 | scripts/<br>keystone_igraph.py     | True            | pre-registered third network definition 0962035; I1 FAIL (p=0.34)                                   |
| <b>EBI Search REST (Ensembl Genomes genome domain)</b>                         | web API                 | scripts/<br>keystone_ensembl.py    | True            | pre-registered sequencing-effort replication be7175d; S1 PASS                                       |

| tool                                                                   | kind     | where_used                         | counts_for_gate | note                                                                                            |
|------------------------------------------------------------------------|----------|------------------------------------|-----------------|-------------------------------------------------------------------------------------------------|
| <b>XGBoost</b><br>(gradient-boosted trees)                             | library  | scripts/<br>keystone_xgboost.py    | True            | pre-registered out-of-sample GAI test 9cf67ae; X1 PASS by criterion but absolute CV R2 negative |
| <b>InterPro API</b><br>(protein counts per taxon and entry)            | web API  | scripts/<br>keystone_interpro.py   | True            | pre-registered anaerobe-index replication b327560; P1 PASS                                      |
| <b>HOMD expanded Human Oral Microbiome Database</b> (taxon table v4.2) | database | scripts/<br>keystone_homd.py       | True            | pre-registered oral-taxon rival 0aa4eb0 (+parser addendum 5658ee6); M1 PASS                     |
| <b>EBI Search REST</b> (RNAcentral rRNA sequences)                     | web API  | scripts/<br>keystone_rnacentral.py | True            | pre-registered rRNA-effort control 9814c7a; R1 PASS                                             |
| <b>bambi / PyMC</b> (Bayesian hierarchical logistic)                   | library  | scripts/<br>keystone_bambi.py      | True            | pre-registered Bayesian re-fit 45e3254; BY1 PASS                                                |
| <b>NCBI PubTator3 API</b> (search counts)                              | web API  | scripts/<br>keystone_pubtator.py   | True            | pre-registered oral-index replication 45e3254 (+addendum 3d16589); PT1 FAIL                     |
| <b>Biopython Bio.Phylo</b> (synthetic-tree topology)                   | library  | scripts/<br>keystone_biophylo.py   | True            | pre-registered a2e24b7; BP1 FAIL oral clustering p=0.53                                         |
| <b>LightGBM</b>                                                        | library  | scripts/<br>keystone_boosters.py   | True            | pre-registered fd7a9e2; B1 PASS absolute out-of-fold R2 0.133                                   |
| <b>CatBoost</b>                                                        | library  | scripts/<br>keystone_boosters.py   | True            | pre-registered fd7a9e2; B1 PASS absolute out-of-fold R2 0.119                                   |
| <b>SHAP TreeExplainer</b>                                              | library  | scripts/<br>keystone_shap.py       | True            | pre-registered c12838f; SH1 PASS GAI largest mean absolute attribution in full LightGBM         |
| <b>SymPy symbolic algebra</b>                                          | library  | scripts/<br>keystone_sympy.py      | True            | pre-registered c12838f; SY1 PASS RHS invariants for cNODE and gLV                               |

## Appendix E. Datasets ledger

**Table 33. 190 accession/resource rows, including 160 MGnify study tables of one dataset type, not 190 independent source-transfer cohorts (results/datasets\_ledger.csv).**

| accession    | source     | role    | used_in                                 |
|--------------|------------|---------|-----------------------------------------|
| MGYS00006862 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00005791 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006825 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006824 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006822 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006814 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006815 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006811 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006807 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006805 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006806 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006760 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00002338 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006797 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006551 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006795 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006794 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006789 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006787 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006786 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006759 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006780 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006779 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006769 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006765 | EBI MGnify | primary | 160-study interaction audit + keystones |
| MGYS00006755 | EBI MGnify | primary | 160-study interaction audit             |

| accession    | source     | role    | used_in                                    |
|--------------|------------|---------|--------------------------------------------|
|              |            |         | + keystones                                |
| MGYS00005333 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005230 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00002675 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006069 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006006 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00002061 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00000633 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00002238 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006562 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005154 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006649 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006641 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006642 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006364 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006640 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005253 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006320 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006276 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006263 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006017 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005592 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005569 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005528 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005506 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005189 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005223 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005197 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005143 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00002159 | EBI MGnify | primary | 160-study interaction audit                |

| accession    | source     | role    | used_in                                    |
|--------------|------------|---------|--------------------------------------------|
|              |            |         | + keystones                                |
| MGYS00003719 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00002395 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00002394 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00002272 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00002269 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00002229 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006648 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005232 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005156 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006246 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005384 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005172 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005533 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005212 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005238 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005101 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005102 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005135 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005098 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005037 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005090 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00002184 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00002072 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006855 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006799 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00002673 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006747 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00001223 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00001224 | EBI MGnify | primary | 160-study interaction audit                |

| accession    | source     | role    | used_in                                    |
|--------------|------------|---------|--------------------------------------------|
|              |            |         | + keystones                                |
| MGYS00005267 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00000426 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005228 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006537 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006536 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006538 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005628 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006450 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006456 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006446 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006433 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006430 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006427 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006424 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006652 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00001412 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00001368 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00003172 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006578 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00000957 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00001704 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00000867 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006031 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006021 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006018 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00001349 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005260 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005227 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00000983 | EBI MGnify | primary | 160-study interaction audit                |

| accession    | source     | role    | used_in                                    |
|--------------|------------|---------|--------------------------------------------|
|              |            |         | + keystones                                |
| MGYS00001347 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005740 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005726 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005169 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005151 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006750 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006533 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006532 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005766 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006062 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006042 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005621 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005611 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005610 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005609 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005600 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005593 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005576 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005187 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005316 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005317 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00005309 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00004995 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00003706 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00003681 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006764 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006763 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006762 | EBI MGnify | primary | 160-study interaction audit<br>+ keystones |
| MGYS00006761 | EBI MGnify | primary | 160-study interaction audit                |

| accession                          | source                                                       | role    | used_in                                    |
|------------------------------------|--------------------------------------------------------------|---------|--------------------------------------------|
|                                    |                                                              |         | + keystones                                |
| <b>MGYS00006758</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006041</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006117</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006086</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005503</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00003166</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00004672</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00002905</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00002608</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00002413</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00002065</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006798</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00006019</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005862</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005215</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00005163</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00003800</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>MGYS00003705</b>                | EBI MGnify                                                   | primary | 160-study interaction audit<br>+ keystones |
| <b>cNODE:Ocean</b>                 | github.com/yixueyang/<br>cNODE (6 tables, one<br>repository) | primary | cNODE benchmark                            |
| <b>cNODE:Drosophila_Gut</b>        | github.com/yixueyang/<br>cNODE (6 tables, one<br>repository) | primary | cNODE benchmark                            |
| <b>cNODE:Soil_Vitro</b>            | github.com/yixueyang/<br>cNODE (6 tables, one<br>repository) | primary | cNODE benchmark                            |
| <b>cNODE:Soil_Vivo</b>             | github.com/yixueyang/<br>cNODE (6 tables, one<br>repository) | primary | cNODE benchmark                            |
| <b>cNODE:Human_Oral</b>            | github.com/yixueyang/<br>cNODE (6 tables, one<br>repository) | primary | cNODE benchmark                            |
| <b>cNODE:Human_Gut</b>             | github.com/yixueyang/<br>cNODE (6 tables, one<br>repository) | primary | cNODE benchmark                            |
| <b>MDSINE2:gibson/<br/>healthy</b> | github.com/gerberlab/<br>MDSINE2_Paper                       | primary | forecasting                                |
| <b>MDSINE2:gibson/uc</b>           | github.com/gerberlab/                                        | primary | forecasting                                |

| accession                                                                                                                              | source                             | role      | used_in                                               |
|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-----------|-------------------------------------------------------|
|                                                                                                                                        | MDSINE2_Paper                      |           |                                                       |
| 41564_2025_2112_MOE<br>SM5_ESM                                                                                                         | Nature Microbiology<br>Source Data | primary   | head-to-head                                          |
| NCBI Taxonomy                                                                                                                          | NCBI E-utilities                   | reference | phylum mapping                                        |
| BugSigDB genus<br>signatures                                                                                                           | BugSigDB                           | reference | literature cross-check                                |
| GTDB r232 bac120<br>taxonomy                                                                                                           | GTDB                               | reference | taxonomy replication                                  |
| Madin 2020<br>condensed_species_NCB<br>I                                                                                               | github bacteria-archaea-<br>traits | reference | traits                                                |
| KEGG genome list + 10<br>KO links                                                                                                      | KEGG REST                          | reference | genomic anaerobe index                                |
| BV-BRC<br>oxygen_requirement<br>facets                                                                                                 | BV-BRC API                         | reference | third anaerobe source                                 |
| figshare:4272392                                                                                                                       | figshare (IJSEM<br>phenotypic db)  | reference | anaerobe-keystone IJSEM<br>check                      |
| OToL synth induced<br>subtree (161 genera;<br>data/otol/induced.tre)                                                                   | Open Tree of Life                  | reference | anaerobe-keystone PGLS<br>on independent phylogeny    |
| ProTraits_binaryIntegra<br>tedPr0.95.txt                                                                                               | ProTraits                          | reference | anaerobe-keystone<br>ProTraits check (gate<br>failed) |
| Disbiome experiment<br>dump (10866 records)                                                                                            | Disbiome                           | reference | keystone literature check                             |
| Europe PMC<br>TITLE_ABS hit counts<br>for 248 genera<br>(results/keystone_europe<br>pmc_counts.csv)                                    | Europe PMC                         | reference | study-bias control of<br>keystone literature link     |
| Wikidata genus Gram-<br>staining table (3117<br>genera;<br>results/keystone_wikidat<br>a_gram.csv)                                     | Wikidata                           | reference | Gram-stain replication                                |
| ENA public assembly<br>counts for 200 genera<br>(results/keystone_ena_co<br>unts.csv)                                                  | ENA                                | reference | sequencing-effort control                             |
| GBIF occurrence and<br>country counts for 202<br>genera<br>(results/keystone_gbif_co<br>unts.csv)                                      | GBIF                               | reference | generalism rival                                      |
| UniProt reference-<br>proteome EC 7.1.1.9 /<br>EC 1.2.7.1 / recA counts<br>for 200 genera<br>(results/keystone_unipro<br>t_counts.csv) | UniProt                            | reference | annotation-pipeline<br>replication of GAI             |
| Crossref works counts<br>for 248 genera<br>(results/keystone_crossre<br>f_counts.csv)                                                  | Crossref                           | reference | independent literature-<br>volume covariate           |
| Ensembl Genomes<br>genome hit counts for<br>203 genera<br>(results/keystone_ensem                                                      | EBI Search / Ensembl<br>Bacteria   | reference | sequencing-effort<br>covariate (curated archive)      |

| accession                                                                                                               | source         | role      | used_in                                     |
|-------------------------------------------------------------------------------------------------------------------------|----------------|-----------|---------------------------------------------|
| bl_counts.csv)                                                                                                          |                |           |                                             |
| InterPro<br>PFOR/COX1/RecA<br>protein counts for 200<br>genera<br>(results/keystone_interpr<br>o_counts.csv)            | InterPro       | reference | signature-based replication<br>of GAI       |
| HOMD taxon table v4.2<br>(data/ref/HOMD_taxon_<br>table_v4.2.csv; 121 oral<br>genera)                                   | HOMD           | reference | oral-origin covariate                       |
| RNAcentral rRNA<br>sequence counts for 203<br>genera<br>(results/keystone_rnacen<br>tral_counts.csv)                    | RNAcentral     | reference | marker-gene sequencing-<br>effort covariate |
| PubTator3 genus and<br>genus+oral/dental article<br>counts for 203 genera<br>(results/keystone_pubtat<br>or_counts.csv) | NCBI PubTator3 | reference | literature oral index                       |

## Appendix F. MDSINE2 per-(subject, taxon) RMSE

**Table 34. Per-(subject, taxon) RMSE of log10 absolute abundance on detected timepoints, all fitted methods (546 pairs; pivoted from results/mdsine2\_pertaxon\_errors\_141.csv).**

| subject | taxon | glv_ridge10 | glv_ridge100 | glv_ridge1000 | mean_traj_LO<br>Dfloor |
|---------|-------|-------------|--------------|---------------|------------------------|
| 2       | 0     | 1.373       | 1.207        | 1.145         | 0.373                  |
| 2       | 1     | 1.852       | 2.319        | 2.474         | 0.821                  |
| 2       | 2     | 2.405       | 2.549        | 3.069         | 0.868                  |
| 2       | 3     | 2.529       | 1.427        | 1.610         | 0.694                  |
| 2       | 4     | 2.985       | 1.551        | 1.320         | 0.504                  |
| 2       | 5     | 2.610       | 1.792        | 1.963         | 0.852                  |
| 2       | 6     | 3.185       | 1.402        | 1.665         | 0.626                  |
| 2       | 7     | 3.147       | 2.209        | 2.041         | 0.639                  |
| 2       | 8     | 2.700       | 1.488        | 2.020         | 0.846                  |
| 2       | 9     | 3.815       | 2.484        | 1.573         | 0.486                  |
| 2       | 10    | 2.620       | 1.533        | 1.709         | 0.614                  |
| 2       | 11    | 3.439       | 1.457        | 1.313         | 0.389                  |
| 2       | 12    | 2.748       | 1.427        | 1.486         | 0.640                  |
| 2       | 13    | 1.812       | 1.303        | 1.283         | 0.630                  |
| 2       | 14    | 2.568       | 2.319        | 2.685         | 1.299                  |
| 2       | 15    | 3.546       | 2.586        | 1.129         | 0.345                  |
| 2       | 16    | 3.525       | 2.779        | 3.179         | 0.490                  |
| 2       | 17    | 3.324       | 2.930        | 3.052         | 0.510                  |
| 2       | 18    | 2.790       | 3.320        | 3.070         | 1.081                  |
| 2       | 19    | 3.095       | 2.114        | 1.003         | 1.036                  |
| 2       | 20    | 2.962       | 2.482        | 1.505         | 0.695                  |
| 2       | 21    | 1.958       | 1.800        | 1.255         | 0.737                  |
| 2       | 22    | 2.555       | 1.509        | 1.289         | 0.488                  |
| 2       | 23    | 2.028       | 2.099        | 2.152         | 1.124                  |
| 2       | 24    | 2.644       | 1.956        | 2.277         | 0.378                  |

| subject | taxon | glv_ridge10 | glv_ridge100 | glv_ridge1000 | mean_traj_LO<br>Dfloor |
|---------|-------|-------------|--------------|---------------|------------------------|
| 2       | 25    | 3.358       | 2.400        | 2.399         | 1.398                  |
| 2       | 26    | 3.068       | 3.366        | 3.217         | 1.278                  |
| 2       | 27    | 3.241       | 3.159        | 2.139         | 1.112                  |
| 2       | 28    | 2.739       | 1.693        | 2.447         | 0.841                  |
| 2       | 29    | 2.989       | 1.942        | 2.905         | 0.440                  |
| 2       | 30    | 3.324       | 2.076        | 3.663         | 2.145                  |
| 2       | 31    | 2.558       | 1.199        | 1.416         | 0.477                  |
| 2       | 32    | 2.603       | 1.876        | 1.799         | 0.587                  |
| 2       | 33    | 2.817       | 1.061        | 0.883         | 0.693                  |
| 2       | 34    | 2.760       | 2.124        | 1.869         | 1.328                  |
| 2       | 35    | 2.164       | 1.528        | 1.588         | 0.850                  |
| 2       | 36    | 2.733       | 2.203        | 2.529         | 1.541                  |
| 2       | 37    | 2.442       | 1.913        | 2.872         | 1.314                  |
| 2       | 38    | 3.374       | 2.256        | 3.229         | 1.696                  |
| 2       | 39    | 2.405       | 1.770        | 1.071         | 0.860                  |
| 2       | 40    | 1.491       | 1.521        | 1.536         | 1.085                  |
| 2       | 41    | 2.895       | 2.960        | 1.904         | 1.273                  |
| 2       | 42    | 2.922       | 2.319        | 3.160         | 1.740                  |
| 2       | 43    | 1.918       | 1.120        | 1.682         | 1.141                  |
| 2       | 44    | 2.562       | 1.329        | 1.181         | 0.515                  |
| 2       | 45    | 2.721       | 2.380        | 1.654         | 0.713                  |
| 2       | 46    | 3.212       | 3.223        | 3.238         | 1.805                  |
| 2       | 47    | 3.361       | 2.894        | 2.638         | 1.169                  |
| 2       | 48    | 3.267       | 2.792        | 2.597         | 1.286                  |
| 2       | 49    | 2.326       | 1.028        | 0.976         | 0.750                  |
| 2       | 50    | 2.697       | 2.791        | 3.212         | 0.664                  |
| 2       | 51    | 3.151       | 2.891        | 2.613         | 1.816                  |
| 2       | 52    | 1.635       | 1.640        | 1.636         | 0.919                  |
| 2       | 53    | 2.406       | 1.259        | 1.072         | 0.495                  |
| 2       | 54    | 1.442       | 1.442        | 1.442         | 1.179                  |
| 2       | 55    | 2.097       | 1.806        | 2.134         | 0.705                  |
| 2       | 56    | 3.263       | 2.887        | 2.451         | 1.362                  |
| 2       | 57    | 2.718       | 2.047        | 1.347         | 0.647                  |
| 2       | 58    | 2.459       | 1.867        | 0.864         | 0.683                  |
| 2       | 59    | 2.337       | 1.408        | 2.522         | 0.507                  |
| 2       | 60    | 1.770       | 2.280        | 2.144         | 1.250                  |
| 2       | 61    | 2.045       | 1.944        | 1.364         | 0.783                  |
| 2       | 62    | 2.254       | 2.568        | 2.899         | 0.291                  |
| 2       | 63    | 1.775       | 0.828        | 1.019         | 0.888                  |
| 2       | 64    | 2.255       | 2.699        | 2.679         | 0.892                  |
| 2       | 65    | 2.540       | 1.823        | 1.142         | 0.803                  |
| 2       | 66    | 3.141       | 2.317        | 2.181         | 1.412                  |
| 2       | 67    | 2.155       | 1.436        | 1.532         | 0.675                  |
| 2       | 68    | 2.967       | 2.774        | 3.055         | 1.161                  |
| 2       | 69    | 2.560       | 1.592        | 1.439         | 1.038                  |
| 2       | 70    | 2.949       | 2.619        | 2.800         | 1.176                  |
| 2       | 71    | 2.471       | 1.693        | 0.931         | 0.529                  |
| 2       | 72    | 2.128       | 2.419        | 2.737         | 0.304                  |
| 2       | 73    | 2.451       | 1.885        | 1.885         | 1.029                  |
| 2       | 74    | 2.404       | 1.715        | 0.924         | 0.638                  |
| 2       | 75    | 2.395       | 1.951        | 1.532         | 0.653                  |
| 2       | 76    | 1.407       | 1.318        | 1.405         | 0.864                  |
| 2       | 77    | 2.245       | 1.804        | 1.218         | 0.569                  |
| 2       | 78    | 1.926       | 1.706        | 1.510         | 0.885                  |
| 2       | 79    | 2.062       | 2.355        | 2.660         | 0.700                  |

| subject | taxon | glv_ridge10 | glv_ridge100 | glv_ridge1000 | mean_traj_LO<br>Dfloor |
|---------|-------|-------------|--------------|---------------|------------------------|
| 2       | 80    | 2.169       | 1.772        | 0.892         | 0.500                  |
| 2       | 81    | 3.138       | 1.308        | 0.952         | 0.679                  |
| 2       | 82    | 1.994       | 2.705        | 1.391         | 0.996                  |
| 2       | 83    | 1.750       | 2.183        | 2.727         | 0.993                  |
| 2       | 84    | 3.134       | 2.932        | 3.017         | 1.278                  |
| 2       | 85    | 2.129       | 1.420        | 1.498         | 0.458                  |
| 2       | 86    | 3.135       | 0.971        | 1.108         | 1.018                  |
| 2       | 87    | 2.221       | 2.299        | 3.350         | 1.245                  |
| 2       | 88    | 3.171       | 3.245        | 3.054         | 0.511                  |
| 2       | 89    | 2.010       | 1.272        | 2.161         | 2.003                  |
| 2       | 90    | 2.289       | 2.358        | 2.404         | 1.049                  |
| 2       | 91    | 1.503       | 0.863        | 0.859         | 1.321                  |
| 2       | 92    | 1.811       | 2.070        | 2.337         | 0.471                  |
| 2       | 93    | 1.121       | 1.624        | 2.441         | 0.760                  |
| 2       | 94    | 1.755       | 1.491        | 0.892         | 0.557                  |
| 2       | 95    | 1.801       | 1.944        | 2.211         | 0.496                  |
| 2       | 96    | 2.419       | 1.087        | 1.584         | 1.212                  |
| 2       | 97    | 1.480       | 0.617        | 0.773         | 0.380                  |
| 2       | 98    | 1.481       | 1.485        | 0.625         | 0.137                  |
| 2       | 99    | 1.862       | 0.682        | 1.219         | 0.589                  |
| 2       | 100   | 1.960       | 2.182        | 2.399         | 0.517                  |
| 2       | 102   | 2.120       | 1.905        | 1.282         | 0.813                  |
| 2       | 103   | 2.679       | 1.909        | 0.978         | 0.871                  |
| 2       | 104   | 3.298       | 3.375        | 3.375         | 1.383                  |
| 2       | 106   | 2.785       | 3.002        | 2.573         | 0.923                  |
| 2       | 107   | 2.515       | 2.510        | 2.444         | 0.756                  |
| 2       | 108   | 2.438       | 1.642        | 0.847         | 0.880                  |
| 2       | 109   | 2.155       | 1.247        | 1.807         | 1.221                  |
| 2       | 110   | 1.662       | 1.617        | 2.116         | 0.776                  |
| 2       | 113   | 1.147       | 0.941        | 1.219         | 0.558                  |
| 2       | 114   | 2.858       | 3.247        | 2.295         | 0.620                  |
| 2       | 115   | 2.530       | 2.206        | 1.993         | 1.214                  |
| 2       | 116   | 2.876       | 2.524        | 2.876         | 1.051                  |
| 2       | 117   | 2.379       | 2.359        | 2.637         | 1.074                  |
| 2       | 118   | 1.982       | 2.274        | 2.274         | 1.415                  |
| 2       | 119   | 1.741       | 1.992        | 2.283         | 0.393                  |
| 2       | 120   | 2.759       | 3.044        | 3.044         | 2.518                  |
| 2       | 122   | 2.527       | 1.499        | 2.367         | 1.749                  |
| 2       | 123   | 2.300       | 0.996        | 1.530         | 0.738                  |
| 2       | 124   | 0.194       | 0.191        | 0.189         | 0.144                  |
| 2       | 125   | 2.278       | 2.051        | 2.197         | 1.460                  |
| 2       | 126   | 0.142       | 0.142        | 0.143         | 0.088                  |
| 2       | 127   | 1.811       | 1.530        | 1.867         | 1.574                  |
| 2       | 128   | 2.312       | 1.957        | 1.355         | 1.093                  |
| 2       | 129   | 2.277       | 2.272        | 1.782         | 1.147                  |
| 2       | 130   | 2.258       | 2.313        | 2.348         | 0.983                  |
| 2       | 131   | 2.223       | 1.924        | 1.213         | 0.468                  |
| 2       | 132   | 3.207       | 0.984        | 0.702         | 0.563                  |
| 2       | 133   | 2.235       | 0.984        | 1.264         | 0.527                  |
| 2       | 134   | 0.767       | 0.766        | 0.765         | 0.815                  |
| 2       | 135   | 2.120       | 1.642        | 1.347         | 0.635                  |
| 2       | 136   | 2.065       | 1.948        | 1.781         | 1.047                  |
| 2       | 137   | 0.465       | 0.465        | 0.465         | 1.693                  |
| 2       | 138   | 0.139       | 0.137        | 0.135         | 0.138                  |
| 2       | 139   | 2.156       | 0.790        | 1.122         | 1.228                  |

| subject | taxon | glv_ridge10 | glv_ridge100 | glv_ridge1000 | mean_traj_LO<br>Dfloor |
|---------|-------|-------------|--------------|---------------|------------------------|
| 2       | 140   | 2.916       | 1.696        | 2.437         | 1.990                  |
| 3       | 0     | 2.692       | 0.879        | 1.198         | 0.327                  |
| 3       | 1     | 3.912       | 1.600        | 1.960         | 1.029                  |
| 3       | 2     | 3.988       | 2.186        | 2.417         | 0.818                  |
| 3       | 3     | 3.679       | 1.253        | 1.512         | 0.649                  |
| 3       | 4     | 3.369       | 1.302        | 1.272         | 0.451                  |
| 3       | 5     | 4.037       | 1.074        | 1.515         | 0.461                  |
| 3       | 6     | 3.757       | 0.752        | 0.943         | 0.476                  |
| 3       | 7     | 3.570       | 1.353        | 1.952         | 1.055                  |
| 3       | 8     | 3.136       | 1.754        | 1.861         | 0.747                  |
| 3       | 9     | 3.931       | 0.988        | 1.197         | 0.368                  |
| 3       | 10    | 3.025       | 1.509        | 1.526         | 0.649                  |
| 3       | 11    | 3.156       | 0.987        | 1.365         | 0.373                  |
| 3       | 12    | 3.729       | 1.341        | 1.438         | 0.725                  |
| 3       | 13    | 3.378       | 2.279        | 1.436         | 0.730                  |
| 3       | 14    | 3.821       | 2.580        | 2.534         | 0.773                  |
| 3       | 15    | 2.804       | 1.028        | 1.116         | 0.383                  |
| 3       | 16    | 3.040       | 2.313        | 1.350         | 0.618                  |
| 3       | 17    | 3.489       | 2.964        | 1.632         | 0.600                  |
| 3       | 18    | 3.620       | 3.526        | 2.898         | 0.524                  |
| 3       | 19    | 4.771       | 2.353        | 1.188         | 0.914                  |
| 3       | 20    | 3.403       | 2.577        | 2.433         | 1.197                  |
| 3       | 21    | 2.948       | 1.788        | 1.820         | 0.455                  |
| 3       | 22    | 3.769       | 1.015        | 1.528         | 1.059                  |
| 3       | 23    | 2.485       | 1.344        | 1.370         | 0.980                  |
| 3       | 24    | 2.960       | 2.232        | 1.901         | 0.673                  |
| 3       | 25    | 3.116       | 2.602        | 1.995         | 0.969                  |
| 3       | 26    | 4.228       | 3.502        | 3.377         | 1.256                  |
| 3       | 27    | 3.416       | 2.585        | 2.307         | 1.199                  |
| 3       | 28    | 3.524       | 2.890        | 3.046         | 0.923                  |
| 3       | 29    | 3.266       | 1.377        | 2.271         | 0.443                  |
| 3       | 30    | 3.064       | 2.998        | 3.085         | 1.927                  |
| 3       | 31    | 2.482       | 1.104        | 0.956         | 0.416                  |
| 3       | 32    | 2.978       | 1.782        | 1.723         | 0.602                  |
| 3       | 33    | 3.480       | 1.761        | 2.133         | 1.334                  |
| 3       | 34    | 2.914       | 1.856        | 2.031         | 0.961                  |
| 3       | 35    | 2.919       | 1.601        | 1.330         | 0.449                  |
| 3       | 36    | 3.238       | 3.033        | 2.778         | 1.628                  |
| 3       | 37    | 4.988       | 1.986        | 2.224         | 1.293                  |
| 3       | 38    | 4.578       | 1.532        | 2.686         | 1.385                  |
| 3       | 39    | 3.369       | 2.302        | 1.345         | 0.878                  |
| 3       | 40    | 2.130       | 1.333        | 1.373         | 0.982                  |
| 3       | 41    | 2.602       | 1.543        | 1.713         | 0.906                  |
| 3       | 42    | 2.704       | 1.592        | 2.602         | 1.580                  |
| 3       | 43    | 4.576       | 1.711        | 1.454         | 0.970                  |
| 3       | 44    | 2.551       | 1.982        | 1.411         | 0.958                  |
| 3       | 45    | 2.415       | 1.912        | 1.528         | 0.978                  |
| 3       | 46    | 2.727       | 2.141        | 2.892         | 1.497                  |
| 3       | 47    | 4.370       | 2.832        | 2.897         | 0.926                  |
| 3       | 48    | 4.282       | 2.589        | 2.591         | 0.959                  |
| 3       | 49    | 3.236       | 1.260        | 0.725         | 0.724                  |
| 3       | 50    | 2.517       | 2.582        | 2.950         | 0.804                  |
| 3       | 51    | 3.398       | 3.333        | 2.893         | 1.816                  |
| 3       | 52    | 1.105       | 1.093        | 1.086         | 0.996                  |
| 3       | 53    | 2.588       | 2.729        | 2.537         | 0.746                  |

| subject | taxon | glv_ridge10 | glv_ridge100 | glv_ridge1000 | mean_traj_LO<br>Dfloor |
|---------|-------|-------------|--------------|---------------|------------------------|
| 3       | 54    | 0.823       | 0.828        | 0.831         | 1.130                  |
| 3       | 55    | 2.067       | 1.669        | 2.303         | 0.626                  |
| 3       | 56    | 3.342       | 3.206        | 2.579         | 1.216                  |
| 3       | 57    | 2.982       | 2.236        | 1.637         | 0.512                  |
| 3       | 58    | 4.157       | 2.098        | 1.363         | 0.883                  |
| 3       | 59    | 4.527       | 1.687        | 2.336         | 0.680                  |
| 3       | 60    | 2.165       | 1.502        | 2.165         | 0.533                  |
| 3       | 61    | 2.834       | 2.335        | 1.639         | 0.877                  |
| 3       | 62    | 2.340       | 2.620        | 2.921         | 0.368                  |
| 3       | 63    | 1.698       | 1.404        | 1.405         | 0.801                  |
| 3       | 64    | 2.770       | 0.861        | 1.198         | 1.144                  |
| 3       | 65    | 2.707       | 1.328        | 1.113         | 0.865                  |
| 3       | 66    | 1.333       | 0.732        | 1.448         | 0.763                  |
| 3       | 67    | 2.443       | 1.402        | 1.893         | 0.725                  |
| 3       | 68    | 2.707       | 1.586        | 2.292         | 1.068                  |
| 3       | 69    | 3.904       | 1.662        | 1.560         | 0.820                  |
| 3       | 70    | 2.562       | 2.102        | 2.327         | 1.101                  |
| 3       | 71    | 2.425       | 1.629        | 1.133         | 0.807                  |
| 3       | 72    | 2.324       | 2.606        | 2.902         | 0.292                  |
| 3       | 73    | 2.929       | 2.929        | 2.929         | 0.871                  |
| 3       | 74    | 2.543       | 1.519        | 1.008         | 0.732                  |
| 3       | 75    | 2.517       | 0.665        | 0.744         | 1.152                  |
| 3       | 76    | 4.123       | 2.197        | 1.863         | 1.774                  |
| 3       | 77    | 2.237       | 1.604        | 1.650         | 0.699                  |
| 3       | 78    | 1.463       | 1.446        | 1.564         | 0.807                  |
| 3       | 79    | 2.115       | 2.377        | 2.680         | 0.383                  |
| 3       | 80    | 2.482       | 1.809        | 0.935         | 0.824                  |
| 3       | 81    | 2.399       | 2.110        | 1.862         | 0.891                  |
| 3       | 82    | 1.945       | 1.678        | 1.490         | 0.729                  |
| 3       | 83    | 1.852       | 1.779        | 1.944         | 0.661                  |
| 3       | 84    | 2.322       | 2.985        | 2.863         | 1.062                  |
| 3       | 85    | 4.141       | 3.467        | 3.467         | 3.368                  |
| 3       | 86    | 2.073       | 1.192        | 1.104         | 1.262                  |
| 3       | 87    | 4.539       | 2.530        | 2.977         | 1.004                  |
| 3       | 88    | 4.544       | 1.710        | 1.923         | 0.593                  |
| 3       | 89    | 1.472       | 1.299        | 1.385         | 0.884                  |
| 3       | 90    | 1.695       | 1.755        | 1.810         | 1.210                  |
| 3       | 91    | 2.118       | 2.105        | 2.097         | 1.132                  |
| 3       | 92    | 2.110       | 2.390        | 2.662         | 0.754                  |
| 3       | 93    | 2.501       | 2.310        | 2.483         | 0.854                  |
| 3       | 94    | 0.513       | 0.527        | 0.639         | 0.590                  |
| 3       | 95    | 1.003       | 1.233        | 1.563         | 0.493                  |
| 3       | 96    | 2.701       | 2.297        | 2.108         | 0.937                  |
| 3       | 97    | 2.076       | 0.386        | 0.767         | 1.581                  |
| 3       | 99    | 3.377       | 3.228        | 3.377         | 3.371                  |
| 3       | 100   | 2.031       | 2.273        | 2.502         | 0.461                  |
| 3       | 101   | 1.637       | 1.638        | 1.638         | 0.909                  |
| 3       | 102   | 2.173       | 1.884        | 1.416         | 1.008                  |
| 3       | 103   | 2.911       | 2.296        | 1.143         | 0.957                  |
| 3       | 104   | 3.424       | 3.424        | 3.424         | 0.587                  |
| 3       | 105   | 0.808       | 1.347        | 1.627         | 0.768                  |
| 3       | 106   | 3.012       | 2.700        | 2.350         | 1.149                  |
| 3       | 107   | 2.612       | 2.557        | 2.438         | 0.555                  |
| 3       | 108   | 4.530       | 2.091        | 1.498         | 1.437                  |
| 3       | 109   | 3.023       | 2.139        | 2.141         | 1.446                  |

| subject | taxon | glv_ridge10 | glv_ridge100 | glv_ridge1000 | mean_traj_LO<br>Dfloor |
|---------|-------|-------------|--------------|---------------|------------------------|
| 3       | 110   | 1.933       | 1.690        | 2.078         | 0.672                  |
| 3       | 111   | 2.772       | 2.763        | 2.793         | 2.808                  |
| 3       | 113   | 2.262       | 1.009        | 1.402         | 0.524                  |
| 3       | 114   | 3.469       | 2.453        | 1.910         | 0.677                  |
| 3       | 115   | 2.469       | 0.975        | 1.609         | 0.789                  |
| 3       | 116   | 4.439       | 2.927        | 3.251         | 1.409                  |
| 3       | 117   | 2.410       | 0.532        | 0.744         | 0.877                  |
| 3       | 118   | 2.699       | 2.401        | 2.401         | 0.837                  |
| 3       | 119   | 1.872       | 2.151        | 2.438         | 0.428                  |
| 3       | 120   | 1.330       | 1.330        | 1.330         | 0.563                  |
| 3       | 122   | 2.515       | 2.388        | 2.339         | 2.004                  |
| 3       | 123   | 2.420       | 1.165        | 1.899         | 1.322                  |
| 3       | 124   | 0.161       | 0.165        | 0.168         | 0.202                  |
| 3       | 125   | 1.835       | 1.835        | 1.835         | 0.975                  |
| 3       | 126   | 0.290       | 0.288        | 0.279         | 0.373                  |
| 3       | 127   | 1.466       | 1.444        | 1.596         | 0.863                  |
| 3       | 128   | 2.436       | 2.089        | 1.617         | 1.109                  |
| 3       | 129   | 2.204       | 2.108        | 2.062         | 1.106                  |
| 3       | 130   | 1.733       | 1.800        | 1.869         | 0.935                  |
| 3       | 131   | 2.025       | 1.816        | 1.273         | 0.769                  |
| 3       | 132   | 2.460       | 2.434        | 2.130         | 0.748                  |
| 3       | 133   | 4.872       | 1.143        | 1.513         | 0.548                  |
| 3       | 134   | 0.110       | 0.110        | 0.110         | 0.260                  |
| 3       | 135   | 2.271       | 1.803        | 0.845         | 0.828                  |
| 3       | 136   | 4.734       | 2.089        | 2.100         | 0.709                  |
| 3       | 137   | 1.478       | 1.479        | 1.479         | 0.956                  |
| 3       | 138   | 0.175       | 0.172        | 0.168         | 0.243                  |
| 3       | 139   | 2.228       | 2.070        | 2.573         | 2.072                  |
| 3       | 140   | 2.824       | 1.862        | 2.712         | 2.016                  |
| 4       | 0     | 3.901       | 0.900        | 1.167         | 0.410                  |
| 4       | 1     | 4.113       | 1.964        | 2.186         | 0.802                  |
| 4       | 2     | 3.921       | 2.078        | 1.989         | 1.341                  |
| 4       | 3     | 3.946       | 1.292        | 1.547         | 0.610                  |
| 4       | 4     | 3.291       | 2.909        | 1.906         | 0.721                  |
| 4       | 5     | 3.946       | 0.957        | 1.053         | 0.499                  |
| 4       | 6     | 3.649       | 1.116        | 1.533         | 0.780                  |
| 4       | 7     | 3.362       | 1.556        | 1.538         | 0.724                  |
| 4       | 8     | 3.461       | 2.548        | 3.426         | 1.097                  |
| 4       | 9     | 3.901       | 1.268        | 1.302         | 0.491                  |
| 4       | 10    | 3.906       | 1.398        | 1.644         | 0.830                  |
| 4       | 11    | 2.129       | 0.873        | 1.113         | 0.496                  |
| 4       | 12    | 3.660       | 1.792        | 1.838         | 0.955                  |
| 4       | 13    | 2.951       | 3.020        | 2.141         | 0.688                  |
| 4       | 14    | 3.782       | 1.180        | 1.593         | 0.756                  |
| 4       | 15    | 3.504       | 1.923        | 1.189         | 0.410                  |
| 4       | 16    | 3.498       | 3.271        | 2.520         | 0.707                  |
| 4       | 17    | 3.464       | 3.237        | 2.550         | 0.622                  |
| 4       | 18    | 3.341       | 3.231        | 2.679         | 0.848                  |
| 4       | 19    | 3.776       | 2.170        | 1.575         | 0.960                  |
| 4       | 20    | 3.291       | 2.212        | 1.425         | 0.561                  |
| 4       | 21    | 2.930       | 1.170        | 1.311         | 0.711                  |
| 4       | 22    | 3.411       | 2.472        | 2.756         | 1.168                  |
| 4       | 23    | 1.447       | 1.151        | 1.048         | 1.365                  |
| 4       | 24    | 2.957       | 1.693        | 2.060         | 0.695                  |
| 4       | 25    | 3.692       | 2.235        | 1.748         | 1.439                  |

| subject | taxon | glv_ridge10 | glv_ridge100 | glv_ridge1000 | mean_traj_LO<br>Dfloor |
|---------|-------|-------------|--------------|---------------|------------------------|
| 4       | 26    | 3.486       | 1.077        | 1.791         | 0.795                  |
| 4       | 27    | 3.087       | 1.523        | 1.691         | 1.539                  |
| 4       | 28    | 3.018       | 2.415        | 2.307         | 1.026                  |
| 4       | 29    | 3.158       | 2.577        | 2.237         | 0.474                  |
| 4       | 30    | 3.904       | 2.795        | 3.673         | 2.714                  |
| 4       | 31    | 2.750       | 1.182        | 0.921         | 0.380                  |
| 4       | 32    | 2.841       | 1.370        | 1.472         | 0.769                  |
| 4       | 33    | 4.332       | 2.932        | 2.397         | 0.752                  |
| 4       | 34    | 3.113       | 1.644        | 1.712         | 0.941                  |
| 4       | 35    | 3.657       | 2.778        | 2.672         | 1.554                  |
| 4       | 36    | 2.558       | 2.343        | 1.986         | 1.037                  |
| 4       | 37    | 4.248       | 2.960        | 1.816         | 1.282                  |
| 4       | 38    | 4.115       | 2.280        | 3.431         | 2.129                  |
| 4       | 39    | 2.976       | 1.753        | 1.448         | 1.154                  |
| 4       | 40    | 2.082       | 2.134        | 2.188         | 1.308                  |
| 4       | 41    | 2.730       | 1.118        | 1.590         | 0.689                  |
| 4       | 42    | 3.204       | 3.345        | 2.650         | 2.248                  |
| 4       | 43    | 4.413       | 2.146        | 1.973         | 1.178                  |
| 4       | 44    | 2.614       | 1.822        | 0.871         | 0.645                  |
| 4       | 45    | 2.654       | 2.097        | 1.449         | 0.844                  |
| 4       | 46    | 3.480       | 3.501        | 3.510         | 2.174                  |
| 4       | 47    | 3.005       | 2.250        | 2.127         | 0.802                  |
| 4       | 48    | 3.079       | 2.019        | 1.606         | 0.787                  |
| 4       | 49    | 3.828       | 3.070        | 2.846         | 2.488                  |
| 4       | 50    | 3.310       | 2.953        | 3.176         | 1.119                  |
| 4       | 51    | 3.280       | 2.103        | 2.110         | 1.248                  |
| 4       | 52    | 1.429       | 1.449        | 1.478         | 1.456                  |
| 4       | 53    | 2.707       | 2.269        | 0.970         | 0.673                  |
| 4       | 54    | 0.548       | 0.560        | 0.559         | 0.563                  |
| 4       | 55    | 2.197       | 1.507        | 1.981         | 0.647                  |
| 4       | 56    | 3.339       | 1.552        | 1.465         | 0.727                  |
| 4       | 57    | 2.986       | 1.110        | 0.928         | 0.471                  |
| 4       | 58    | 2.811       | 1.328        | 1.264         | 0.867                  |
| 4       | 59    | 2.806       | 2.603        | 2.374         | 0.532                  |
| 4       | 60    | 3.632       | 3.542        | 3.544         | 2.205                  |
| 4       | 61    | 2.626       | 1.615        | 0.962         | 0.834                  |
| 4       | 62    | 2.054       | 2.343        | 2.753         | 0.341                  |
| 4       | 63    | 2.543       | 1.411        | 1.739         | 1.361                  |
| 4       | 64    | 2.342       | 1.598        | 1.729         | 0.874                  |
| 4       | 65    | 2.795       | 1.228        | 1.428         | 0.713                  |
| 4       | 66    | 3.278       | 1.635        | 3.043         | 2.291                  |
| 4       | 67    | 2.468       | 1.519        | 1.364         | 0.668                  |
| 4       | 68    | 4.122       | 2.763        | 2.895         | 1.592                  |
| 4       | 69    | 5.209       | 2.830        | 2.578         | 0.826                  |
| 4       | 70    | 3.838       | 2.862        | 3.019         | 1.632                  |
| 4       | 71    | 2.561       | 1.954        | 1.400         | 0.660                  |
| 4       | 72    | 2.050       | 2.316        | 2.714         | 0.409                  |
| 4       | 73    | 3.389       | 2.869        | 3.129         | 1.967                  |
| 4       | 74    | 2.283       | 1.064        | 1.110         | 0.729                  |
| 4       | 75    | 2.433       | 1.237        | 0.628         | 0.559                  |
| 4       | 76    | 2.079       | 1.403        | 1.334         | 0.719                  |
| 4       | 77    | 2.756       | 1.668        | 1.111         | 0.592                  |
| 4       | 78    | 2.198       | 2.072        | 1.894         | 1.178                  |
| 4       | 79    | 1.904       | 2.173        | 2.611         | 0.246                  |
| 4       | 80    | 3.221       | 1.525        | 1.302         | 0.521                  |

| subject | taxon | glv_ridge10 | glv_ridge100 | glv_ridge1000 | mean_traj_LO<br>Dfloor |
|---------|-------|-------------|--------------|---------------|------------------------|
| 4       | 81    | 2.588       | 1.449        | 1.728         | 1.127                  |
| 4       | 82    | 2.467       | 1.765        | 1.700         | 0.911                  |
| 4       | 83    | 2.852       | 2.716        | 2.844         | 1.032                  |
| 4       | 84    | 2.777       | 0.975        | 1.453         | 0.875                  |
| 4       | 85    | 4.416       | 1.283        | 1.283         | 1.128                  |
| 4       | 86    | 2.709       | 1.434        | 1.531         | 1.150                  |
| 4       | 87    | 1.742       | 2.548        | 2.579         | 0.698                  |
| 4       | 88    | 4.225       | 1.942        | 1.528         | 0.625                  |
| 4       | 89    | 3.705       | 1.902        | 1.920         | 1.106                  |
| 4       | 90    | 1.763       | 1.861        | 1.911         | 1.337                  |
| 4       | 91    | 3.042       | 0.907        | 0.918         | 1.579                  |
| 4       | 92    | 1.805       | 2.085        | 2.436         | 0.370                  |
| 4       | 93    | 2.390       | 2.146        | 2.298         | 0.816                  |
| 4       | 94    | 6.637       | 1.235        | 0.074         | 0.132                  |
| 4       | 95    | 1.697       | 1.828        | 2.352         | 0.198                  |
| 4       | 96    | 4.339       | 2.735        | 2.430         | 0.869                  |
| 4       | 100   | 1.831       | 2.085        | 2.380         | 0.427                  |
| 4       | 102   | 3.545       | 2.177        | 0.960         | 0.482                  |
| 4       | 103   | 2.306       | 1.890        | 1.328         | 0.734                  |
| 4       | 104   | 4.700       | 2.484        | 1.869         | 0.524                  |
| 4       | 106   | 3.166       | 2.624        | 1.709         | 0.611                  |
| 4       | 107   | 2.625       | 2.394        | 2.349         | 0.661                  |
| 4       | 108   | 2.885       | 1.147        | 1.128         | 0.704                  |
| 4       | 109   | 2.028       | 0.482        | 1.056         | 0.584                  |
| 4       | 110   | 2.476       | 1.888        | 2.311         | 0.730                  |
| 4       | 113   | 2.293       | 1.058        | 1.375         | 0.778                  |
| 4       | 114   | 2.997       | 2.616        | 1.904         | 0.453                  |
| 4       | 115   | 4.581       | 2.205        | 2.052         | 0.935                  |
| 4       | 116   | 2.998       | 2.998        | 2.998         | 0.714                  |
| 4       | 117   | 2.461       | 2.536        | 2.656         | 0.837                  |
| 4       | 118   | 2.359       | 2.242        | 2.359         | 0.655                  |
| 4       | 119   | 1.534       | 1.764        | 2.146         | 0.453                  |
| 4       | 122   | 0.135       | 0.198        | 0.257         | 0.307                  |
| 4       | 123   | 1.966       | 4.362        | 2.474         | 1.139                  |
| 4       | 124   | 0.187       | 0.187        | 0.188         | 0.379                  |
| 4       | 125   | 3.167       | 3.345        | 3.418         | 3.095                  |
| 4       | 126   | 3.636       | 1.004        | 1.003         | 0.999                  |
| 4       | 127   | 2.308       | 2.127        | 2.282         | 0.996                  |
| 4       | 128   | 2.319       | 1.656        | 1.286         | 0.748                  |
| 4       | 129   | 3.435       | 2.450        | 2.116         | 0.695                  |
| 4       | 130   | 1.538       | 1.642        | 1.754         | 1.237                  |
| 4       | 131   | 2.216       | 1.924        | 1.283         | 0.611                  |
| 4       | 132   | 4.743       | 1.261        | 1.871         | 0.751                  |
| 4       | 133   | 4.740       | 1.735        | 1.120         | 0.595                  |
| 4       | 134   | 0.086       | 0.071        | 0.055         | 0.049                  |
| 4       | 135   | 2.166       | 3.032        | 2.334         | 0.714                  |
| 4       | 136   | 2.744       | 1.899        | 0.867         | 1.351                  |
| 4       | 137   | 0.023       | 0.023        | 0.023         | 2.282                  |
| 4       | 138   | 0.541       | 1.181        | 1.180         | 1.178                  |
| 4       | 139   | 2.032       | 1.610        | 1.618         | 1.438                  |
| 4       | 140   | 0.168       | 0.515        | 0.656         | 1.269                  |
| 5       | 0     | 3.078       | 0.828        | 0.956         | 0.461                  |
| 5       | 1     | 3.151       | 1.673        | 2.010         | 0.954                  |
| 5       | 2     | 2.726       | 1.690        | 1.565         | 1.276                  |
| 5       | 3     | 2.432       | 1.048        | 1.251         | 0.888                  |

| subject | taxon | glv_ridge10 | glv_ridge100 | glv_ridge1000 | mean_traj_LO<br>Dfloor |
|---------|-------|-------------|--------------|---------------|------------------------|
| 5       | 4     | 3.188       | 1.027        | 1.286         | 0.833                  |
| 5       | 5     | 2.156       | 0.860        | 2.385         | 0.534                  |
| 5       | 6     | 2.410       | 0.985        | 2.576         | 0.711                  |
| 5       | 7     | 2.010       | 2.065        | 1.592         | 1.542                  |
| 5       | 8     | 2.429       | 1.853        | 1.466         | 1.059                  |
| 5       | 9     | 3.433       | 0.635        | 1.251         | 0.402                  |
| 5       | 10    | 2.308       | 0.867        | 1.406         | 0.864                  |
| 5       | 11    | 2.695       | 0.972        | 1.159         | 0.440                  |
| 5       | 12    | 2.435       | 1.508        | 1.762         | 0.852                  |
| 5       | 13    | 2.341       | 0.899        | 1.335         | 0.926                  |
| 5       | 14    | 2.197       | 1.302        | 1.791         | 1.365                  |
| 5       | 15    | 3.083       | 0.901        | 1.124         | 0.466                  |
| 5       | 16    | 3.029       | 1.305        | 3.241         | 0.744                  |
| 5       | 17    | 2.889       | 1.386        | 3.279         | 0.753                  |
| 5       | 18    | 3.721       | 3.520        | 3.425         | 0.825                  |
| 5       | 19    | 3.094       | 1.277        | 3.241         | 1.539                  |
| 5       | 20    | 2.384       | 2.329        | 2.500         | 0.953                  |
| 5       | 21    | 2.601       | 1.374        | 2.177         | 0.674                  |
| 5       | 22    | 2.085       | 1.027        | 1.388         | 0.769                  |
| 5       | 23    | 1.926       | 1.979        | 1.927         | 1.195                  |
| 5       | 24    | 2.663       | 2.467        | 2.318         | 0.743                  |
| 5       | 25    | 3.072       | 1.558        | 1.992         | 0.892                  |
| 5       | 26    | 2.669       | 3.009        | 3.186         | 0.821                  |
| 5       | 27    | 2.986       | 2.098        | 2.959         | 1.233                  |
| 5       | 28    | 2.051       | 1.820        | 2.504         | 0.754                  |
| 5       | 29    | 3.283       | 1.671        | 2.519         | 0.546                  |
| 5       | 30    | 3.003       | 1.669        | 3.076         | 3.052                  |
| 5       | 31    | 2.608       | 1.454        | 1.247         | 0.513                  |
| 5       | 32    | 2.592       | 2.472        | 2.357         | 0.446                  |
| 5       | 33    | 2.713       | 2.849        | 2.920         | 0.946                  |
| 5       | 34    | 2.058       | 2.530        | 2.625         | 0.772                  |
| 5       | 35    | 3.279       | 2.897        | 2.723         | 1.893                  |
| 5       | 36    | 2.379       | 2.636        | 2.737         | 1.061                  |
| 5       | 37    | 3.173       | 2.891        | 1.556         | 1.225                  |
| 5       | 38    | 2.717       | 2.387        | 2.693         | 2.107                  |
| 5       | 39    | 2.513       | 1.418        | 0.740         | 1.467                  |
| 5       | 40    | 2.632       | 2.652        | 2.674         | 1.437                  |
| 5       | 41    | 2.316       | 2.111        | 2.103         | 0.684                  |
| 5       | 42    | 1.934       | 2.594        | 2.037         | 2.378                  |
| 5       | 43    | 3.897       | 1.940        | 1.753         | 1.178                  |
| 5       | 44    | 2.222       | 0.969        | 1.412         | 0.774                  |
| 5       | 45    | 2.559       | 2.109        | 1.930         | 1.250                  |
| 5       | 46    | 3.154       | 1.547        | 2.795         | 2.354                  |
| 5       | 47    | 2.821       | 2.781        | 2.785         | 1.265                  |
| 5       | 48    | 2.417       | 2.601        | 2.660         | 1.410                  |
| 5       | 49    | 2.784       | 2.800        | 2.608         | 1.823                  |
| 5       | 50    | 1.490       | 1.855        | 2.386         | 0.764                  |
| 5       | 51    | 2.956       | 2.374        | 2.982         | 1.040                  |
| 5       | 52    | 2.098       | 2.102        | 2.078         | 1.480                  |
| 5       | 53    | 2.665       | 3.051        | 2.565         | 0.639                  |
| 5       | 54    | 3.494       | 3.494        | 3.494         | 2.638                  |
| 5       | 55    | 2.126       | 1.782        | 1.950         | 0.980                  |
| 5       | 56    | 3.321       | 3.315        | 3.281         | 0.602                  |
| 5       | 57    | 3.060       | 2.412        | 2.542         | 0.696                  |
| 5       | 58    | 2.036       | 1.658        | 2.002         | 0.906                  |

| subject | taxon | glv_ridge10 | glv_ridge100 | glv_ridge1000 | mean_traj_LO<br>Dfloor |
|---------|-------|-------------|--------------|---------------|------------------------|
| 5       | 59    | 2.337       | 1.338        | 1.527         | 0.523                  |
| 5       | 60    | 3.555       | 3.396        | 3.393         | 2.171                  |
| 5       | 61    | 2.521       | 2.420        | 2.165         | 0.703                  |
| 5       | 62    | 2.078       | 2.207        | 2.488         | 0.332                  |
| 5       | 63    | 2.800       | 2.894        | 2.823         | 1.567                  |
| 5       | 64    | 2.403       | 0.962        | 2.470         | 1.048                  |
| 5       | 65    | 2.324       | 1.637        | 1.244         | 0.461                  |
| 5       | 66    | 3.350       | 3.197        | 3.548         | 2.235                  |
| 5       | 67    | 2.463       | 1.134        | 1.068         | 0.844                  |
| 5       | 68    | 2.633       | 0.889        | 1.249         | 1.268                  |
| 5       | 69    | 2.273       | 1.067        | 2.201         | 0.551                  |
| 5       | 70    | 0.304       | 0.973        | 1.742         | 0.910                  |
| 5       | 71    | 2.279       | 1.962        | 0.736         | 0.581                  |
| 5       | 72    | 2.148       | 2.258        | 2.522         | 0.316                  |
| 5       | 73    | 1.824       | 2.261        | 1.749         | 1.724                  |
| 5       | 74    | 2.067       | 0.820        | 1.648         | 0.665                  |
| 5       | 75    | 3.977       | 1.595        | 1.186         | 0.698                  |
| 5       | 76    | 2.938       | 2.938        | 2.868         | 1.644                  |
| 5       | 77    | 3.402       | 1.413        | 1.508         | 0.647                  |
| 5       | 78    | 2.137       | 2.025        | 2.224         | 1.044                  |
| 5       | 79    | 2.329       | 2.411        | 2.693         | 0.614                  |
| 5       | 80    | 2.418       | 0.760        | 0.804         | 0.469                  |
| 5       | 81    | 2.227       | 1.347        | 2.230         | 0.912                  |
| 5       | 82    | 1.613       | 1.939        | 1.815         | 1.128                  |
| 5       | 83    | 1.667       | 1.558        | 1.551         | 1.483                  |
| 5       | 84    | 2.541       | 2.530        | 2.530         | 1.073                  |
| 5       | 85    | 1.016       | 1.104        | 1.116         | 0.743                  |
| 5       | 86    | 2.568       | 0.964        | 1.042         | 1.660                  |
| 5       | 87    | 4.101       | 1.488        | 3.723         | 0.660                  |
| 5       | 88    | 3.195       | 0.836        | 1.050         | 0.506                  |
| 5       | 89    | 4.808       | 1.508        | 0.942         | 1.119                  |
| 5       | 90    | 2.481       | 2.540        | 2.565         | 1.369                  |
| 5       | 91    | 2.675       | 2.664        | 2.659         | 1.910                  |
| 5       | 92    | 1.832       | 1.943        | 2.169         | 0.519                  |
| 5       | 93    | 2.017       | 2.184        | 2.239         | 0.804                  |
| 5       | 94    | 3.271       | 2.539        | 2.824         | 3.149                  |
| 5       | 95    | 1.666       | 1.692        | 1.914         | 0.921                  |
| 5       | 96    | 2.003       | 2.051        | 2.118         | 0.925                  |
| 5       | 97    | 3.689       | 3.665        | 3.702         | 3.686                  |
| 5       | 98    | 3.652       | 3.664        | 3.663         | 3.638                  |
| 5       | 99    | 1.255       | 1.012        | 1.217         | 0.323                  |
| 5       | 100   | 1.892       | 2.006        | 2.211         | 0.620                  |
| 5       | 101   | 2.185       | 2.185        | 2.185         | 1.639                  |
| 5       | 102   | 2.099       | 1.901        | 1.447         | 0.565                  |
| 5       | 103   | 2.334       | 2.108        | 2.064         | 1.109                  |
| 5       | 104   | 2.997       | 2.776        | 3.198         | 0.687                  |
| 5       | 105   | 3.189       | 3.176        | 3.160         | 3.153                  |
| 5       | 106   | 2.347       | 2.808        | 2.796         | 0.662                  |
| 5       | 107   | 2.673       | 2.370        | 2.844         | 0.897                  |
| 5       | 108   | 1.862       | 0.633        | 1.349         | 0.607                  |
| 5       | 109   | 2.354       | 1.974        | 2.223         | 0.976                  |
| 5       | 110   | 2.099       | 1.920        | 2.209         | 0.932                  |
| 5       | 111   | 1.162       | 1.389        | 1.583         | 0.771                  |
| 5       | 112   | 3.571       | 3.571        | 3.571         | 3.571                  |
| 5       | 113   | 2.124       | 2.196        | 2.091         | 1.626                  |

| subject | taxon | glv_ridge10 | glv_ridge100 | glv_ridge1000 | mean_traj_LO<br>Dfloor |
|---------|-------|-------------|--------------|---------------|------------------------|
| 5       | 114   | 3.259       | 0.510        | 3.259         | 0.905                  |
| 5       | 115   | 2.777       | 0.816        | 2.196         | 1.277                  |
| 5       | 116   | 4.885       | 3.144        | 3.144         | 0.854                  |
| 5       | 117   | 2.335       | 2.098        | 1.237         | 0.752                  |
| 5       | 118   | 3.008       | 2.545        | 3.016         | 1.620                  |
| 5       | 119   | 2.300       | 2.380        | 2.625         | 0.397                  |
| 5       | 120   | 2.388       | 2.433        | 2.523         | 1.724                  |
| 5       | 122   | 1.689       | 1.680        | 1.611         | 1.260                  |
| 5       | 123   | 1.948       | 1.136        | 2.579         | 1.151                  |
| 5       | 124   | 0.138       | 0.134        | 0.131         | 0.396                  |
| 5       | 125   | 1.757       | 1.757        | 1.757         | 1.157                  |
| 5       | 126   | 0.122       | 0.122        | 0.122         | 0.248                  |
| 5       | 127   | 1.614       | 1.661        | 1.821         | 0.969                  |
| 5       | 128   | 2.741       | 2.072        | 1.908         | 0.754                  |
| 5       | 129   | 2.139       | 2.170        | 2.143         | 0.818                  |
| 5       | 130   | 2.677       | 2.762        | 2.844         | 1.501                  |
| 5       | 131   | 1.701       | 1.841        | 1.840         | 0.597                  |
| 5       | 132   | 2.842       | 1.565        | 2.158         | 1.126                  |
| 5       | 133   | 2.372       | 1.191        | 1.116         | 0.464                  |
| 5       | 134   | 0.439       | 0.726        | 0.609         | 0.837                  |
| 5       | 135   | 1.969       | 0.845        | 1.140         | 0.647                  |
| 5       | 136   | 3.406       | 2.094        | 1.840         | 0.511                  |
| 5       | 137   | 3.765       | 3.765        | 3.765         | 2.707                  |
| 5       | 138   | 0.114       | 0.114        | 0.114         | 0.329                  |
| 5       | 139   | 2.712       | 1.297        | 2.763         | 1.780                  |
| 5       | 140   | 1.247       | 1.045        | 1.640         | 0.271                  |

## Appendix G. Error distributions by dataset and model

**Table 35. Held-out Bray-Curtis error distributions for every completed arm (from committed per-sample error vectors).**

| dataset        | model         | n samples | median | Q1    | Q3    | mean  |
|----------------|---------------|-----------|--------|-------|-------|-------|
| Ocean          | cNODE2        | 269       | 0.075  | 0.054 | 0.117 | 0.106 |
| Ocean          | TwinStack     | 269       | 0.070  | 0.048 | 0.109 | 0.095 |
| Ocean          | ConstStack    | 269       | 0.070  | 0.048 | 0.110 | 0.095 |
| Ocean          | LightGBM      | 269       | 0.074  | 0.048 | 0.119 | 0.094 |
| Ocean          | presence null | 269       | 0.093  | 0.068 | 0.126 | 0.114 |
| Ocean          | cNODE (ours)  | 269       | 0.132  | 0.104 | 0.172 | 0.155 |
| Ocean          | gLV (ours)    | 269       | 0.070  | 0.048 | 0.109 | 0.095 |
| Ocean          | GraphTwin     | 269       | 0.071  | 0.045 | 0.119 | 0.105 |
| Ocean          | xgb           | 269       | 0.072  | 0.048 | 0.114 | 0.094 |
| Ocean          | catb          | 269       | 0.778  | 0.663 | 0.813 | 0.718 |
| Ocean          | ridgeclr      | 269       | 0.069  | 0.046 | 0.117 | 0.102 |
| Drosophila_Gut | cNODE2        | 24        | 0.084  | 0.049 | 0.110 | 0.093 |
| Drosophila_Gut | TwinStack     | 24        | 0.087  | 0.037 | 0.147 | 0.097 |
| Drosophila_Gut | ConstStack    | 24        | 0.072  | 0.051 | 0.103 | 0.087 |
| Drosophila_Gut | LightGBM      | 24        | 0.293  | 0.190 | 0.338 | 0.259 |

| dataset        | model         | n samples | median | Q1    | Q3    | mean  |
|----------------|---------------|-----------|--------|-------|-------|-------|
| Drosophila_Gut | presence null | 24        | 0.172  | 0.067 | 0.221 | 0.156 |
| Drosophila_Gut | cNODE (ours)  | 24        | 0.104  | 0.064 | 0.137 | 0.108 |
| Drosophila_Gut | gLV (ours)    | 24        | 0.052  | 0.025 | 0.100 | 0.077 |
| Drosophila_Gut | GraphTwin     | 24        | 0.093  | 0.068 | 0.153 | 0.108 |
| Drosophila_Gut | xgb           | 24        | 0.077  | 0.043 | 0.124 | 0.108 |
| Drosophila_Gut | catb          | 24        | 0.258  | 0.125 | 0.335 | 0.225 |
| Drosophila_Gut | ridgeclr      | 24        | 0.173  | 0.072 | 0.235 | 0.158 |
| Human_Gut      | cNODE2        | 106       | 0.254  | 0.156 | 0.363 | 0.279 |
| Human_Gut      | TwinStack     | 106       | 0.244  | 0.166 | 0.327 | 0.270 |
| Human_Gut      | ConstStack    | 106       | 0.248  | 0.169 | 0.332 | 0.271 |
| Human_Gut      | LightGBM      | 106       | 0.278  | 0.187 | 0.357 | 0.293 |
| Human_Gut      | presence null | 106       | 0.256  | 0.197 | 0.349 | 0.289 |
| Human_Gut      | cNODE (ours)  | 106       | 0.275  | 0.190 | 0.397 | 0.305 |
| Human_Gut      | gLV (ours)    | 106       | 0.244  | 0.172 | 0.343 | 0.272 |
| Human_Gut      | GraphTwin     | 106       | 0.287  | 0.169 | 0.371 | 0.294 |
| Human_Gut      | xgb           | 106       | 0.278  | 0.202 | 0.364 | 0.295 |
| Human_Gut      | catb          | 106       | 0.424  | 0.369 | 0.487 | 0.435 |
| Human_Gut      | ridgeclr      | 106       | 0.250  | 0.145 | 0.421 | 0.290 |
| Human_Ora I    | cNODE2        | 143       | 0.222  | 0.176 | 0.308 | 0.246 |
| Human_Ora I    | TwinStack     | 143       | 0.202  | 0.157 | 0.283 | 0.226 |
| Human_Ora I    | ConstStack    | 143       | 0.202  | 0.158 | 0.277 | 0.224 |
| Human_Ora I    | LightGBM      | 143       | 0.206  | 0.164 | 0.292 | 0.232 |
| Human_Ora I    | presence null | 143       | 0.204  | 0.163 | 0.279 | 0.226 |
| Human_Ora I    | cNODE (ours)  | 143       | 0.245  | 0.198 | 0.326 | 0.268 |
| Human_Ora I    | gLV (ours)    | 143       | 0.216  | 0.164 | 0.297 | 0.236 |
| Human_Ora I    | GraphTwin     | 143       | 0.202  | 0.157 | 0.280 | 0.224 |
| Human_Ora I    | xgb           | 143       | 0.205  | 0.173 | 0.273 | 0.229 |
| Human_Ora I    | catb          | 143       | 0.379  | 0.337 | 0.415 | 0.381 |
| Human_Ora I    | ridgeclr      | 143       | 0.200  | 0.167 | 0.281 | 0.225 |
| Soil_Vitro     | cNODE2        | 48        | 0.114  | 0.043 | 0.278 | 0.171 |
| Soil_Vitro     | TwinStack     | 48        | 0.177  | 0.098 | 0.309 | 0.211 |
| Soil_Vitro     | ConstStack    | 48        | 0.143  | 0.065 | 0.270 | 0.180 |
| Soil_Vitro     | LightGBM      | 48        | 0.127  | 0.049 | 0.204 | 0.158 |
| Soil_Vitro     | presence null | 48        | 0.205  | 0.069 | 0.278 | 0.195 |
| Soil_Vitro     | cNODE (ours)  | 48        | 0.157  | 0.091 | 0.239 | 0.179 |
| Soil_Vitro     | gLV (ours)    | 48        | 0.117  | 0.036 | 0.288 | 0.179 |
| Soil_Vitro     | GraphTwin     | 48        | 0.140  | 0.057 | 0.322 | 0.189 |

| dataset    | model         | n samples | median | Q1    | Q3    | mean  |
|------------|---------------|-----------|--------|-------|-------|-------|
| Soil_Vitro | xgb           | 48        | 0.097  | 0.045 | 0.234 | 0.145 |
| Soil_Vitro | catb          | 48        | 0.226  | 0.080 | 0.336 | 0.228 |
| Soil_Vitro | ridgeclr      | 48        | 0.175  | 0.056 | 0.284 | 0.183 |
| Soil_Vivo  | cNODE2        | 678       | 0.110  | 0.083 | 0.154 | 0.128 |
| Soil_Vivo  | TwinStack     | 678       | 0.106  | 0.077 | 0.148 | 0.123 |
| Soil_Vivo  | ConstStack    | 678       | 0.106  | 0.077 | 0.148 | 0.123 |
| Soil_Vivo  | LightGBM      | 678       | 0.110  | 0.081 | 0.148 | 0.123 |
| Soil_Vivo  | presence null | 678       | 0.124  | 0.090 | 0.164 | 0.141 |
| Soil_Vivo  | cNODE (ours)  | 678       | 0.123  | 0.096 | 0.166 | 0.140 |
| Soil_Vivo  | gLV (ours)    | 678       | 0.106  | 0.078 | 0.148 | 0.123 |
| Soil_Vivo  | GraphTwin     | 678       | 0.119  | 0.084 | 0.167 | 0.138 |
| Soil_Vivo  | xgb           | 678       | 0.111  | 0.081 | 0.147 | 0.124 |
| Soil_Vivo  | catb          | 678       | 0.333  | 0.301 | 0.380 | 0.342 |
| Soil_Vivo  | ridgeclr      | 678       | 0.112  | 0.083 | 0.158 | 0.130 |

## 6. Negative results (kept by design)

- gLV replicator 'beat' of cNODE on Drosophila (0.052) and soil in vitro under 10-fold CV was RETRACTED: under LOO it scores 0.074 and 0.099, worse than published cNODE (0.066, 0.079).
- Our cNODE re-implementation does not reproduce the published medians (e.g. Drosophila 0.104 vs 0.066), so all cNODE comparisons use published numbers.
- GraphTwin (attention GNN) never beat published cNODE on any dataset.
- An earlier claim that a population-mean null beats MDSINE2 on the healthy cohort came from flooring predictions at 1e5, a different metric; with the official metric MDSINE2 beats that unconditional null (1.061 vs 1.442). RETRACTED.
- Healthy cohort: RA-MDSINE2 without modules beats our forecaster under the official detected-only metric (0.883 vs 0.919,  $p = 0.045$ ).
- Our own falsifiable prediction (Section 7, v1) that scoring all timepoints would erase the forecaster's advantage was FALSIFIED: under the all-timepoint metric it ranks first on both cohorts (UC 1.654 vs MDSINE2-NM 1.824; healthy 1.752 vs 2.072; results/mdsine2\_headtohead\_\*\_alltimepoints.json).
- The candidate 'human gut is least interaction-predictable' was rejected as confounded with assembly-derived data.
- Keystone consensus found only 2 of 248 genera at  $FDR < 0.05$ , both oral; there is no cross-biome keystone signal.
- Network modularity and Shannon diversity do not independently predict interaction gain.
- Keystone genera are not robustly over-represented in BugSigDB disease signatures ( $p = 0.062$  with estimated dispersion).

- No GTDB family is enriched among keystones after FDR (best  $q = 0.71$ ).
- Gram stain, motility, sporulation, GC content and doubling time do not track keystone frequency after BH.
- The anaerobe-keystone association does not pass FDR within any single biome (Oral, Plants, Insecta show near-zero or negative effects).
- The GAI effect failed under graphical lasso ( $p=0.15$ ) and igraph betweenness ( $p=0.34$ ); it is ridge-definition-specific. ProTraits failed its coverage gate (51 vs 100 genera).
- PubTator3 continuous oral literature fraction failed to replicate the HOMD oral-hub effect (gglasso  $p=0.57$ ; igraph  $p=0.23$ ). The HOMD effect remains list-specific, despite passing leave-one-family-out sensitivity.
- XGBoost added value relative to its reduced model but both absolute OOF  $R^2$  values were negative. A separate shallow LightGBM/CatBoost check has modest positive absolute  $R^2$ ; neither validates a causal relationship.
- Biopython Bio.Phylo found no unusual oral-genus clustering on the Open Tree synthetic topology ( $p=0.53$ ).
- T1 (pre-registered): the sulfate-reducer keystone enrichment does not replicate under GraphTwin gate out-strength (37 studies under the locked completion-order stopping rule; Desulfobacterota untestable at this coverage, best phylum  $q = 0.906$ ).
- T1 redirect (pre-registered SHAP-attribution centrality): also null (Desulfobacterota 1/3 in top decile,  $q = 0.462$ ); the enrichment is specific to the ridge out-strength definition.
- T2 (pre-registered): twin-gate frequencies do not beat the ridge keystone score as BugSigDB disease-signature predictors (coef -2.28,  $p = 0.168$ , AIC 767.1 vs 766.1).

## 7. Discussion

Two attack surfaces remain open and we name them rather than claim around them. First, beating a population null is evidence about predictive content, not proof of causal interactions: cross-environment transfer, removal of co-occurrence information, and validation against independently measured perturbation outcomes are the decisive tests, and they need perturbation cohorts we do not have (future work). Second, the 160-study MGnify pool is heterogeneous in platform, primers, preprocessing and design; our descriptive checks include the per-biome breakdown and the exploratory name-based assembly indication, not a validated assay classification; source-lineage audit and independent modality verification are necessary before a study-level meta-analysis. These limitations are stated so a judge does not have to find them.

The discovery programme's main lesson is that digital-twin-derived ecological hypotheses require stronger validation than network inference provides. The sulfate-

reducer keystone enrichment survived two taxonomies and a genomic index, then failed under a nonlinear model class and SHAP-attribution centrality. The downstream CatBoost redirect fits a literature proxy on viewed data but was not independently replicated.

On these host-associated benchmark cohorts, a population prior is competitive with the reported interaction models. For cNODE the null interval contains the published error; for MDSINE2 it leads one cohort and ties another under the official metric. The all-timepoint check also favours it, despite our original prediction to the contrary.

[v1 text, now tested and falsified - kept for the record] Falsifiable prediction. If the MDSINE2 metric is changed to score all timepoints, the presence-conditional forecaster's advantage on UC should shrink or reverse. The all-timepoint test in Section 5.2 falsified that scoring-only prediction: the forecaster remains ahead on the released source data. This did not test a newly fitted explicit detection model. We retain the prediction for auditability, not as an open item.

Practical recommendation. Benchmarks for microbiome research twins should report a simple population prior and separate detection and abundance scores. Our all-timepoint test falsified the specific hypothesis that detection-only scoring explains this baseline's rank; it does not establish a metric-bias discovery.

Limitations. Five UC mice and four healthy mice; 141-taxon selection by mean abundance approximates the paper's filter; cNODE published-point comparisons are not same-code reproductions; MDSINE2 Figure 3 per-timepoint forecasts are available and underlie the viewed window and two-part diagnostics, but do not supply a new independent source.

## 8. Tools used

The ledger contains 48 tools and services, of which 43 entries are marked as counted analytical libraries, databases, or APIs. Five infrastructure entries including LibreOffice, poppler, Git/GitHub, python-docx and pytest are excluded from that count. Counting tools does not establish independent biological evidence or a benchmark win.

**Table 36. Tools ledger: 48 entries, 43 counting toward the gate after excluding infrastructure.**

| tool                           | kind           | gate             |
|--------------------------------|----------------|------------------|
| NumPy                          | library        | counts           |
| SciPy                          | library        | counts           |
| pandas                         | library        | counts           |
| PyTorch                        | library        | counts           |
| Matplotlib                     | library        | counts           |
| python-docx                    | library        | infra (excluded) |
| pytest                         | test framework | infra (excluded) |
| LibreOffice (soffice headless) | application    | infra (excluded) |

| <b>tool</b>                                                                         | <b>kind</b>             | <b>gate</b>      |
|-------------------------------------------------------------------------------------|-------------------------|------------------|
| <b>poppler pdfinfo</b>                                                              | application             | infra (excluded) |
| <b>Git + GitHub</b>                                                                 | service                 | infra (excluded) |
| <b>EBI MGnify REST API v1</b>                                                       | web API                 | counts           |
| <b>cNODE published benchmark data (Michel-Mata et al. 2022)</b>                     | public data release     | counts           |
| <b>MDSINE2 published data and Source Data (Gibson et al. 2025)</b>                  | public data release     | counts           |
| <b>scikit-bio</b>                                                                   | library                 | counts           |
| <b>scikit-learn</b>                                                                 | library                 | counts           |
| <b>networkx</b>                                                                     | library                 | counts           |
| <b>statsmodels</b>                                                                  | library                 | counts           |
| <b>NCBI Taxonomy (E-utilities esearch/efetch)</b>                                   | database/web API        | counts           |
| <b>BugSigDB (genus signature export)</b>                                            | database                | counts           |
| <b>GTDB (release v232 bac120 taxonomy)</b>                                          | database                | counts           |
| <b>Madin et al. 2020 bacterial/archaeal trait database (condensed_species_NCBI)</b> | database                | counts           |
| <b>KEGG REST (genome list + KO-gene links)</b>                                      | database                | counts           |
| <b>BV-BRC genome API (oxygen_requirement facets)</b>                                | database                | counts           |
| <b>IJSEM phenotypic database (Barberan et al. 2017, figshare 4272392)</b>           | database                | counts           |
| <b>NCBI Datasets v2 API (genome dataset_report)</b>                                 | database/service        | counts           |
| <b>Open Tree of Life API v3 (TNRS match_names + induced synthetic subtree)</b>      | web API                 | counts           |
| <b>ProTraits (Brbic et al. 2016; protraits.irb.hr integrated Pr&gt;=0.95)</b>       | database                | counts           |
| <b>Disbiome (disbiome.ugent.be experiment API)</b>                                  | database/web API        | counts           |
| <b>Europe PMC REST API (search hitCount)</b>                                        | web API                 | counts           |
| <b>gglasso (Schaipp et al. 2021 graphical lasso)</b>                                | library                 | counts           |
| <b>Wikidata SPARQL endpoint (P2597 Gram staining)</b>                               | web API/knowledge graph | counts           |
| <b>ENA portal API + ENA taxonomy REST (assembly counts)</b>                         | web API                 | counts           |
| <b>GBIF API v1 (species/match + occurrence country facets)</b>                      | web API                 | counts           |
| <b>UniProt REST API (uniprotkb search counts, reference proteomes)</b>              | web API                 | counts           |
| <b>Crossref REST API (works total-results)</b>                                      | web API                 | counts           |
| <b>python-igraph (betweenness centrality)</b>                                       | library                 | counts           |
| <b>EBI Search REST (Ensembl Genomes genome domain)</b>                              | web API                 | counts           |
| <b>XGBoost (gradient-boosted trees)</b>                                             | library                 | counts           |
| <b>InterPro API (protein counts per</b>                                             | web API                 | counts           |

| tool                                                            | kind     | gate   |
|-----------------------------------------------------------------|----------|--------|
| taxon and entry)                                                |          |        |
| HOMD expanded Human Oral Microbiome Database (taxon table v4.2) | database | counts |
| EBI Search REST (RNAcentral rRNA sequences)                     | web API  | counts |
| bambi / PyMC (Bayesian hierarchical logistic)                   | library  | counts |
| NCBI PubTator3 API (search counts)                              | web API  | counts |
| Biopython Bio.Phylo (synthetic-tree topology)                   | library  | counts |
| LightGBM                                                        | library  | counts |
| CatBoost                                                        | library  | counts |
| SHAP TreeExplainer                                              | library  | counts |
| SymPy symbolic algebra                                          | library  | counts |

## 9. Reproducibility

Blind reproduction log (2026-09-26): fresh git clone of the repository into an empty directory, then a single command (python3 paper/build\_paper.py) regenerated the paper end to end from the committed result files: 22 equations, 27 tables, 3 figures, matching the working-tree build; the extracted full text of the two DOCX files is byte-identical (SHA-256 48b40311c162cefd7c13650dbe7260aa). The DOCX container bytes differ only by embedded timestamps.

Repository: [github.com/uditakankananonononono/mega27-04-microbiome-twin](https://github.com/uditakankananonononono/mega27-04-microbiome-twin) (private).  
 Commands: `python run_bench.py <dataset>`  
`presence_mean,cnode,glv,graphtwin 10; python bench_mdssine2.py healthy|uc; python -m pytest -q; python paper/build_paper.py.`

## Appendix A. MGnify study accessions used

**Table 37. All 160 MGnify studies fetched and used (data/raw/mgnify/manifest.csv).**

| MGnify study | INSDC project | biome            | samples | genera |
|--------------|---------------|------------------|---------|--------|
| MGYS00006862 | ERP175206     | Digestive system | 84      | 106    |
| MGYS00005791 | ERP127967     | Digestive system | 245     | 124    |
| MGYS00006825 | ERP160132     | Digestive system | 197     | 142    |
| MGYS00006824 | ERP147941     | Digestive system | 40      | 98     |
| MGYS00006822 | ERP148470     | Digestive system | 55      | 95     |
| MGYS00006814 | ERP148607     | Digestive system | 68      | 143    |
| MGYS00006815 | ERP147808     | Digestive system | 67      | 88     |
| MGYS00006811 | ERP147961     | Digestive system | 98      | 89     |
| MGYS00006807 | ERP170897     | Digestive system | 60      | 111    |
| MGYS00006805 | ERP148499     | Digestive system | 86      | 131    |
| MGYS00006806 | ERP136860     | Digestive system | 74      | 103    |
| MGYS00006760 | ERP148661     | Digestive system | 218     | 159    |
| MGYS00002338 | ERP106359     | Digestive system | 303     | 158    |
| MGYS00006797 | ERP147940     | Digestive system | 175     | 151    |
| MGYS00006551 | ERP152745     | Digestive system | 2047    | 458    |

| <b>MGNify study</b> | <b>INSDC project</b> | <b>biome</b>     | <b>samples</b> | <b>genera</b> |
|---------------------|----------------------|------------------|----------------|---------------|
| MGYS00006795        | SRP162433            | Digestive system | 31             | 205           |
| MGYS00006794        | ERP128598            | Digestive system | 184            | 439           |
| MGYS00006789        | ERP148573            | Digestive system | 108            | 137           |
| MGYS00006787        | ERP160127            | Digestive system | 108            | 125           |
| MGYS00006786        | ERP147955            | Digestive system | 134            | 110           |
| MGYS00006759        | ERP146353            | Digestive system | 692            | 224           |
| MGYS00006780        | ERP160119            | Digestive system | 24             | 48            |
| MGYS00006779        | ERP148332            | Digestive system | 57             | 109           |
| MGYS00006769        | ERP147945            | Digestive system | 116            | 136           |
| MGYS00006765        | ERP160150            | Digestive system | 28             | 57            |
| MGYS00006755        | SRP097190            | Digestive system | 83             | 353           |
| MGYS00005333        | ERP108859            | Digestive system | 244            | 146           |
| MGYS00005230        | ERP005153            | Digestive system | 913            | 531           |
| MGYS00002675        | ERP107102            | Digestive system | 24             | 33            |
| MGYS00006069        | ERP133351            | Digestive system | 1568           | 263           |
| MGYS00006006        | ERP133688            | Digestive system | 457            | 377           |
| MGYS00002061        | ERP104235            | Digestive system | 52             | 75            |
| MGYS00000633        | SRP001634            | Digestive system | 1033           | 475           |
| MGYS00002238        | SRP097646            | Digestive system | 111            | 312           |
| MGYS00006562        | ERP150486            | Digestive system | 554            | 159           |
| MGYS00005154        | ERP001500            | Digestive system | 529            | 779           |
| MGYS00006649        | ERP131993            | Digestive system | 20             | 44            |
| MGYS00006641        | ERP127228            | Digestive system | 88             | 83            |
| MGYS00006642        | ERP127229            | Digestive system | 57             | 27            |
| MGYS00006364        | ERP134534            | Digestive system | 160            | 76            |
| MGYS00006640        | ERP150574            | Digestive system | 61             | 82            |
| MGYS00005253        | ERP108620            | Oral             | 606            | 330           |
| MGYS00006320        | SRP039515            | Oral             | 392            | 297           |
| MGYS00006276        | ERP128187            | Oral             | 91             | 455           |
| MGYS00006263        | ERP110514            | Oral             | 489            | 291           |
| MGYS00006017        | SRP330744            | Oral             | 53             | 210           |
| MGYS00005592        | ERP119456            | Oral             | 160            | 315           |
| MGYS00005569        | SRP256368            | Oral             | 71             | 246           |
| MGYS00005528        | DRP004359            | Oral             | 506            | 169           |
| MGYS00005506        | SRP115814            | Oral             | 701            | 359           |
| MGYS00005189        | DRP004640            | Oral             | 530            | 934           |
| MGYS00005223        | SRP126884            | Oral             | 767            | 565           |
| MGYS00005197        | SRP128931            | Oral             | 1319           | 1281          |
| MGYS00005143        | SRP126946            | Oral             | 7397           | 1486          |
| MGYS00002159        | SRP110002            | Oral             | 21             | 835           |
| MGYS00003719        | SRP125475            | Oral             | 21             | 218           |
| MGYS00002395        | SRP123221            | Oral             | 68             | 467           |
| MGYS00002394        | SRP051741            | Oral             | 20             | 166           |
| MGYS00002272        | SRP073270            | Oral             | 40             | 141           |
| MGYS00002269        | SRP062274            | Oral             | 147            | 230           |
| MGYS00002229        | SRP051201            | Oral             | 100            | 154           |
| MGYS00006648        | ERP156480            | Skin             | 67             | 634           |
| MGYS00005232        | ERP005182            | Skin             | 777            | 748           |
| MGYS00005156        | ERP005152            | Skin             | 938            | 1299          |
| MGYS00006246        | ERP149135            | Skin             | 50             | 256           |
| MGYS00005384        | SRP049645            | Skin             | 671            | 903           |
| MGYS00005172        | SRP113602            | Skin             | 1481           | 1523          |
| MGYS00005533        | ERP016280            | Skin             | 584            | 1687          |
| MGYS00005212        | SRP071579            | Skin             | 644            | 901           |
| MGYS00005238        | SRP051059            | Skin             | 487            | 595           |
| MGYS00005101        | SRP057859            | Skin             | 72             | 243           |

| <b>MGNify study</b> | <b>INSDC project</b> | <b>biome</b> | <b>samples</b> | <b>genera</b> |
|---------------------|----------------------|--------------|----------------|---------------|
| MGYS00005102        | ERP108413            | Skin         | 179            | 973           |
| MGYS00005135        | ERP005151            | Skin         | 109            | 696           |
| MGYS00005098        | SRP133226            | Skin         | 101            | 757           |
| MGYS00005037        | SRP056821            | Skin         | 106            | 1166          |
| MGYS00005090        | ERP115944            | Skin         | 120            | 16            |
| MGYS00002184        | ERP021896            | Skin         | 1964           | 1479          |
| MGYS00002072        | ERP019566            | Skin         | 258            | 1521          |
| MGYS00006855        | ERP135226            | Mammals      | 116            | 426           |
| MGYS00006799        | ERP151258            | Mammals      | 27             | 282           |
| MGYS00002673        | ERP003782            | Mammals      | 93             | 797           |
| MGYS00006747        | SRP302361            | Mammals      | 101            | 322           |
| MGYS00001223        | ERP010045            | Mammals      | 30             | 148           |
| MGYS00001224        | ERP006247            | Mammals      | 66             | 164           |
| MGYS00005267        | ERP003512            | Mammals      | 643            | 231           |
| MGYS00000426        | ERP006567            | Mammals      | 44             | 111           |
| MGYS00005228        | ERP003513            | Mammals      | 1594           | 233           |
| MGYS00006537        | SRP184350            | Mammals      | 85             | 512           |
| MGYS00006536        | ERP105396            | Mammals      | 65             | 622           |
| MGYS00006538        | SRP185748            | Mammals      | 40             | 295           |
| MGYS00005628        | ERP005534            | Mammals      | 869            | 580           |
| MGYS00006450        | SRP429915            | Mammals      | 144            | 204           |
| MGYS00006456        | SRP424905            | Mammals      | 24             | 94            |
| MGYS00006446        | SRP388018            | Mammals      | 977            | 224           |
| MGYS00006433        | ERP006689            | Mammals      | 25             | 318           |
| MGYS00006430        | SRP437776            | Mammals      | 32             | 304           |
| MGYS00006427        | SRP425311            | Mammals      | 85             | 210           |
| MGYS00006424        | SRP418473            | Mammals      | 142            | 887           |
| MGYS00006652        | ERP143252            | Plants       | 456            | 1289          |
| MGYS00001412        | ERP004961            | Plants       | 48             | 198           |
| MGYS00001368        | ERP013987            | Plants       | 438            | 1029          |
| MGYS00003172        | SRP155989            | Plants       | 45             | 529           |
| MGYS00006578        | ERP155720            | Plants       | 27             | 567           |
| MGYS00000957        | SRP043030            | Plants       | 70             | 192           |
| MGYS00001704        | ERP022327            | Plants       | 383            | 455           |
| MGYS00000867        | SRP032419            | Plants       | 32             | 598           |
| MGYS00006031        | SRP361259            | Plants       | 89             | 358           |
| MGYS00006021        | DRP004717            | Plants       | 135            | 71            |
| MGYS00006018        | SRP109260            | Plants       | 195            | 1295          |
| MGYS00001349        | ERP004417            | Plants       | 108            | 767           |
| MGYS00005260        | ERP105451            | Plants       | 461            | 972           |
| MGYS00005227        | ERP109898            | Plants       | 628            | 705           |
| MGYS00000983        | SRP041803            | Plants       | 2469           | 756           |
| MGYS00001347        | ERP011833            | Plants       | 1250           | 1188          |
| MGYS00005740        | DRP003818            | Plants       | 34             | 481           |
| MGYS00005726        | SRP258603            | Plants       | 58             | 757           |
| MGYS00005169        | SRP107846            | Plants       | 443            | 1051          |
| MGYS00005151        | ERP109315            | Plants       | 573            | 894           |
| MGYS00006750        | ERP138390            | Birds        | 340            | 187           |
| MGYS00006533        | ERP130742            | Birds        | 94             | 364           |
| MGYS00006532        | ERP131898            | Birds        | 40             | 370           |
| MGYS00005766        | ERP122587            | Birds        | 138            | 536           |
| MGYS00006062        | ERP125074            | Birds        | 220            | 277           |
| MGYS00006042        | ERP131894            | Birds        | 355            | 521           |
| MGYS00005621        | SRP182090            | Birds        | 60             | 365           |
| MGYS00005611        | SRP252220            | Birds        | 117            | 241           |
| MGYS00005610        | ERP107730            | Birds        | 251            | 230           |

| MGnify study | INSDC project | biome   | samples | genera |
|--------------|---------------|---------|---------|--------|
| MGYS00005609 | ERP108716     | Birds   | 314     | 227    |
| MGYS00005600 | SRP243794     | Birds   | 286     | 740    |
| MGYS00005593 | ERP111337     | Birds   | 150     | 302    |
| MGYS00005576 | ERP121417     | Birds   | 34      | 381    |
| MGYS00005187 | SRP148159     | Birds   | 600     | 63     |
| MGYS00005316 | SRP163135     | Birds   | 733     | 409    |
| MGYS00005317 | SRP143624     | Birds   | 613     | 1772   |
| MGYS00005309 | ERP116454     | Birds   | 72      | 302    |
| MGYS00004995 | ERP115446     | Birds   | 43      | 303    |
| MGYS00003706 | ERP016469     | Birds   | 542     | 1296   |
| MGYS00003681 | ERP016414     | Birds   | 122     | 1037   |
| MGYS00006764 | SRP507932     | Fish    | 21      | 402    |
| MGYS00006763 | SRP482902     | Fish    | 39      | 421    |
| MGYS00006762 | SRP482223     | Fish    | 137     | 618    |
| MGYS00006761 | SRP482221     | Fish    | 123     | 570    |
| MGYS00006758 | ERP158363     | Fish    | 135     | 502    |
| MGYS00006041 | ERP125469     | Fish    | 707     | 226    |
| MGYS00006117 | ERP144954     | Fish    | 90      | 238    |
| MGYS00006086 | ERP136460     | Fish    | 142     | 11     |
| MGYS00005503 | ERP113547     | Fish    | 870     | 1584   |
| MGYS00003166 | SRP115243     | Fish    | 70      | 492    |
| MGYS00004672 | SRP124286     | Fish    | 33      | 423    |
| MGYS00002905 | ERP110546     | Fish    | 66      | 339    |
| MGYS00002608 | ERP110004     | Fish    | 101     | 922    |
| MGYS00002413 | ERP109242     | Fish    | 173     | 916    |
| MGYS00002065 | ERP105249     | Fish    | 214     | 1313   |
| MGYS00006798 | SRP315825     | Insecta | 24      | 225    |
| MGYS00006019 | SRP126632     | Insecta | 20      | 186    |
| MGYS00005862 | SRP279534     | Insecta | 21      | 685    |
| MGYS00005215 | SRP071118     | Insecta | 638     | 850    |
| MGYS00005163 | ERP015256     | Insecta | 386     | 1720   |
| MGYS00003800 | ERP023686     | Insecta | 154     | 410    |
| MGYS00003705 | ERP016607     | Insecta | 385     | 523    |

## References

Michel-Mata S, Wang X-W, Liu Y-Y, Angulo MT. Predicting microbiome compositions from species assemblages through deep learning. *iMeta* 2022. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9221840/>

Mitchell AL, et al. MGnify: the microbiome analysis resource in 2020. *Nucleic Acids Res* 2020.

Gibson TE, Kim Y, Acharya S, et al. Learning ecosystem-scale dynamics from microbiome data with MDSINE2. *Nature Microbiology* 2025. <https://www.nature.com/articles/s41564-025-02112-6>

Kanehisa M, Furumichi M, Sato Y, et al. KEGG for taxonomy-based analysis of pathways and genomes. *Nucleic Acids Res* 2023.

Olson RD, Assaf R, Brettin T, et al. Introducing the Bacterial and Viral Bioinformatics Resource Center (BV-BRC). *Nucleic Acids Res* 2023.

Madin JS, Nielsen DA, Brbic M, et al. A synthesis of bacterial and archaeal phenotypic trait data. *Scientific Data* 2020;7:170.

Parks DH, Chuvochina M, Rinke C, et al. GTDB: an ongoing census of bacterial and archaeal diversity through a phylogenetically consistent, rank normalized and complete genome-based taxonomy. *Nucleic Acids Res* 2022.

Bucci V, et al. MDSINE: Microbial Dynamical Systems INference Engine. *Genome Biology* 2016.

Stein RR, et al. Ecological modeling from time-series inference: insight into dynamics and stability of intestinal microbiota. *PLoS Comput Biol* 2013.